

Solutions Manual

Mathematical Programming and Operations Research — Book 1:
Linear and Integer Programming

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1 Chapter 1: Mathematical Programming

Coverage assessment. The eleven exercises track the chapter closely. The four warm-ups drill the classification taxonomy of Sections 1.5 through 1.8 (LP, BIP/ILP, NLP, MINLP), one exercise per class, which is exactly the skill the chapter builds. The core problems return to the Section 1.1 shirt store and to the model components of Section 1.3 (variables, objective, constraints), and the concept exercises cover the divisibility issue, the history of the word programming, and the Section 1.2 success stories. The challenge problem closes the loop by turning the warm-up scenario into a full solvable LP, so the ladder flows cleanly from one-star recognition to three-star formulation. Gaps are minor: nothing exercises the machine learning subsections (loss minimization, k-means) or the optimization-under-uncertainty section, both of which are explicitly out of scope for the book, so this is defensible. All exercises appear original to the book; no licensing concerns.

Exercise 1.1 (Classify It: Production Plan) ★

This is a **linear program (LP)**. The two features that justify the classification are: (i) the type of functions, since the objective $2x_1 + 3x_2$ and the constraint left-hand sides $x_1 + 2x_2$ and $x_1 + x_2$ are all linear functions of the decision variables; and (ii) the type of variables, since x_1 and x_2 are continuous quantities (liters of juice) restricted only by linear inequalities and non-negativity, with no integrality requirement. Linearity rules out NLP and MINLP; continuity rules out ILP, BIP, and MILP. Written out, the model is

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 800 \\ & x_1 + x_2 \leq 600 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Exercise 1.2 (Classify It: Choosing Projects) ★

This is a **binary integer program (BIP)**, a special case of ILP: the objective $\sum_i c_i x_i$ and the budget constraint $\sum_i a_i x_i \leq b$ are linear, and every variable is restricted to the discrete set $\{0, 1\}$. The binary restriction is what rules out LP.

If the city may fund fractions of projects, replace $x_i \in \{0, 1\}$ with $0 \leq x_i \leq 1$. All functions are then linear over continuous variables and the model becomes an **LP**. (This LP is the linear relaxation of the BIP; its optimal value is always at least as large as the BIP value, since every BIP-feasible point is LP-feasible.)

Exercise 1.3 (Classify It: Fitting a Line) ★

This is a **nonlinear program (NLP)**. The decision variables m and q are continuous and unconstrained, so no integer class applies, but the objective

$$\sum_{i=1}^N (y_i - (mt_i + q))^2$$

is *not* linear in m and q : expanding any one term produces $m^2 t_i^2 + q^2 + 2mqt_i + \dots$, so the objective is a quadratic function of the decision variables. (The model $f(t) = mt + q$ is linear in t , but classification is always with respect to the *decision variables*, here m and q .) In fact this is a convex quadratic minimization, the classical least-squares problem, so it sits in the easy convex corner of NLP and even has a closed-form solution.

Exercise 1.4 (Classify It: Warehouses with Congestion)

★

This is a **mixed-integer nonlinear program (MINLP)**. Ruling out the simpler classes one ingredient at a time:

- The quadratic handling costs x_j^2 in the objective rule out LP, ILP/BIP, and MILP, all of which require linear functions.
- The binary open/close variables $y_j \in \{0, 1\}$ rule out LP and NLP, which allow only continuous variables.
- The model mixes continuous flows $x_j \geq 0$ with binary y_j , so it is not a pure integer model.

Nonlinear functions plus a mix of integer and continuous variables is precisely the definition of MINLP. (The linking requirement that flow is allowed only when $y_j = 1$ is typically written $x_j \leq My_j$ for a large constant M ; that constraint itself is linear.)

Exercise 1.5 (Shirt Store: Variables and One Constraint)

★★

1. Let $x_T, x_F, x_S \geq 0$ be the number of shirts ordered on Thursday, Friday, and Saturday, and let $s_F, s_S \geq 0$ be the inventory on hand at the end of Friday and at the end of Saturday.
2. Thursday's order is everything the store has available on Friday (the store starts empty and orders arrive one day after placement). Whatever is not sold to Friday's 5 customers is Friday's ending inventory:

$$x_T = 5 + s_F.$$

3. The non-negativity constraint $s_F \geq 0$, combined with the balance equation above, is what prevents a Friday stockout: together they force $x_T \geq 5$, so the store must order at least enough on Thursday to cover Friday's demand. Without $s_F \geq 0$, the model could "borrow" shirts by letting the inventory go negative.

Exercise 1.6 (Spot the Pieces: Food Truck)

★★

A model answer in words:

- **Decision variables:** the number of tacos to prepare (units: tacos) and the number of burritos to prepare (units: burritos) for tomorrow.
- **Objective function:** maximize total profit (units: dollars), which is \$2 per taco times the number of tacos plus \$3.50 per burrito times the number of burritos.
- **Constraints:**
 - Meat: 0.1 kg per taco times the number of tacos plus 0.25 kg per burrito times the number of burritos must not exceed 30 kg of available meat (units: kilograms).
 - Labor: prep time per taco times the number of tacos plus prep time per burrito times the number of burritos must not exceed 10 hours (units: hours). The per-item prep times are data the owner must supply.
 - Non-negativity: both production quantities are at least zero.

QA flag

Underspecified data (issue type b): the exercise gives the 10-hour labor limit but no prep time per taco or per burrito, so the labor constraint cannot be written numerically. This is workable for an identify-the-pieces exercise (the student can name the constraint and observe that data is missing), but if that is not the intent, one added sentence fixes it, for example "each taco takes 3 minutes of prep and each burrito 5 minutes." Recommendation: either add the prep times or add a sentence asking students to identify what data is missing, making the omission clearly deliberate.

Exercise 1.7 (Spot the Pieces: Tutoring Center)

★★

Decision variables: for each of the 12 tutors, the number of one-hour sessions that tutor works next week (units: sessions, a whole number between 0 and 8). **Objective:** maximize the total number of sessions covered (units: sessions). **Constraints:** (i) each tutor works at most 8 sessions; (ii) the total number of sessions is at least 40; (iii) total spending, tutor i 's rate p_i dollars per session times the sessions assigned to tutor i , summed over tutors, is at most the \$900 budget (units: dollars); (iv) session counts are non-negative integers.

Model class: this is an **integer linear program (ILP)**. All functions (objective, budget, session counts) are linear, but the decision variables must be whole numbers of sessions, since a tutor cannot be hired for a fraction of a one-hour session. The integrality requirement is exactly what keeps this from being an LP, and it is what makes the problem potentially much harder to solve.

Exercise 1.8 (Half a Bus)

★★

1. Rounding is likely acceptable for the shirts: 1045.3 rounded to 1045 changes the plan by about 0.03 percent, and shirts are cheap, so the cost impact and the feasibility risk are negligible. For the buses, rounding is risky: 12.5 to 12 or 13 is a 4 percent change in a decision where each unit is expensive, and rounding down may leave demand uncovered while rounding up may waste a large capital cost. The rounded plan may be far from optimal, or not even feasible.
2. Divisibility is the LP assumption that fractional variable values make sense. When they do not, and the numbers involved are small enough that rounding is unsafe, we must move from the LP class to the MILP class and impose integrality explicitly. The price is computational: LPs are solvable in polynomial time, while MILP is NP-hard, so insisting on integer variables moves the problem into a class where solve times can grow explosively with size.

Exercise 1.9 (Why "Programming"?)

★★

In this subject, "programming" means planning or scheduling: a "program" was a plan of activities, as in a military logistics program or a concert program, and the term was coined in the 1940s before computer programming was common. Linear programming therefore means constructing an optimal plan using linear relationships; it has nothing to do with writing computer code in any particular style.

Exercise 1.10 (Reading a Success Story)

★★

Answers vary; the goal is to translate a headline into model components. A model answer for UPS ORION:

- (a) **Decision variables:** for each driver, the choice of route, in practice binary variables indicating whether the driver travels directly from one stop to another, which together encode the order in which that driver's packages are delivered.
- (b) **Objective:** minimize total cost of the routes, dominated by total miles driven (equivalently fuel and time), summed over all drivers.
- (c) **One constraint the model must respect:** every package on the truck is delivered exactly once (each stop is visited); other acceptable answers include delivery time windows for scheduled packages, vehicle capacity, and driver shift lengths.

For American Airlines yield management, acceptable answers are variables for how many seats to sell in each fare class on each flight, objective of maximizing expected revenue, and the constraint that seats sold cannot exceed aircraft capacity. For Netherlands Railways, variables are departure and arrival times and platform/rolling-stock assignments, the objective mixes profit and service quality, and constraints include track capacity and minimum transfer times.

Exercise 1.11 (Shirt Store: The Full Model)

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1. Let $x_T, x_F, x_S \geq 0$ be the shirts ordered Thursday, Friday, and Saturday (arriving the next day), and $s_F, s_S, s_{Sun} \geq 0$ the end-of-day inventories. Each day, arriving shirts plus carried inventory must cover that day's demand, with the excess carried forward:

$$\begin{aligned} \min \quad & 8x_T + 9x_F + 10x_S + 0.5s_F + 0.5s_S \\ \text{s.t.} \quad & x_T = 5 + s_F && \text{(Friday balance)} \\ & s_F + x_F = 10 + s_S && \text{(Saturday balance)} \\ & s_S + x_S = 7 + s_{Sun} && \text{(Sunday balance)} \\ & x_T, x_F, x_S, s_F, s_S, s_{Sun} \geq 0. \end{aligned}$$

2. Compare the ways to cover each day's demand. A Saturday shirt ordered Thursday costs $8 + 0.5 = \$8.50$ (one night of holding), cheaper than \$9 ordered Friday. A Sunday shirt ordered Thursday costs $8 + 0.5 + 0.5 = \$9$, cheaper than \$9.50 (ordered Friday, held one night) and \$10 (ordered Saturday). So the optimal plan orders everything Thursday:

$$x_T = 22, \quad x_F = x_S = 0, \quad s_F = 17, \quad s_S = 7, \quad s_{Sun} = 0,$$

with total cost $22 \cdot 8 + 0.5(17 + 7) = \188 . (Verified with an LP solver: the optimum is exactly 188 at this point.)

3. Integrality does not matter for this instance. The demands are integers and the optimal LP solution is already integer, so adding the requirement $x, s \in \mathbb{Z}$ would not change the answer. In general, inventory-balance models with integer demands like this one have integer optimal solutions, so the LP is safe here; the divisibility assumption only becomes a real issue when the LP optimum comes out fractional.

2 Chapter 2: Modeling: Linear Programming

Coverage assessment. The fourteen exercises drill the two skills the chapter teaches: recognizing linearity (2.1, 2.2, 2.11, tied to Definition 2.1 and the Section 2.2 assumptions) and formulating word problems as LPs (2.3 through 2.10 and 2.14, tied to the worked examples of Section 2.3: production mix, diet, work scheduling). The assumptions of Section 2.2.1 get direct treatment in 2.12 (certainty) and 2.13 (all four), and 2.10(c) plants the LP-bound-implies-integer-bound idea that pays off in the integer programming chapters. The ladder flows well: one-star pattern-matching against worked examples, two-star fresh formulations, a three-star twist (a resource you can buy). Gaps: the knapsack, capital allocation, transportation, and assignment subsections (2.3.1 through 2.3.4) have no matching exercises in this chapter, so those model types go unpracticed here; presumably they recur in the integer programming chapters, but one binary knapsack and one small transportation exercise would round this set out. Licensing: the chapter's exercise section carries a footnote attributing adapted exercises to Sekhon and Bloom (LibreTexts, CC BY 4.0), which covers the oven store, two-machine, factory scheduling, and quiz grading problems; the rest appear original. No verbatim matches to closed-license textbooks found.

Exercise 2.1 (Identify Linear Functions) ★

Only item (4) is linear.

1. $x^2 + 3y + 2z$: not linear; the term x^2 is a variable raised to a power other than one.
2. $\sqrt{x_1 x_2 x_3}$: not linear; the variables are multiplied together and put under a root.
3. $\sum_{i=1}^n a_i 2^{x_i}$: not linear; each variable appears in an exponent, and 2^{x_i} is an exponential function of x_i .
4. $\sum_{i=1}^n a_i x_i$: linear; it is exactly the form $c_1 x_1 + \cdots + c_n x_n$ of Definition 2.1, with constant coefficients a_i .

The test throughout: every term must be a constant times a single variable to the first power.

Exercise 2.2 (Identify Linear Function) ★

Function (2) is linear.

1. $2x_1 + x_2 x_3 - 7x_3$: not linear because of the product $x_2 x_3$ of two decision variables.
2. $\sqrt{2}x_1 + x_2 + 3^{1.6}x_3$: linear. The coefficients $\sqrt{2}$ and $3^{1.6}$ look exotic, but they are *constants*; each term is still a constant times a single variable. Linearity is about how the *variables* enter, not about how pretty the coefficients are.

QA flag

Mildly duplicative (issue type d): Exercise 2.2 repeats the skill of Exercise 2.1 with two more items. It does add one new lesson (irrational constant coefficients are fine) that 2.1 does not test. Recommendation: merge these two items into Exercise 2.1 as parts (5) and (6) and retire 2.2, or keep as is if the redundancy is intentional for drill.

Exercise 2.3 (Hoodies and Posters)

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Let x_1 be the number of hoodies and x_2 the number of posters produced per week. The profit coefficients are revenue minus material cost: $\$30 - \$18 = \$12$ per hoodie and $\$8 - \$3 = \$5$ per poster. The LP, mirroring the screen printing example, is

$$\begin{array}{ll} \max & z = 12x_1 + 5x_2 \\ \text{s.t.} & 3x_1 + x_2 \leq 120 & (\text{printing hours}) \\ & x_1 + x_2 \leq 70 & (\text{finishing hours}) \\ & x_1 \leq 35 & (\text{market limit}) \\ & x_1, x_2 \geq 0. \end{array}$$

The optimal solution is $x_1 = 25$ hoodies and $x_2 = 45$ posters, where the printing and finishing constraints intersect, with weekly profit $12(25) + 5(45) = \$525$ (verified by solver). Note that producing the maximum 35 hoodies is not optimal: it leaves only $120 - 105 = 15$ printing hours for posters, and profit drops to $12(35) + 5(15) = \$495$.

Exercise 2.4 (Smoothie Powders)

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Let $x_A, x_B, x_C \geq 0$ be the scoops of powders A, B, C in the daily mix. Following the diet problem template:

$$\begin{array}{ll} \min & z = 1.20x_A + 0.80x_B + 0.60x_C \\ \text{s.t.} & 6x_A + 4x_B + 2x_C \geq 24 & (\text{protein, g}) \\ & 3x_A + 2x_B + 4x_C \geq 18 & (\text{fiber, g}) \\ & 2x_A + 3x_B + x_C \geq 12 & (\text{iron, mg}) \\ & x_A, x_B, x_C \geq 0. \end{array}$$

Solving with software gives $x_A = x_B = x_C = 2$, cost $2(1.20 + 0.80 + 0.60) = \5.20 , as claimed. All three nutritional constraints are tight there: protein $12 + 8 + 4 = 24$, fiber $6 + 4 + 8 = 18$, iron $4 + 6 + 2 = 12$, so the optimal mix wastes nothing.

Exercise 2.5 (Chemical Manufacturing)

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Let x_1 and x_2 be the number of hours per day that Process 1 and Process 2 run. Each hour of Process 1 costs \$4 and yields $(3, 1, 1)$ units of chemicals (A, B, C); each hour of Process 2 costs \$1 and yields $(1, 1, 0)$:

$$\begin{array}{ll} \min & z = 4x_1 + x_2 \\ \text{s.t.} & 3x_1 + x_2 \geq 10 & (\text{chemical A}) \\ & x_1 + x_2 \geq 5 & (\text{chemical B}) \\ & x_1 \geq 3 & (\text{chemical C}) \\ & x_1, x_2 \geq 0. \end{array}$$

Since only Process 1 makes chemical C, its demand forces $x_1 \geq 3$. The solver gives $x_1 = 3, x_2 = 2$ with minimum daily cost \$14: three hours of Process 1 cover chemical C and most of A, and two hours of the cheap Process 2 finish off the demands for A and B (A: $9 + 2 = 11 \geq 10$ with slack, B: $3 + 2 = 5$ tight).

Exercise 2.6 (Oven Store Profit Maximization)

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Let $x_1, x_2, x_3 \geq 0$ be the numbers of small, medium, and large ovens sold per month:

$$\begin{array}{ll} \max & z = 50x_1 + 100x_2 + 150x_3 \\ \text{s.t.} & 3x_1 + 5x_2 + 6x_3 \leq 90 & (\text{storage, cu ft}) \\ & x_1 + x_2 + x_3 \leq 200 & (\text{sales hours}) \\ & x_1 + x_2 + 2x_3 \leq 280 & (\text{installer hours}) \\ & x_1, x_2, x_3 \geq 0. \end{array}$$

Storage is the scarce resource: profit per cubic foot is $50/3 \approx 16.7$ for small, $100/5 = 20$ for medium, and $150/6 = 25$ for large ovens, so the store should fill its storage with large ovens. The solver confirms the optimum $x_1 = x_2 = 0$, $x_3 = 15$ with maximum profit \$2250. The sales constraint (15 of 200 hours) and installation constraint (30 of 280 hours) are far from binding; only storage binds.

QA flag

Book fix applied (issue type c, ugly data): the exercise previously gave 100 cubic feet of storage, which makes the LP optimum $x_3 = 50/3 \approx 16.67$ ovens with profit \$2500, a fractional number of ovens in a problem that asks "how many of each type should be sold." The book now says 90 cubic feet, giving the clean integer optimum above (re-verified by solver). Instructors using older printings should adjust. Separately, the sales and installer constraints can never bind with any of these storage levels; they are harmless as which-constraints-matter distractors, but could be tightened if a two-binding-constraint instance is wanted.

Exercise 2.7 (Manufacturing Profit Problem)

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Let $x_A, x_B, x_C \geq 0$ be the monthly production of products A, B, and C. Each machine's monthly hours give one constraint:

$$\begin{array}{ll} \max & z = 20x_A + 30x_B + 40x_C \\ \text{s.t.} & x_A + 2x_B + 2x_C \leq 180 & (\text{Machine I}) \\ & 2x_A + 2x_B + 4x_C \leq 300 & (\text{Machine II}) \\ & x_A, x_B, x_C \geq 0. \end{array}$$

An optimal solution is $x_A = 120$, $x_B = 30$, $x_C = 0$ with maximum profit \$3300 (verified by solver). Both machine constraints bind: $120 + 60 = 180$ and $240 + 60 = 300$. Product C is not worth making: it earns \$40 but uses as much Machine I time as B and twice the Machine II time of A, so its profit does not justify the machine hours it consumes.

Exercise 2.8 (Factory Production Scheduling)

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Let x_1 and x_2 be the number of days that Factory I and Factory II operate. Minimize operating cost subject to filling the order:

$$\begin{array}{ll} \min & z = 4000x_1 + 5000x_2 \\ \text{s.t.} & 10x_1 + 20x_2 \geq 1000 \quad (\text{product A}) \\ & 20x_1 + 20x_2 \geq 1600 \quad (\text{product B}) \\ & 20x_1 + 10x_2 \geq 700 \quad (\text{product C}) \\ & x_1, x_2 \geq 0. \end{array}$$

The optimal solution is $x_1 = 60$ days and $x_2 = 20$ days with minimum cost $4000(60) + 5000(20) = \$340,000$ (verified by solver). The constraints for products A ($600+400 = 1000$) and B ($1200+400 = 1600$) are binding, while product C is overproduced ($1400 \geq 700$).

Exercise 2.9 (Quiz Grading Minimization)

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Let $x_1, x_2, x_3 \geq 0$ be the numbers of objective, recall, and recall-plus quizzes. Grading times per quiz are 1, 2, and 3 minutes:

$$\begin{array}{ll} \min & z = x_1 + 2x_2 + 3x_3 \quad (\text{grading minutes}) \\ \text{s.t.} & x_1 + x_2 + x_3 \geq 20 \quad (\text{at least 20 quizzes}) \\ & 10x_1 + 30x_2 + 60x_3 \geq 720 \quad (\text{preparation minutes}) \\ & 5x_1 + 6x_2 + 7x_3 \geq 130 \quad (\text{total score points}) \\ & x_1, x_2, x_3 \geq 0. \end{array}$$

The solver gives the optimum $x_1 = 12$ objective, $x_2 = 0$ recall, and $x_3 = 10$ recall-plus quizzes, with minimum grading time $12 + 0 + 30 = 42$ minutes. Check: quizzes $12 + 10 = 22 \geq 20$ (slack), preparation $120 + 600 = 720$ (tight), score $60 + 70 = 130$ (tight). Intuition: objective quizzes are the cheapest way to add score and count, while recall-plus quizzes are the most grading-efficient way to generate preparation time; the middle option is dominated by mixing the two extremes.

Exercise 2.10 (Campus Café Scheduling)

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1. Let $x_i \geq 0$ be the number of baristas who start their four consecutive working days on day i , for $i = 1$ (Mon) through 7 (Sun), with the week wrapping around. A barista starting on day i works days $i, i+1, i+2, i+3 \pmod{7}$, so day d is covered by those who started on days $d-3, d-2, d-1, d$:

$$\begin{array}{ll} \min & z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \\ \text{s.t.} & x_5 + x_6 + x_7 + x_1 \geq 5 \quad (\text{Mon}) \\ & x_6 + x_7 + x_1 + x_2 \geq 4 \quad (\text{Tue}) \\ & x_7 + x_1 + x_2 + x_3 \geq 3 \quad (\text{Wed}) \\ & x_1 + x_2 + x_3 + x_4 \geq 4 \quad (\text{Thu}) \\ & x_2 + x_3 + x_4 + x_5 \geq 6 \quad (\text{Fri}) \\ & x_3 + x_4 + x_5 + x_6 \geq 8 \quad (\text{Sat}) \\ & x_4 + x_5 + x_6 + x_7 \geq 7 \quad (\text{Sun}) \\ & x_1, \dots, x_7 \geq 0. \end{array}$$

2. The solver reports the LP optimal value $z_{LP} = 28/3 \approx 9.33$, attained for instance at the fractional point $(x_1, \dots, x_7) = (0, 2/3, 5/3, 5/3, 2, 8/3, 2/3)$.
3. Any whole-number staffing plan is feasible for the LP, so its total cannot beat the LP optimum: every integer plan uses at least $28/3$ baristas, and being an integer, at least $\lceil 28/3 \rceil = 10$. A plan achieving 10 (verified feasible, and verified optimal by an integer solver): $x_3 = 3$ (start Wed), $x_4 = 1$ (Thu), $x_5 = 2$ (Fri), $x_6 = 2$ (Sat), $x_7 = 2$ (Sun), all others 0. Coverage by day: Mon $6 \geq 5$, Tue $4 \geq 4$, Wed $5 \geq 3$, Thu $4 \geq 4$, Fri $6 \geq 6$, Sat $8 \geq 8$, Sun $7 \geq 7$.

Exercise 2.11 (Identify Linear Inequalities) ★★

Only (4) and (5) are linear as written: both compare $\sum_i a_i x_i$, a linear function with constant coefficients, to a constant. An equation like (4) is welcome in a linear program; it is equivalent to the pair of inequalities $\sum_i a_i x_i \geq b$ and $\sum_i a_i x_i \leq b$.

Constraint (1), $\frac{x_1}{x_1 + x_2 + x_3} \geq 0.5$, is not linear because the variables appear in a ratio, but it *can* be made linear: if we know $x_1 + x_2 + x_3 > 0$ (for instance when all variables are non-negative and at least one is positive), multiplying both sides by the denominator gives $x_1 \geq 0.5(x_1 + x_2 + x_3)$, that is, $0.5x_1 - 0.5x_2 - 0.5x_3 \geq 0$.

Constraints (2) and (3) cannot be repaired: $\sin x_1$ and 2^{x_i} are genuinely nonlinear functions of the variables, and no algebraic rearrangement removes them.

Exercise 2.12 (Stochastic Objective) ★★

Replace each random cost coefficient by its expected value and optimize the expected objective. If c_i has a known distribution with mean $\bar{c}_i = \mathbb{E}[c_i]$, then by linearity of expectation the expected cost of any solution $\mathbf{x}$ is

$$\mathbb{E}\left[\sum_i c_i x_i\right] = \sum_i \mathbb{E}[c_i] x_i = \sum_i \bar{c}_i x_i,$$

which is again a linear function of $\mathbf{x}$ with constant coefficients. So the LP

$$\max \left\{ \sum_i \bar{c}_i x_i : A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq 0 \right\}$$

finds the solution with the best expected objective value. As in the hint, a uniformly random payment between \$1 and \$2 is worth \$1.50 in expectation, and that single number is all the LP needs. (This works because the randomness is only in the objective; random coefficients inside the *constraints* are much more delicate, which is the subject of stochastic programming.)

Exercise 2.13 (Which Assumption Breaks?) ★★

1. **Proportionality.** The per-unit cost changes from \$5 to \$4 at 100 units, so the cost contribution of the purchase variable is not proportional to its value; it is piecewise linear.
2. **Additivity.** The combined effect of the two campaigns is less than the sum of their separate effects, so contributions are not independent and additive; there is an interaction term.
3. **Divisibility.** 3.6 trucks is not an implementable decision; trucks come in whole numbers, so the fractional answer signals that divisibility fails and an integer model may be needed.
4. **Certainty.** The yield coefficients depend on future rainfall, so the model data are not known with certainty.

Follow-up for (a): essentially no harm. If purchases always exceed 100 units, the cost is $5(100) + 4(q - 100) = 4q + 100$ in that range: a linear function of q plus a constant. The constant does not affect which solution is optimal, so an LP using a marginal cost of \$4 (plus the fixed \$100 in the reported objective) is exact on the region where the model is used. Assumption violations only matter where the model is actually evaluated.

Exercise 2.14 (Production with Rented Labor)

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1. Let $x_1, x_2, x_3 \geq 0$ be the weekly production of A, B, C, and r the labor hours rented. Rented hours appear in the objective as a cost ($-9r$) and in the labor constraint as extra capacity:

$$\begin{aligned}
 \max \quad & z = 30x_1 + 45x_2 + 24x_3 - 9r \\
 \text{s.t.} \quad & 2x_1 + 4x_2 + x_3 \leq 160 && \text{(machine hours)} \\
 & 3x_1 + 3x_2 + 2x_3 \leq 180 + r && \text{(labor hours)} \\
 & 0 \leq r \leq 30 \\
 & x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

2. The solver gives $x_1 = 0$, $x_2 = 22$, $x_3 = 72$, $r = 30$, with net profit $45(22) + 24(72) - 9(30) = 990 + 1728 - 270 = \2448 . Both resource constraints bind: machine $4(22) + 72 = 160$ and labor $3(22) + 2(72) = 210 = 180 + 30$.
3. Product C turns 2 labor hours into \$24, that is, \$12 of profit per labor hour (and it is machine-light, using only 1 machine hour per unit). As long as some product earns more than \$9 per additional labor hour, each rented hour costing \$9 can be converted into strictly more than \$9 of profit, so an optimal plan keeps renting until the agency limit of 30 hours is reached. In dual language: the shadow price of labor exceeds the \$9 rental rate, so the rental upper bound is binding.

3 Chapter 3: Software – Excel

Coverage and flow. The chapter has a single content section (Using Excel Solver, covering spreadsheet functions, the recommended color-coded layout, the Solver dialog, solving methods, and the Answer/Sensitivity reports), and the nine exercises drill it thoroughly. The two warm-ups rerun workbooks from Chapter 4's examples with new data, the core problems walk through diet, production, and transportation models built from scratch, two exercises specifically exercise the Sensitivity Report (shadow prices, allowable ranges), one concept exercise targets the solving-method dropdown, and the challenge exercise isolates the Make Unconstrained Variables Non-Negative checkbox. Every part of the Solver workflow described in the text is touched at least once; sensitivity interpretation is drilled hardest, which is appropriate since it previews Chapter 7. Honest verdict: coverage is complete and well graded from one to three stars; the only soft spot is that no exercise asks students to use the plain Excel functions (SUMPRODUCT, COUNTIF, absolute references) outside of a Solver model, and the diet exercise's data provenance should be checked (see the flag under Exercise 3.3).

Exercise 3.1 (Solver Rerun: Minimum-Cost Network Flow)

★

The model is the minimum-cost network flow LP from the book's warehouse example, with only the objective coefficients changed. With x_{ij} the flow from warehouse i to store j :

$$\begin{aligned} \min \quad & 3x_{11} + 7x_{12} + 4x_{21} + 2x_{22} \\ \text{s.t.} \quad & x_{11} + x_{12} \leq 20, \quad x_{21} + x_{22} \leq 30, \\ & x_{11} + x_{21} = 25, \quad x_{12} + x_{22} = 25, \\ & 0 \leq x_{11} \leq 15, \quad 0 \leq x_{12} \leq 10, \quad 0 \leq x_{21} \leq 20, \quad 0 \leq x_{22} \leq 20. \end{aligned}$$

Solver setup: the objective cell holds =SUMPRODUCT(costs, flows), the variable cells are the four flow cells, and the constraints are row sums $\leq$ supply, column sums = demand, and flows $\leq$ capacities, solved with Simplex LP and non-negativity checked.

(a) The optimal flows are $x_{11} = 15$, $x_{12} = 5$, $x_{21} = 10$, $x_{22} = 20$, with cost $3(15) + 7(5) + 4(10) + 2(20) = 45 + 35 + 40 + 40 = 160$, matching the stated check value.

(b) Routes $W_1 \rightarrow S_1$ (flow 15, capacity 15) and $W_2 \rightarrow S_2$ (flow 20, capacity 20) are at capacity. These are the two cheapest routes, so the model saturates them first and covers the remaining demand with the more expensive routes.

Exercise 3.2 (Solver Rerun: Production Planning over 10 Periods)

★

(a) Only the demand cells change; the model is otherwise the ten-period production LP from the book (production costs 10, 12, ..., 28, holding cost 5, initial inventory $s_0 = 5$, inventory-balance constraints $s_{t-1} + x_t = d_t + s_t$). *Solver setup:* objective cell =SUMPRODUCT(prodcost, x) + 5*SUM(s), variable cells $x_1, \dots, x_{10}$ and $s_1, \dots, s_{10}$, constraints one balance equation per period, Simplex LP, non-negative variables. The optimal plan is chase production with the initial inventory used in period 1:

$$x = (1, 9, 7, 8, 5, 10, 7, 6, 8, 4), \quad s_t = 0 \text{ for all } t,$$

with total cost $10(1) + 12(9) + 14(7) + 16(8) + 18(5) + 20(10) + 22(7) + 24(6) + 26(8) + 28(4) = 1252$, matching the check value.

(b) Producing one unit in period t and holding it to period $t + 1$ costs $c_t + 5$, while waiting and producing it in period $t + 1$ costs $c_{t+1} = c_t + 2$. Since the holding cost (5) exceeds the cost escalation between consecutive periods (2), carrying inventory is never worthwhile, so the optimal plan produces each period's demand in that period ($x_t = d_t$, except period 1 where the free initial inventory of 5 units offsets demand).

Exercise 3.3 (Diet Problem in Excel)

★★

Let $x_1, \dots, x_5 \geq 0$ be daily servings of raw carrots, baked potatoes, wheat bread, cheddar cheese, and peanut butter. The LP is

$$\begin{aligned} \min \quad & 0.14x_1 + 0.12x_2 + 0.20x_3 + 0.75x_4 + 0.15x_5 \\ \text{s.t.} \quad & 23x_1 + 171x_2 + 65x_3 + 112x_4 + 188x_5 \geq 2000 && \text{(calories)} \\ & 0.1x_1 + 0.2x_2 + 9.3x_4 + 16x_5 \geq 50 && \text{(fat)} \\ & 0.6x_1 + 3.7x_2 + 2.2x_3 + 7x_4 + 7.7x_5 \geq 100 && \text{(protein)} \\ & 6x_1 + 30x_2 + 13x_3 + 2x_5 \geq 250 && \text{(carbohydrate)} \\ & x_1, \dots, x_5 \geq 0. \end{aligned}$$

Solver setup: lay out the nutrition table as data (blue), one yellow cell per food for servings, a green objective cell =SUMPRODUCT(prices, servings), and four green cells =SUMPRODUCT(nutrientrow, servings) constrained $\geq$ the requirements; solve with Simplex LP and non-negativity checked.

The optimal diet is potatoes and peanut butter only:

$$x_2 \approx 7.7147 \text{ servings of baked potatoes,} \quad x_5 \approx 9.2800 \text{ servings of peanut butter,}$$

with $x_1 = x_3 = x_4 = 0$ and minimum daily cost $\approx \$2.32$ (exactly \$2.3178). The protein and carbohydrate constraints are binding (100 g and 250 g exactly); calories (≈ 3064) and fat (≈ 150 g) are comfortably oversatisfied. A screenshot should show the yellow serving cells with these values and the green cost cell at 2.32.

QA flag

Licensing: this exact food table (raw carrots, baked potatoes, wheat bread, cheddar cheese, peanut butter, with these prices and nutrient values and the 2000/50/100/250 requirements) circulates widely in university course materials and closely matches the diet-problem exercise found in at least one commercial optimization textbook. Recommend verifying the original source; if it cannot be traced to an openly licensed source, regenerate the table with fresh (verified) nutrition data, which would not change the pedagogical content. The numbers themselves are fine: the LP is feasible, non-degenerate, and has a clean two-food solution.

Exercise 3.4 (Production Planning in Excel)

★★

Let x_1 and x_2 be the numbers of widgets and gadgets produced. The LP is

$$\begin{aligned} \max \quad & z = 8x_1 + 6x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 40 && \text{(machine time)} \\ & x_1 + 2x_2 \leq 30 && \text{(raw material)} \\ & x_1, x_2 \geq 0. \end{aligned}$$

(a) *Solver setup:* data cells (blue) hold the profit vector (8,6), the constraint matrix, and the right-hand sides (40,30); two yellow variable cells hold x_1, x_2 ; green cells hold the objective =SUMPRODUCT(profits, vars) and the two constraint left-hand sides. Add both $\leq$ constraints, choose Max, Simplex LP, and check Make Unconstrained Variables Non-Negative.

(b) Solver returns $x_1 = 50/3 \approx 16.67$ widgets and $x_2 = 20/3 \approx 6.67$ gadgets with maximum profit $520/3 \approx \$173.33$. Both constraints are binding.

(c) The Sensitivity Report's Variable Cells table shows the widget objective coefficient 8 with allowable increase 4 (up to 12) and allowable decrease 5 (down to 3); geometrically the objective slope must stay between the slopes of the two binding constraints. Raising the widget profit to \$9 is within the allowable increase, so the optimal plan does not change; only the profit rises, to $9(50/3) + 6(20/3) = \$190$. (The report's Constraints table shows shadow prices $10/3$ for machine time and $4/3$ for raw material, useful cross-checks.)

Exercise 3.5 (Transportation Problem in Excel)

★★

(a) Let $x_{ij} \geq 0$ be units shipped from warehouse i to store j . Lay out a 2×3 grid of yellow variable cells mirroring the blue cost table, with green row sums (compared to supply) and green column

sums (compared to demand):

$$\begin{aligned} \min \quad & 4x_{11} + 8x_{12} + x_{13} + 6x_{21} + 3x_{22} + 5x_{23} \\ \text{s.t.} \quad & x_{11} + x_{12} + x_{13} \leq 60, \quad x_{21} + x_{22} + x_{23} \leq 50, \\ & x_{11} + x_{21} = 30, \quad x_{12} + x_{22} = 40, \quad x_{13} + x_{23} = 40, \\ & x_{ij} \geq 0. \end{aligned}$$

Solver setup: objective cell =SUMPRODUCT(costgrid, shipgrid), variable cells the six grid cells, constraints row sums $\leq (60, 50)$ and column sums = $(30, 40, 40)$, Simplex LP, non-negativity checked.

(b) Total supply equals total demand (110), so both warehouses ship everything. The optimal plan is

$$x_{11} = 20, \quad x_{13} = 40, \quad x_{21} = 10, \quad x_{22} = 40, \quad x_{12} = x_{23} = 0,$$

with minimum cost $4(20) + 1(40) + 6(10) + 3(40) = 80 + 40 + 60 + 120 = \300 . Each warehouse saturates its cheapest route (W1 to S3 at \$1, W2 to S2 at \$3) and store S1's demand is split 20/10 to balance the supplies.

Exercise 3.6 (Product Mix with a Sensitivity Report)

★★

Let $x_1, x_2, x_3 \geq 0$ be the numbers of tables, chairs, and desks made per week:

$$\begin{aligned} \max \quad & 70x_1 + 50x_2 + 90x_3 \\ \text{s.t.} \quad & 4x_1 + 3x_2 + 6x_3 \leq 240 && \text{(carpentry)} \\ & 2x_1 + x_2 + 3x_3 \leq 100 && \text{(finishing)} \\ & 30x_1 + 20x_2 + 40x_3 \leq 2000 && \text{(wood)}. \end{aligned}$$

(a) *Solver setup:* three yellow variable cells, green objective =SUMPRODUCT(profits, vars), three green resource-usage cells constrained $\leq (240, 100, 2000)$, Max, Simplex LP. The optimum is $x_1 = 30$ tables, $x_2 = 40$ chairs, $x_3 = 0$ desks, with maximum profit $70(30) + 50(40) = \$4100$, matching the check value.

(b) Sensitivity Report, Constraints table: shadow price \$15 per carpentry hour, \$5 per finishing hour, \$0 per board-foot of wood. Carpentry ($4 \cdot 30 + 3 \cdot 40 = 240$) and finishing ($2 \cdot 30 + 40 = 100$) are binding; wood is not (1700 of 2000 used, slack 300).

(c) No. An extra finishing hour is worth its shadow price, \$5, which is less than the \$10 asking price, so buying finishing time at \$10 per hour would lose \$5 per hour purchased.

(d) The desk's reduced cost is $90 - (6 \cdot 15 + 3 \cdot 5) = 90 - 105 = -15$, so the report lists an allowable increase of 15 on the desk coefficient (up to \$105). At \$100 the plan is unchanged (still 30 tables, 40 chairs, no desks, profit \$4100). At \$110, which exceeds the allowable increase, the plan changes: re-solving gives $x_1 = 0$, $x_2 = 40$, $x_3 = 20$ with profit \$4200, so desks enter the mix.

Exercise 3.7 (Unbalanced Transportation with a Sensitivity Report)

★★

(a) Let $x_{ij} \geq 0$ ship from plant i to city j . Supply is 160 against demand 150, so supplies are $\leq$ and demands are $=$:

$$\begin{aligned} \min \quad & 4x_{11} + 5x_{12} + 7x_{13} + 6x_{21} + 3x_{22} + 2x_{23} + 5x_{31} + 8x_{32} + 4x_{33} \\ \text{s.t.} \quad & x_{11} + x_{12} + x_{13} \leq 70, \quad x_{21} + x_{22} + x_{23} \leq 50, \quad x_{31} + x_{32} + x_{33} \leq 40, \\ & x_{11} + x_{21} + x_{31} = 50, \quad x_{12} + x_{22} + x_{32} = 60, \quad x_{13} + x_{23} + x_{33} = 40, \quad x_{ij} \geq 0. \end{aligned}$$

Solver setup: 3×3 yellow grid, objective =SUMPRODUCT(costs, flows), row sums $\leq$ supplies, column sums = demands, Simplex LP. The optimal plan is

$$x_{11} = 50, \quad x_{12} = 10, \quad x_{22} = 50, \quad x_{33} = 40 \quad (\text{all other routes } 0),$$

with minimum cost $4(50) + 5(10) + 3(50) + 4(40) = 200 + 50 + 150 + 160 = \560 , matching the check value.

(b) Plant 1 ships only 60 of its 70 units, so 10 units stay at P1. (P2 and P3 ship their full capacities.)

(c) Sensitivity Report shadow prices on the supply constraints: P1 gets 0, P2 gets -2 , P3 gets 0 (P3's constraint is binding but degenerate here, so its reported price is 0). The nonzero price means one extra unit of capacity at plant 2 would reduce total cost by \$2: P2 has the cheapest route into city 2, and extra P2 capacity would displace a unit of P1's \$5 shipping with a \$3 shipment. The sign is negative because this is a minimization: relaxing a binding $\leq$ supply constraint can only help, lowering the objective.

Exercise 3.8 (Simplex LP versus GRG Nonlinear) ★★

(a) Simplex LP is designed for problems whose objective and constraint cells are all linear functions of the variable cells (linear programs). GRG Nonlinear (generalized reduced gradient) is designed for problems where those functions are smooth (differentiable) but nonlinear.

(b) For a linear program, Simplex LP guarantees a globally optimal solution and produces the standard LP Sensitivity Report with shadow prices, reduced costs, and allowable increase/decrease ranges. GRG is a local-search method: on a general nonlinear problem it only guarantees a locally optimal point (it happens that for LPs local and global coincide, but Solver does not exploit that), it can be slower and dependent on the starting point, and its sensitivity output reports Lagrange multipliers and reduced gradients without the allowable-range information that makes the LP report useful.

(c) Solver runs a linearity test on the model before (and after) solving. Because $=B2*B3$ multiplies two variable cells, the constraint is not a linear function of the variables, and Solver stops with the message that the linearity conditions required by this LP Solver are not satisfied. The fix is either to reformulate the model linearly or to switch to GRG Nonlinear.

(d) Any smooth nonlinear objective works, for example minimizing portfolio variance $f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}$ (a quadratic), or maximizing $\sqrt{x_1} + \sqrt{x_2}$. These are differentiable, so GRG applies, but they are not linear, so Simplex LP rejects them.

Exercise 3.9 (When a Variable Must Go Negative) ★★★

The model is

$$\max 0.05x_A + 0.12x_B \quad \text{s.t.} \quad x_A + x_B = 100, \quad 0 \leq x_B \leq 140, \quad x_A \text{ free.}$$

Solver setup: two yellow cells for x_A, x_B , objective $=0.05*x_A+0.12*x_B$, constraints $x_A+x_B=100$ and $x_B \leq 140$, Simplex LP.

(a) With Make Unconstrained Variables Non-Negative checked, Solver silently imposes $x_A \geq 0$ and $x_B \geq 0$, so the best it can do is $x_A = 0, x_B = 100$, with return $0.05(0) + 0.12(100) = 12$, that is \$12,000.

(b) With the box unchecked and $x_B \geq 0$ added explicitly, Solver finds $x_A = -40, x_B = 140$: short \$40,000 of fund A to push fund B to its margin cap. The return is $0.05(-40) + 0.12(140) = -2 + 16.8 = 14.8$, that is \$14,800.

(c) The checkbox adds a lower bound of zero to every decision variable that does not already have an explicit lower bound in the constraint list. Short selling is exactly a negative holding, so that hidden bound cuts off the true optimum; the box must be unchecked for x_A to go negative. The risk is that unchecking removes the automatic bound from all variables at once: any variable that should be non-negative (here x_B) must now receive its own explicit ≥ 0 constraint, or Solver may return meaningless negative values for it.

4 Chapter 4: Modeling with Compact Notation

Coverage and flow. The chapter builds compact summation models (production planning, assignment, network flow, max flow, multicommodity flow, transportation, multi-period investment), and its fourteen exercises track that arc well. Production planning is drilled by 4.1 (warm-up) and 4.10 (capacity constraints and prebuilding); assignment by 4.2, 4.4 (a 12 by 12 data-driven instance), 4.9 (forbidden pairs), and the fairness pair 4.12/4.14; minimum-cost flow and transshipment by 4.3, 4.7, 4.8, and the conceptual 4.11; max flow by 4.5 and 4.6; multicommodity flow by 4.13. The two concept exercises (4.11 on aggregate flow balance, 4.12 on integrality) are genuine highlights that connect the models to later theory. Gaps: the multi-period capital investment section and the source-sink multicommodity formulation have no exercises, the max-flow min-cut theorem is exercised only implicitly (a cut certificate appears in 4.6), and nothing drills the school-bus/transportation-table material directly. Honest verdict: coverage of the core modeling sections is thorough and nicely graded; one exercise on the investment model would round it out. One data error was found and fixed in the book (Exercise 4.7, see below).

Exercise 4.1 (Production Planning over Five Periods)

★

(a) With production $x_t \geq 0$ and end-of-period inventory $s_t \geq 0$:

$$\begin{aligned} \min \quad & 8x_1 + 10x_2 + 12x_3 + 14x_4 + 16x_5 + 3(s_1 + s_2 + s_3 + s_4 + s_5) \\ \text{s.t.} \quad & 2 + x_1 = 5 + s_1 \\ & s_1 + x_2 = 7 + s_2 \\ & s_2 + x_3 = 6 + s_3 \\ & s_3 + x_4 = 8 + s_4 \\ & s_4 + x_5 = 4 + s_5 \\ & x_t, s_t \geq 0 \quad \text{for } t = 1, \dots, 5. \end{aligned}$$

(b) The optimal plan is chase production, $x = (3, 7, 6, 8, 4)$ with all $s_t = 0$: the initial inventory of 2 covers part of period 1's demand. The cost is $8(3) + 10(7) + 12(6) + 14(8) + 16(4) = 24 + 70 + 72 + 112 + 64 = 342$, matching the check value.

(c) Producing a unit one period early costs the earlier production cost plus the holding cost of 3; waiting costs only 2 more, since the production cost rises by 2 per period. Because $3 > 2$, prebuilding is never worthwhile and no inventory is carried.

Exercise 4.2 (Machine Assignment with New Costs)

★

The model is the standard assignment problem with binary x_{ij} (1 if machine i does job j), objective $\min \sum_{i,j} c_{ij}x_{ij}$, one constraint $\sum_j x_{ij} = 1$ per machine and one constraint $\sum_i x_{ij} = 1$ per job. With

the given cost matrix the optimal assignment is

$$0 \rightarrow 1 \text{ (3)}, \quad 1 \rightarrow 0 \text{ (5)}, \quad 2 \rightarrow 2 \text{ (2)}, \quad 3 \rightarrow 3 \text{ (3)},$$

for total cost $3 + 5 + 2 + 3 = 13$, matching the check value. Enumerating all $4! = 24$ assignments confirms optimality; the second-best assignment costs 16.

Exercise 4.3 (Warehouse Shipping with New Data) ★

The model from the book's warehouse example with the new data:

$$\begin{aligned} \min \quad & 4x_{11} + 6x_{12} + 3x_{21} + 2x_{22} \\ \text{s.t.} \quad & x_{11} + x_{12} \leq 20, \quad x_{21} + x_{22} \leq 30, \\ & x_{11} + x_{21} = 20, \quad x_{12} + x_{22} = 30, \\ & 0 \leq x_{11} \leq 15, \quad 0 \leq x_{12} \leq 10, \quad 0 \leq x_{21} \leq 20, \quad 0 \leq x_{22} \leq 20. \end{aligned}$$

(a) Total demand (50) equals total supply, so both warehouses ship everything. The optimal flow is

$$x_{11} = 10, \quad x_{12} = 10, \quad x_{21} = 10, \quad x_{22} = 20,$$

with cost $4(10) + 6(10) + 3(10) + 2(20) = 40 + 60 + 30 + 40 = 170$, matching the check value.

(b) Routes $W_1 \rightarrow S_2$ (flow 10, capacity 10) and $W_2 \rightarrow S_2$ (flow 20, capacity 20) are at capacity: Store 2's demand of 30 exactly saturates both routes into it, and the remaining flows are then forced.

QA flag

Duplication note: this exercise and Exercise 3.1 both rerun the same warehouse network from the same example, differing only in costs and demands. That is defensible (3.1 drills the Excel workbook, 4.3 drills writing the model), but if the chapters are ever assigned together, consider changing the network topology in one of them so students do not solve essentially the same instance twice.

Exercise 4.4 (Fair Housing Assignment for 12 Families) ★★

Model: binary x_{ij} with $x_{ij} = 1$ if family i gets apartment j , objective $\min \sum_{i,j} P_{ij}x_{ij}$ where P_{ij} is family i 's rank of apartment j , with $\sum_j x_{ij} = 1$ for each family and $\sum_i x_{ij} = 1$ for each apartment. In Excel: a 12×12 yellow grid of variables next to the blue preference table from the CSV, objective `=SUMPRODUCT(prefs, assign)`, row and column sums constrained to 1, variables ≥ 0 (binary is not needed; the assignment LP returns a 0-1 solution, see Exercise 4.12), Simplex LP.

(a, b) The minimum total preference score is **17**, achieved by

Family	Apartment	Rank	Family	Apartment	Rank
1	12	2	7	5	1
2	4	1	8	6	1
3	11	1	9	10	1
4	8	2	10	9	2
5	1	1	11	3	1
6	7	1	12	2	3

Eight families receive a first choice, three a second choice, and one a third choice. (Alternative optima with total 17 may exist; any assignment Solver returns with total 17 is correct.)

(c) The worst preference score assigned is 3 (Family 12 receives Apartment 2, its third choice).

Exercise 4.5 (Maximum Network Flow to a Single Node)

★★

Variables. For each arc (i, j) in the picture let $x_{ij} \geq 0$ be the units of supplies sent from i to j .

Objective. Maximize the total amount reaching the hub:

$$\max \quad x_{D_1, \text{Hub}} + x_{D_2, \text{Hub}} + x_{D_3, \text{Hub}}.$$

Constraints. Three families:

- Truck capacities: $x_{ij} \leq u_{ij}$ for every arc, with u as labeled ($x_{W_1, D_1} \leq 100$, $x_{W_2, D_1} \leq 150$, $x_{W_2, D_2} \leq 70$, $x_{W_3, D_2} \leq 120$, $x_{W_4, D_1} \leq 120$, $x_{W_4, D_3} \leq 100$, $x_{W_5, D_3} \leq 90$, $x_{D_1, \text{Hub}} \leq 200$, $x_{D_1, D_2} \leq 80$, $x_{D_2, \text{Hub}} \leq 180$, $x_{D_3, \text{Hub}} \leq 160$).
- Flow conservation at the distribution nodes (nothing is stored there):

$$x_{W_1, D_1} + x_{W_2, D_1} + x_{W_4, D_1} = x_{D_1, D_2} + x_{D_1, \text{Hub}},$$

$$x_{W_2, D_2} + x_{W_3, D_2} + x_{D_1, D_2} = x_{D_2, \text{Hub}},$$

$$x_{W_4, D_3} + x_{W_5, D_3} = x_{D_3, \text{Hub}}.$$

- Warehouse inventory: total outflow from each warehouse is at most its inventory, for example $x_{W_2, D_1} + x_{W_2, D_2} \leq 300$ and $x_{W_4, D_1} + x_{W_4, D_3} \leq 250$.

Solving the LP gives a maximum delivery of **540** units: all three hub arcs run at capacity ($200 + 180 + 160$), which is also an upper bound since every unit must enter the hub on one of them. One optimal plan: $x_{W_2, D_1} = 80$, $x_{W_4, D_1} = 120$, $x_{W_2, D_2} = 60$, $x_{W_3, D_2} = 120$, $x_{W_4, D_3} = 70$, $x_{W_5, D_3} = 90$, $x_{D_1, D_2} = 0$, hub arcs at 200, 180, 160 (many alternative optima exist, for example ones that use W_1).

Important modeling caution: the warehouse constraints must be inequalities. Writing them as equalities (shipping out all inventory) is infeasible, since for instance W_1 holds 200 units but its only outgoing arc has capacity 100.

QA flag

Wording: the exercise says “you do not want to leave any inventory at the distribution centers,” but the nodes holding inventory are the warehouses, and Exercise 4.11(3) later refers back to this exercise as if it had said warehouses. Recommend rewording to “supplies may not be stored at the distribution centers (whatever enters must leave), and warehouses may ship up to their inventory,” which is what the intended model does. As written, a literal student might try equality constraints at the warehouses and hit the infeasibility discussed above (which 4.11 then exploits, so keep the discussion, just make the wording consistent).

Exercise 4.6 (Max Flow)

★★

Let $x_{ij} \geq 0$ be the flow on arc (i, j) . Maximize the flow into t subject to conservation at the four internal nodes and the capacities:

$$\begin{aligned}
 \max \quad & x_{v_3t} + x_{v_4t} \\
 \text{s.t.} \quad & x_{sv_1} + x_{v_2v_1} = x_{v_1v_3} + x_{v_1v_2} & (v_1) \\
 & x_{sv_2} + x_{v_1v_2} + x_{v_3v_2} = x_{v_2v_4} + x_{v_2v_1} & (v_2) \\
 & x_{v_1v_3} + x_{v_4v_3} = x_{v_3t} + x_{v_3v_2} & (v_3) \\
 & x_{v_2v_4} = x_{v_4t} + x_{v_4v_3} & (v_4) \\
 & x_{sv_2} \leq 8, \quad x_{sv_1} \leq 11, \quad x_{v_1v_2} \leq 10, \quad x_{v_2v_1} \leq 1, \quad x_{v_2v_4} \leq 11, \\
 & x_{v_1v_3} \leq 12, \quad x_{v_3v_2} \leq 4, \quad x_{v_4v_3} \leq 7, \quad x_{v_4t} \leq 4, \quad x_{v_3t} \leq 15, \\
 & x_{ij} \geq 0.
 \end{aligned}$$

In Excel, one variable cell per arc, the four conservation constraints as equalities, capacities as $\leq$ bounds, Simplex LP.

The maximum flow is **19**. One optimal flow: $x_{sv_1} = 11$, $x_{sv_2} = 8$, $x_{v_1v_3} = 8$, $x_{v_1v_2} = 3$, $x_{v_2v_4} = 11$, $x_{v_4t} = 4$, $x_{v_4v_3} = 7$, $x_{v_3t} = 15$, all other arcs 0. Optimality is certified without any theory: the two arcs into t have total capacity $15 + 4 = 19$, so no flow can exceed 19, and this flow achieves it (a minimum cut).

Exercise 4.7 (Min Cost Network Flow)

★★

Reading the picture (capacity in black, cost in red, blue labels are net demands with positive meaning demand and negative meaning supply), the data are: demands $d_{v_1} = 6$, $d_{v_6} = 10$; supplies 2, 5, 3, 6 at v_5, v_2, v_3, v_4 ; and arcs (capacity, cost): (v_5, v_2) : (8, 7), (v_5, v_1) : (11, 4), (v_2, v_4) : (11, 7), (v_4, v_6) : (4, 2), (v_1, v_3) : (12, 9), (v_3, v_6) : (15, 2), (v_3, v_2) : (4, 1), (v_4, v_3) : (7, 8), (v_2, v_1) : (4, 2), (v_1, v_2) : (10, 1). The model, with x_{ij} the flow on arc (i, j) :

$$\begin{aligned}
 \min \quad & \sum_{(i,j)} c_{ij}x_{ij} \\
 \text{s.t.} \quad & \sum_{i:(i,k)} x_{ik} - \sum_{j:(k,j)} x_{kj} = d_k \quad \text{for all nodes } k \quad (d_{v_1} = 6, d_{v_6} = 10, d_{v_5} = -2, \\
 & d_{v_2} = -5, d_{v_3} = -3, d_{v_4} = -6) \\
 & 0 \leq x_{ij} \leq u_{ij} \quad \text{for all arcs.}
 \end{aligned}$$

The optimal cost is **67**, with flows

$$x_{v_5v_1} = 2, \quad x_{v_2v_1} = 4, \quad x_{v_2v_4} = 1, \quad x_{v_4v_6} = 4, \quad x_{v_4v_3} = 3, \quad x_{v_3v_6} = 6,$$

and all other arcs zero. (Check: $2 \cdot 4 + 4 \cdot 2 + 1 \cdot 7 + 4 \cdot 2 + 3 \cdot 8 + 6 \cdot 2 = 8 + 8 + 7 + 8 + 24 + 12 = 67$.) A nice feature of the instance: node v_1 's demand of 6 exactly matches its maximum possible net inflow (2 from v_5 , whose supply is 2, plus the capacity 4 of arc (v_2, v_1)), so every feasible flow has $x_{v_5v_1} = 2$ and $x_{v_2v_1} = 4$, and the LP's only real freedom is how v_6 's demand is routed.

QA flag

Fixed in the book (2026-07-13): as originally drawn, arc (v_2, v_1) had capacity 1, making the instance infeasible under either sign convention (node v_1 demands 6 but could receive at most $2 + 1 = 3$; flipping signs instead makes v_6 a supply node with no outgoing arcs). The book figure now shows capacity 4 on that arc, the exercise states the label convention explicitly, and a check value (minimum cost 67) was added. Verified with scipy.

Exercise 4.8 (Transshipment Through Regional Hubs)

★★

(a) Ten arc variables: $x_{A,H_1}, x_{A,H_2}, x_{B,H_1}, x_{B,H_2}$ from factories to hubs, and x_{H_h,S_j} from each hub to each store. The model:

$$\begin{aligned}
 \min \quad & 2x_{A,H_1} + 4x_{A,H_2} + 3x_{B,H_1} + x_{B,H_2} \\
 & + 4x_{H_1,S_1} + 6x_{H_1,S_2} + 7x_{H_1,S_3} + 5x_{H_2,S_1} + 3x_{H_2,S_2} + 2x_{H_2,S_3} \\
 \text{s.t.} \quad & x_{A,H_1} + x_{A,H_2} \leq 40, \quad x_{B,H_1} + x_{B,H_2} \leq 60 && (\text{supply}) \\
 & x_{A,H_1} + x_{B,H_1} = x_{H_1,S_1} + x_{H_1,S_2} + x_{H_1,S_3} && (\text{balance at } H_1) \\
 & x_{A,H_2} + x_{B,H_2} = x_{H_2,S_1} + x_{H_2,S_2} + x_{H_2,S_3} && (\text{balance at } H_2) \\
 & x_{H_1,S_j} + x_{H_2,S_j} = d_j \quad (d = 30, 45, 25) && (\text{demand}) \\
 & x \geq 0.
 \end{aligned}$$

The two hub-balance equalities are the explicit statement that hubs neither create nor store product.

(b) Total demand (100) equals total supply, so both factories ship everything. An optimal plan:

$$x_{A,H_1} = 30, \quad x_{A,H_2} = 10, \quad x_{B,H_2} = 60, \quad x_{H_1,S_1} = 30, \quad x_{H_2,S_2} = 45, \quad x_{H_2,S_3} = 25,$$

with minimum cost $30(2) + 10(4) + 60(1) + 30(4) + 45(3) + 25(2) = 60 + 40 + 60 + 120 + 135 + 50 = \mathbf{465}$. The structure is intuitive: B uses its cheap arc into H_2 , H_2 serves the stores it reaches cheaply (S_2, S_3), and S_1 is served through H_1 .

Exercise 4.9 (Assignment with Forbidden Pairs)

★★

(a) Without the certification rules, the optimal assignment is

$$1 \rightarrow 2 (8), \quad 2 \rightarrow 1 (7), \quad 3 \rightarrow 3 (6), \quad 4 \rightarrow 5 (6), \quad 5 \rightarrow 4 (6),$$

with minimum cost $8 + 7 + 6 + 6 + 6 = 33$, matching the check value. Note that it uses both forbidden pairs, worker 1 on job 2 and worker 4 on job 5.

(b) Adding $x_{12} = 0$ (and $x_{45} = 0$) simply removes those variables from the feasible region, so no assignment using the pair can be selected. Replacing c_{12} by a huge cost M leaves the pair feasible but makes any assignment using it cost at least M , which exceeds the cost of every assignment avoiding it, so the optimizer never picks it. Both approaches leave the assignment structure (and its integrality property) intact; the explicit constraint is cleaner numerically, while the big-cost trick needs no new constraints.

(c) With the forbidden pairs enforced, the optimum becomes

$$1 \rightarrow 5 (11), \quad 2 \rightarrow 1 (7), \quad 3 \rightarrow 3 (6), \quad 4 \rightarrow 2 (7), \quad 5 \rightarrow 4 (6),$$

with cost $11 + 7 + 6 + 7 + 6 = 37$, matching the check value. The certification rules cost the company $37 - 33 = 4$.

Exercise 4.10 (Multi-Period Inventory with a Production Cap)

★★

(a) The model is the production planning LP with the added bounds $x_t \leq 120$:

$$\min \sum_{t=1}^6 (c_t x_t + 2s_t) \quad \text{s.t.} \quad s_{t-1} + x_t = d_t + s_t \quad (t = 1, \dots, 6), \quad 0 \leq x_t \leq 120, \quad s_t \geq 0, \quad s_0 = 0,$$

with $c = (20, 20, 22, 25, 24, 23)$ and $d = (80, 120, 60, 140, 90, 110)$.

(b) A period-4 unit produced in period t costs c_t plus 2 per period held: period 3 costs $22 + 2 = 24$, period 2 costs $20 + 4 = 24$, period 1 costs $20 + 6 = 26$. Periods 2 and 3 tie as cheapest at 24, and both beat producing in period 4 itself (25). However, period 2's capacity is fully consumed by its own demand ($d_2 = 120$ equals the cap), so all prebuilding must happen in period 3.

(c) The optimal plan is

$$x = (80, 120, 120, 80, 90, 110), \quad s = (0, 0, 60, 0, 0, 0),$$

with minimum cost **13,450**, matching the check value. This confirms part (b) with a twist worth pointing out to students: since prebuilt units cost 24 versus 25 in period 4, the LP does not prebuild just the 20 units forced by the cap ($140 - 120$); it fills period 3 to capacity and prebuilds 60 units, producing only 80 in period 4. (Because period 2 is at capacity, there are no ties to exploit and this plan is essentially unique.)

Exercise 4.11 (Why Everything Ships Somewhere)

★★

(a) Fix an arc $(i, j) \in A$. The variable x_{ij} appears in exactly two balance constraints: in node i 's constraint with coefficient $+1$ (it is an outflow of i) and in node j 's with coefficient -1 (an inflow of j). Summing the constraints over all $k \in V$ therefore cancels every variable, leaving

$$0 = \sum_{k \in V} d_k.$$

So if total net supply does not equal total net demand, the summed equation $0 = \sum_k d_k$ is violated by every $\mathbf{x}$, and the model is infeasible. Balance of supply and demand is a necessary condition for feasibility.

(b) At an intermediate node the constraint says inflow minus outflow is zero: whatever arrives must depart. Flow cannot be created (outflow exceeding inflow) or destroyed (inflow exceeding outflow) at such a node. Chaining this along any route, every unit that leaves a supply node travels through intermediate nodes unchanged until it reaches a node with positive demand, where the balance constraint lets it be absorbed.

(c) In Exercise 4.5 the warehouses hold $200 + 300 + 150 + 250 + 180 = 1080$ units, but the arcs into the hub can absorb at most $200 + 180 + 160 = 540$. Writing the warehouse constraints as equalities declares every warehouse's net supply to be its full inventory, so the d_k sum to 1080 minus whatever the hub absorbs; no choice of hub demand up to 540 makes the total zero while respecting the capacities, and by part (a) the equality model cannot be feasible (part (b) says the surplus cannot simply vanish at the warehouses). Standard fixes: state warehouse supply as an inequality (outflow at most inventory), or keep equalities and add a dummy node that absorbs the surplus through zero-cost, high-capacity arcs from each warehouse. Both let the model choose how much of the inventory actually moves.

Exercise 4.12 (Integer Answers for Free)

★★

(a) Model answer: the feasible region of the assignment LP is a polytope whose corner points all happen to have 0-1 coordinates, because the constraint matrix is totally unimodular. The simplex method always stops at a corner point, so even though we only asked for $x_{ij} \geq 0$, the answer it returns is automatically a legitimate 0-1 assignment. We get integrality for free from the geometry, not from imposing it.

(b) The transportation constraint matrix has the same network structure, so the same result applies: when all supplies s_i and demands d_j are integers, every corner point of the transportation polytope is integral, and there is always an optimal shipping plan in whole units. Solving the LP with the simplex method returns integer shipments with no integrality constraints needed.

(c) In the two-team example (both teams rank Project 2 first with scores 2 and 1), every integer assignment gives one team its score-2 project and the other its score-1 project, so the min-max value over integer assignments is 2. The LP with the added constraints $z \leq \sum_j C_{ij}x_{ij}$ (or $z \geq$, for min-max) does strictly better: splitting each project half-and-half gives both teams 1.5. Since the LP optimum (1.5) is not attained by any integer point, the optimal corner of this LP is fractional, so the formulation no longer has the integrality property. The added z rows destroy the network (totally unimodular) structure of the constraint matrix.

Exercise 4.13 (Multicommodity Rerouting Under a Tighter Capacity)

★★★

(a) With the original capacities the commodities do not compete: commodity 1 sends all 10 units along $1 \rightarrow 2 \rightarrow 4$ at per-unit cost $2 + 1 = 3$ (cost 30), and commodity 2 sends its 5 units along $1 \rightarrow 3 \rightarrow 4$ at per-unit cost $2 + 1 = 3$ (cost 15). Total minimum cost 45, as claimed; all arc loads (10, 5, 10, 5) are within capacity.

(b) With $u_{24} = 8$, commodity 1 can push only 8 units through $2 \rightarrow 4$; the LP reroutes its other 2 units along $1 \rightarrow 3 \rightarrow 4$ at per-unit cost $3 + 2 = 5$. Commodity 2 stays on $1 \rightarrow 3 \rightarrow 4$ (its cost there is 3 versus 5 via node 2). Optimal flows: commodity 1 sends 8 on $1 \rightarrow 2 \rightarrow 4$ and 2 on $1 \rightarrow 3 \rightarrow 4$; commodity 2 sends 5 on $1 \rightarrow 3 \rightarrow 4$. Cost $8(3) + 2(5) + 5(3) = 24 + 10 + 15 = 49$, matching the check value. Arc (2, 4) is at its capacity of 8; arcs (1, 3) and (3, 4) carry 7 units each, within capacity.

(c) Example: two nodes s and t joined by two parallel routes, a cheap route with cost 1 and capacity 10, and an expensive route with cost 3 and capacity 10. Commodities 1 and 2 each must send 8 units from s to t . Solved separately, each commodity routes all 8 units on the cheap route, loading it to $16 > 10$: infeasible. The joint LP sees the shared capacity constraint $f_{\text{cheap},1} + f_{\text{cheap},2} \leq 10$ and resolves the conflict by rationing the cheap route: it carries 10 units total (split between the commodities in any proportion) and the remaining 6 units go on the expensive route, for total cost $10(1) + 6(3) = 28$. The coupling constraint forces the commodities to negotiate over the scarce cheap capacity, which is exactly what makes multicommodity flow harder than solving each commodity alone.

Exercise 4.14 (Max-Min Fair Project Assignment)

★★★

(a) Enumerating the $3! = 6$ assignments, the maximum total satisfaction is achieved by team 1 $\rightarrow$ project 1, team 2 $\rightarrow$ project 2, team 3 $\rightarrow$ project 3, with total $8 + 8 + 7 = 23$ (the unique maximizer), matching the check value.

(b) The same assignment maximizes the minimum: its worst-off team scores $\min(8, 8, 7) = 7$. No assignment achieves a minimum of 8: team 2 scores at least 8 only on project 2, and team 3

scores at least 8 only on project 2 (scores 6, 9, 7), so both would need project 2, which is impossible. Hence the best achievable minimum is 7.

(c) The max-min LP

$$\max z \quad \text{s.t.} \quad z \leq \sum_j S_{ij}x_{ij} \quad (i = 1, 2, 3), \quad \sum_j x_{ij} = 1, \quad \sum_i x_{ij} = 1, \quad x_{ij} \geq 0$$

has optimal value $z = 7.5$. One optimal solution is the doubly stochastic matrix

$$x = \begin{bmatrix} 5/6 & 0 & 1/6 \\ 1/6 & 3/4 & 1/12 \\ 0 & 1/4 & 3/4 \end{bmatrix},$$

which gives every team an expected satisfaction of exactly 7.5. Fractions help here because the max-min objective rewards evening out the teams: splitting projects lets the worst-off team absorb some of another team's surplus (team 1 trades away part of project 1's value to lift teams 2 and 3). In part (a) the objective $\sum_{ij} S_{ij}x_{ij}$ is linear over the assignment polytope, whose corners are integral, so some integer assignment is always optimal and fractions cannot improve the total. The added constraints $z \leq \sum_j S_{ij}x_{ij}$ break that structure (see Exercise 4.12(c)), so the LP optimum genuinely exceeds the best integer value.

5 Chapter 5: Graphically Solving Linear Programs

Coverage assessment. The 17 exercises track the chapter's four learning outcomes closely: warm-ups (5.1–5.4) drill sketching and the basic level-curve sweep, the core problems (5.5–5.10) hit each qualitative outcome (multiple optima, infeasibility, finite optimum, unboundedness, plus one modeling formulation), and the concepts and challenge groups (5.11–5.17) ask students to reason about and construct each phenomenon. Difficulty ramps sensibly from one to three stars. The main redundancy is that Exercises 5.12, 5.13, and 5.17 all ask the student to construct an objective with infinitely many optima; each uses a different feasible region and 5.17 adds a tie-breaking twist, so the overlap is tolerable but the author may wish to trim one. Several exercises (5.2, 5.7, 5.8, 5.10, 5.15) derive from Kevin Cheung's CC BY-SA 4.0 text; the attribution currently lives in a commented-out block near the end of the source file rather than a visible footnote. Two data errors in the book were found and corrected directly (see flags at Exercises 5.10 and 5.16).

QA flag

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Exercise 5.1 (Feasible Region Analysis)

★

The correct statement is (b): the feasible region is an unbounded region.

The region is the set of points on or above the line $x + y = 5$ that also lie in the first quadrant. It is nonempty (for example (5, 0) and (3, 2) are feasible), so (d) fails, and it contains more than

one point, so (c) fails. It is not bounded: the ray (t, t) for $t \geq 2.5$ satisfies $t + t \geq 5$ and $t \geq 0$, so feasible points continue infinitely in the direction $(1, 1)$ (indeed in any nonnegative direction). Hence (a) fails and (b) holds.

Exercise 5.2 (Sketch Feasible Region) ★

The boundary line $x - 2y = 2$ passes through $(2, 0)$ and $(0, -1)$. Writing the constraint as $y \geq \frac{1}{2}x - 1$, the feasible set is the closed half-plane consisting of the boundary line and everything above it. To check the side, test the origin: $0 - 0 = 0 \leq 2$, so the origin's side (above the line) is the feasible side. The sketch is the line through $(2, 0)$ and $(0, -1)$ with the upper half-plane shaded. (This matches the book's Selected Solution.)

Exercise 5.3 (Graphical Method 1 Practice) ★

Following Graphical Method 1:

Feasible region. The constraints $x + y \leq 4$, $x \leq 3$, $x, y \geq 0$ cut out a quadrilateral with vertices

$$(0, 0), \quad (3, 0), \quad (3, 1), \quad (0, 4).$$

The vertex $(3, 1)$ is the intersection of $x + y = 4$ and $x = 3$.

Level curves. The lines $3x + 2y = z$ have slope $-3/2$ and move in the direction of the gradient $(3, 2)$ (up and to the right) as z increases.

Sweep. Objective values at the vertices are $z(0, 0) = 0$, $z(3, 0) = 9$, $z(3, 1) = 11$, $z(0, 4) = 8$. Sweeping the level curves in the direction $(3, 2)$, the last line to touch the region passes through $(3, 1)$, where both $x + y \leq 4$ and $x \leq 3$ are binding.

Answer. The unique optimal solution is $(x^*, y^*) = (3, 1)$ with optimal value $z^* = 11$. (Verified numerically; matches the book's Selected Solution.)

Exercise 5.4 (Graphical Method for Lemonade Vendor) ★

The feasible region of

$$\max 3x + 2y \quad \text{s.t.} \quad x + 3y \leq 6, \quad 2x + y \leq 4, \quad x, y \geq 0$$

is the quadrilateral with vertices $(0, 0)$, $(2, 0)$, $(1.2, 1.6)$, and $(0, 2)$; the vertex $(1.2, 1.6)$ solves $x + 3y = 6$ and $2x + y = 4$ simultaneously (from the second equation $y = 4 - 2x$; substituting, $x + 12 - 6x = 6$ gives $x = 1.2$, $y = 1.6$).

The level curves $3x + 2y = z$ have slope $-3/2$; the objective values at the vertices are

$$z(0, 0) = 0, \quad z(2, 0) = 6, \quad z(1.2, 1.6) = 6.8, \quad z(0, 2) = 4.$$

Sweeping the level lines in the gradient direction $(3, 2)$, the last one to touch the region does so only at $(1.2, 1.6)$. The optimal plan is 1.2 units of lemonade and 1.6 units of lemon juice, for a maximum profit of $z^* = 6.8$ dollars.

Exercise 5.5 (Graphical Method) ★★

Note first that the second constraint is $6x_1 + 10x_2 \geq 90$, which after dividing by 2 reads $3x_1 + 5x_2 \geq 45$: its boundary is parallel to the level curves of the objective $3x_1 + 5x_2$. This is the signal for multiple optima.

Region. The constraints $3x_1 + 2x_2 \geq 36$, $6x_1 + 10x_2 \geq 90$, $x_1, x_2 \geq 0$ give an unbounded region lying above both lines. Its vertices are

$$(0, 18), \quad (10, 3), \quad (15, 0),$$

where $(10, 3)$ is the intersection of $3x_1 + 2x_2 = 36$ and $6x_1 + 10x_2 = 90$, and $(15, 0)$ is where $6x_1 + 10x_2 = 90$ meets the x_1 -axis (note $3(15) + 2(0) = 45 \geq 36$, so it is feasible).

Sweep. We minimize, so we sweep the level lines $3x_1 + 5x_2 = z$ opposite to the gradient $(3, 5)$, that is, down and to the left. The last position at which a level line still touches the region is $z = 45$, and there the level line lies along the entire edge of the constraint $6x_1 + 10x_2 = 90$ between $(10, 3)$ and $(15, 0)$.

Two optimal solutions. $(x_1, x_2) = (10, 3)$ and $(x_1, x_2) = (15, 0)$, both with $z^* = 3(10) + 5(3) = 3(15) + 5(0) = 45$. (Verified numerically.) In fact every point of the segment between them, $\{(10 + 5t, 3 - 3t) : t \in [0, 1]\}$, is optimal.

Exercise 5.6 (Infeasible Region Graph) ★★

Graphing the constraints: $x_1 + x_2 \leq 6$ shades below the line through $(6, 0)$ and $(0, 6)$; $x_1 - x_2 \geq 1$ shades below and to the right of the line $x_2 = x_1 - 1$; $x_2 - x_1 \geq 3$ shades above and to the left of the parallel line $x_2 = x_1 + 3$. The second and third boundary lines are parallel (both have slope 1) and their feasible half-planes point away from each other.

Algebraically, adding $x_1 - x_2 \geq 1$ and $x_2 - x_1 \geq 3$ gives $0 \geq 4$, a contradiction, so no point can satisfy both. There are **zero** feasible solutions: the problem is infeasible, regardless of the objective function. (Verified numerically: the solver reports infeasibility. This matches, and slightly expands, the book's Selected Solution.)

Exercise 5.7 (Determine Optimal Value) ★★

Note there are no nonnegativity constraints, so the region is just the intersection of the two half-planes $2x + y \geq 4$ and $x + 3y \geq 1$, an unbounded “wedge” whose single vertex is the intersection of the two boundary lines: from $y = 4 - 2x$ and $x + 3y = 1$ we get $x + 12 - 6x = 1$, so $x = \frac{11}{5}$, $y = -\frac{2}{5}$.

Sweeping the level lines $x + y = z$ opposite to the gradient $(1, 1)$, the last line touching the wedge passes through this vertex, so

$$z^* = \frac{11}{5} - \frac{2}{5} = \frac{9}{5}.$$

To certify this without a picture (as in the book's Selected Solution): multiplying the first constraint by $\frac{2}{5}$ and the second by $\frac{1}{5}$ and adding yields

$$\frac{2}{5}(2x + y) + \frac{1}{5}(x + 3y) = x + y \geq \frac{2}{5}(4) + \frac{1}{5}(1) = \frac{9}{5},$$

so no feasible point does better than $\frac{9}{5}$, and the feasible point $(\frac{11}{5}, -\frac{2}{5})$ attains it. The optimal value is $\frac{9}{5}$. (Verified numerically: optimum 1.8 at $(2.2, -0.4)$.) The multipliers $\frac{2}{5}, \frac{1}{5}$ foreshadow duality.

Exercise 5.8 (Show Unbounded Problem) ★★

Consider the family of points $(x, y) = (t, 0)$ for $t \geq 3$. They are feasible: $2t - 0 = 2t \geq 0$ and $t + 0 = t \geq 3$. The objective value there is $-t + 0 = -t$, and as $t \rightarrow \infty$ the objective tends to $-\infty$. Since we are minimizing, the objective has no lower bound on the feasible region, so the problem is unbounded. (Verified numerically: the solver reports unboundedness. The book's Selected Solution gives the same ray parametrized by the level value z .)

Geometrically, $(1, 0)$ is a direction of the feasible region along which the level lines $-x + y = z$ can be swept forever with z decreasing.

Exercise 5.9 (Bounded Solution on Unbounded Region) ★★

No, the problem does not have a finite solution; it is unbounded.

Following the Complete Graphical Method: the region $(x_1 - x_2 \leq 3, x_1 + 2x_2 \geq 4, x_1, x_2 \geq 0)$ is the unbounded region of Example LPUnboundFeasibleRegion1, extending without limit up and to the right. The gradient of the objective is $\nabla z = (-2, 1)$, which points up and to the left, into the unbounded “upward” part of the region. Concretely, the vertical ray

$$(x_1, x_2) = (0, 2 + t), \quad t \geq 0,$$

is feasible for every $t \geq 0$ (check: $0 - (2 + t) = -2 - t \leq 3$ and $0 + 2(2 + t) = 4 + 2t \geq 4$), and along it

$$z = -2(0) + (2 + t) = 2 + t \rightarrow \infty.$$

The level curves can be swept indefinitely while still meeting the region, so by step 5(a) of the Complete Graphical Method the problem is unbounded. (Verified numerically.)

QA flag

The exercise title “Bounded Solution on Unbounded Region” suggests the answer should be a finite optimum, but the correct answer for $\max(-2x_1 + x_2)$ over this region is that the problem is *unbounded* (direction $(0, 1)$ is feasible and improving). If the intended answer was a finite optimum, a suitable objective would be $\max(-2x_1 - x_2)$ or $\min(3x_1 + x_2)$ as in the book’s Example on finite optima. Judgment call: either retitile the exercise (e.g., “Finite or Unbounded?”) or change the objective; the exercise is answerable as stated.

Exercise 5.10 (Diet Problem) ★★

Let $x_i \geq 0$ denote the amount of Product i purchased, $i = 1, 2, 3$. Each column of the table contributes its nutrient amounts, and the requirements are 0.40 mg of M and 0.30 mg of N :

$$\begin{aligned} \min \quad & 27x_1 + 31x_2 + 24x_3 \\ \text{s.t.} \quad & 0.16x_1 + 0.21x_2 + 0.11x_3 \geq 0.40 && (\text{nutrient } M) \\ & 0.19x_1 + 0.13x_2 + 0.15x_3 \geq 0.30 && (\text{nutrient } N) \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

If only whole units can be bought, add $x_1, x_2, x_3 \in \mathbb{Z}$, making it an integer program.

Although the exercise only asks for the formulation, the LP optimum (verified numerically) is $x^* = (\frac{110}{191}, \frac{280}{191}, 0) \approx (0.576, 1.466, 0)$ with cost $\frac{11650}{191} \approx \60.99 ; both nutrient constraints are binding and Product 3 is not purchased.

QA flag

Fixed in the book: the Selected Solution for this exercise had the right-hand sides swapped (≥ 0.30 on the M row and ≥ 0.40 on the N row), contradicting the problem statement, which requires 0.40 mg of M and 0.30 mg of N . Both displayed formulations in the book’s Selected Solutions were corrected to ≥ 0.40 (M row) and ≥ 0.30 (N row) on 2026-07-13 and re-verified.

Exercise 5.11 (Why Must an Optimum Be at a Vertex?)

★★

1. Think of the level lines of $z = c_1x_1 + c_2x_2$ sweeping across the region in the gradient direction.
 - *Interior point.* If $\mathbf{x}$ is strictly inside the region, a small disk around $\mathbf{x}$ is feasible. Moving from $\mathbf{x}$ a small step in the gradient direction (c_1, c_2) stays feasible and strictly increases z (for a maximization). So an interior point is never strictly better than all boundary points; we may keep moving until we hit the boundary.
 - *Point strictly inside an edge.* Along an edge, the objective is a linear function of the single parameter that traces the edge. A linear function of one variable on a segment attains its maximum at an endpoint of the segment. The endpoints of an edge are vertices, so some vertex of that edge has an objective value at least as large as any point inside the edge. (If the objective is constant along the edge, both endpoints tie with the interior points.)

Combining the two observations: from any feasible point we can move to the boundary without decreasing z , and from any edge point we can move to an endpoint (a vertex) without decreasing z . Hence some vertex is at least as good as any given feasible point, so an optimal vertex exists.

2. Boundedness is used when we claim edges are *segments with endpoints*: on a bounded region every edge has two endpoints, and moving along the boundary always terminates at a vertex. On an unbounded region an “edge” may be a ray with no far endpoint, and the improvement path may go on forever. Example with no optimal solution at all: maximize $z = x + y$ subject to $x, y \geq 0$. The region (the whole first quadrant) is unbounded, and along the feasible ray (t, t) the objective $2t \rightarrow \infty$; no feasible point is optimal because every feasible point is beaten by another.

This is deliberately a picture-level argument; the rigorous version is the algebraic proof in the next chapter, consistent with the book’s lighter treatment of proofs here.

Exercise 5.12 (Infinite Optimal Solutions)

★★

Take the objective $\min z = x_1 + 2x_2$ over the region $x_1 - x_2 \leq 3$, $x_1 + 2x_2 \geq 4$, $x_1, x_2 \geq 0$.

The level curves $x_1 + 2x_2 = z$ are parallel to the boundary of the constraint $x_1 + 2x_2 \geq 4$. Sweeping them downward (opposite the gradient $(1, 2)$, since we minimize), the last position that meets the region is $z = 4$, where the level line lies along the whole edge between the vertices $(0, 2)$ and $(\frac{10}{3}, \frac{1}{3})$.

The set of optima is exactly that edge:

$$\{(x_1, x_2) : x_1 + 2x_2 = 4, 0 \leq x_1 \leq \frac{10}{3}\},$$

with optimal value $z^* = 4$ (verified numerically). Note the direction matters: *maximizing* $x_1 + 2x_2$ over this region is unbounded. Any positive multiple of $(1, 2)$, minimized, gives the same answer.

Exercise 5.13 (Infinitely Many Optimal Solutions)

★★

Choose the objective parallel to one of the resource constraints. Two natural choices:

Choice 1. $\max z = x + 3y$ (parallel to the lemon constraint $x + 3y \leq 6$). The last level line touches the region along the entire edge between $(0, 2)$ and $(1.2, 1.6)$, so

$$z^* = 6, \quad \text{optimal set} = \{(x, y) : x + 3y = 6, 0 \leq x \leq 1.2\},$$

the segment from $(0, 2)$ to $(1.2, 1.6)$.

Choice 2. $\max z = 2x + y$ (parallel to the water constraint $2x + y \leq 4$). Then $z^* = 4$ and the optimal set is the segment from $(1.2, 1.6)$ to $(2, 0)$, that is, $\{(x, y) : 2x + y = 4, 1.2 \leq x \leq 2\}$.

(Both verified numerically: the solver returns $z^* = 6$ and $z^* = 4$ respectively, at vertices of the corresponding edges.) In either case the optimal level curve coincides with an edge of the feasible region, which is precisely the tie condition in Graphical Method 2, so there are infinitely many optimal solutions.

Exercise 5.14 (Bounded Optimum on Unbounded Region) ★★

Region. The constraints $-x_1 + x_2 \leq 5$, $x_1 + x_2 \geq 2$, $x_1, x_2 \geq 0$ give an unbounded region with vertices $(0, 5)$, $(0, 2)$, and $(2, 0)$; it extends without bound up and to the right (any direction $\mathbf{d} = (d_1, d_2) \geq 0$ with $d_2 \leq d_1$ is a feasible direction, e.g. $(1, 0)$ and $(1, 1)$).

Objective. Take

$$\max z = -2x_1 - x_2.$$

Its gradient $(-2, -1)$ points down and to the left, away from every unbounded direction of the region: for any feasible direction $\mathbf{d} \geq 0$, we have $-2d_1 - d_2 \leq 0$, so moving into the unbounded part never increases z .

Sweep. Sweeping the level lines $-2x_1 - x_2 = z$ in the direction $(-2, -1)$, the last line touching the region passes through the vertex $(0, 2)$. Objective values at the vertices: $z(0, 5) = -5$, $z(0, 2) = -2$, $z(2, 0) = -4$. The finite optimum is

$$z^* = -2 \quad \text{at} \quad (x_1^*, x_2^*) = (0, 2),$$

verified numerically. The sketch shows the level lines leaving the region at $(0, 2)$ even though the region itself is unbounded. (Any objective $\max c_1x_1 + c_2x_2$ with $c_1, c_2 < 0$ and (c_1, c_2) not parallel to $(1, 1)$ gives a unique finite optimum in the same way; $\max(-x_1 - x_2)$ gives a finite optimum attained on the whole edge from $(0, 2)$ to $(2, 0)$.)

Exercise 5.15 (Unbounded Objective Direction) ★★★

1. For $(x_1, x_2) = (t, t)$ with $t \geq 0$:

$$-t + t = 0 \leq 3, \quad t - 2t = -t \leq 2, \quad t \geq 0, \quad t \geq 0,$$

so all four constraints hold for every $t \geq 0$. The objective value is

$$z(t) = -2t + t = -t,$$

and $z(t) \rightarrow -\infty$ as $t \rightarrow \infty$. Since we minimize, the objective is unbounded below along this ray.

2. For $(x_1, x_2) = (2t + 2, t)$ with $t \geq 0$:

$$-(2t + 2) + t = -t - 2 \leq 3, \quad (2t + 2) - 2t = 2 \leq 2 \text{ (tight)}, \quad 2t + 2 \geq 0, \quad t \geq 0,$$

so the point is feasible for all $t \geq 0$; it travels along the boundary line $x_1 - 2x_2 = 2$. The objective value is

$$z(t) = -2(2t + 2) + t = -3t - 4,$$

which again tends to $-\infty$ as $t \rightarrow \infty$. This gives a second certificate of unboundedness, along a different ray. (Unboundedness verified numerically.)

The two rays have directions $(1, 1)$ and $(2, 1)$; any feasible direction $\mathbf{d}$ with $\mathbf{c}^\top \mathbf{d} < 0$ certifies unboundedness of a minimization.

Exercise 5.16 (Bounded and Unbounded Optimization)

★★★

1. *Minimum.* The region has vertices $(0, 6)$, $(0, 4)$, $(1, 2)$, and $(5, 0)$, where $(1, 2)$ is the intersection of $2x + y = 4$ and $x + 2y = 5$, and it is unbounded to the upper right. Objective values: $z(0, 6) = 12$, $z(0, 4) = 8$, $z(1, 2) = 6$, $z(5, 0) = 10$. Sweeping the level lines $2x + 2y = z$ downward (we minimize, gradient $(2, 2)$), the last contact is at the vertex $(1, 2)$. The minimum value is

$$z^* = 2(1) + 2(2) = 6 \quad \text{at } (x, y) = (1, 2),$$

a bounded feasible point at which $2x + y \geq 4$ and $x + 2y \geq 5$ are binding. (Verified numerically.)

2. For $(x, y) = (5 + t, 0)$ with $t \geq 0$: $-(5 + t) + 0 \leq 12$, $2(5 + t) = 10 + 2t \geq 4$, $(5 + t) + 0 = 5 + t \geq 5$, and $x, y \geq 0$. So the ray is feasible, and

$$z(t) = 2(5 + t) + 0 = 10 + 2t \rightarrow \infty \quad \text{as } t \rightarrow \infty,$$

so the maximization is unbounded.

3. For $(x, y) = (0, 6) + t(2, 1) = (2t, 6 + t)$ with $t \geq 0$:

$$-2t + 2(6 + t) = 12 \leq 12 \text{ (tight for all } t), \quad 2(2t) + (6 + t) = 6 + 5t \geq 4, \quad 2t + 2(6 + t) = 12 + 4t \geq 5,$$

and $x = 2t \geq 0$, $y = 6 + t \geq 0$. The ray travels along the boundary $-x + 2y = 12$. The objective value is

$$z(t) = 2(2t) + 2(6 + t) = 12 + 6t \rightarrow \infty \quad \text{as } t \rightarrow \infty,$$

a second demonstration of unboundedness. (All parts verified numerically.)

QA flag

Fixed in the book: part (c) originally used the direction $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ from $(0, 6)$, but $(t, 6 + 2t)$ violates $-x + 2y \leq 12$ for every $t > 0$ (the left side is $12 + 3t$), so the claimed feasibility was false. The boundary direction of that constraint is $(2, 1)$; the book was corrected to $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ on 2026-07-13 and re-verified (the ray $(2t, 6 + t)$ keeps $-x + 2y = 12$ tight and all other constraints satisfied). There was no book Selected Solution to sync.

Exercise 5.17 (From an Optimal Edge to a Unique Optimum)

★★★

1. Consider

$$\max z = x + y \quad \text{s.t. } x + y \leq 4, \quad x \leq 3, \quad x, y \geq 0,$$

the feasible region of Exercise 5.3, with vertices $(0, 0)$, $(3, 0)$, $(3, 1)$, $(0, 4)$. The level curves $x + y = z$ are parallel to the constraint boundary $x + y = 4$. Sweeping in the gradient direction $(1, 1)$, the last level curve to touch the region is $x + y = 4$, and it contains the entire edge from $(0, 4)$ to $(3, 1)$. By the tie-checking step of Graphical Method 2 (optimal level curve parallel to an edge adjacent to an optimal vertex), every point of

$$\{(x, y) : x + y = 4, \quad 0 \leq x \leq 3\}$$

is optimal, with $z^* = 4$: the optimal set is a whole edge, not a single vertex.

2. Change the y -coefficient from 1 to 2, giving $\max z = x + 2y$ over the same region. Now the level curves have slope $-1/2$, no longer parallel to any edge of the region. Vertex values: $z(0,0) = 0$, $z(3,0) = 3$, $z(3,1) = 5$, $z(0,4) = 8$. The unique optimum is $(0,4)$ with $z^* = 8$. The tie disappears because along the former optimal edge $x + y = 4$ the new objective $x + 2y = x + 2(4 - x) = 8 - x$ is strictly decreasing in x , so the edge's endpoint $(0,4)$ strictly beats every other point of the edge.

Any construction with these two features (objective parallel to a binding edge, then a single perturbed coefficient) earns full credit.

6 Chapter 6: Formal Mathematical Statements

Coverage assessment. The 12 exercises give a well-balanced pass over the chapter: warm-ups (6.1–6.5) drill vectors, combinations, rank, convexity recognition, and matrix form of an LP; core problems (6.6–6.9) work concretely with convex combinations, halfspaces, and the vertex/extreme point machinery on the same polyhedron used to illustrate the Representation Theorem, which is a nice touch; the concepts group (6.10–6.11) probes the proofs (union versus intersection, and exactly where boundedness enters the vertex theorem); and the single challenge problem (6.12) is the one genuine proof exercise. Proof coverage is intentionally light, matching the chapter's own warning box that students need not reproduce the proofs; this is appropriate and the author has indicated this is acceptable for theory chapters. No data errors were found; the four book Selected Solutions (6.1, 6.6, 6.9, 6.10) check out and are reused below. No exercise here appears to derive from outside sources; content and exercises look original to this book. One minor pedagogical note: in Exercise 6.2 part 1, every vector in $\mathbb{R}^2$ is a linear combination of $\mathbf{v}^1, \mathbf{v}^2$, so all three answers are “yes”; this is arguably the point of the exercise, so it is not flagged.

Exercise 6.1 (Vector Operations) ★

Working componentwise (matches the book's Selected Solution):

1. $\mathbf{v} + \mathbf{u} = [3 + (-1), -2 + 0, 4 + 2] = [2, -2, 6]$.
2. $2\mathbf{v} = [6, -4, 8]$.
3. $\mathbf{v} \cdot \mathbf{u} = (3)(-1) + (-2)(0) + (4)(2) = -3 + 0 + 8 = 5$.

Exercise 6.2 (Linear and Convex Combinations) ★

1. All three are linear combinations of $\mathbf{v}^1 = [1, 0]$ and $\mathbf{v}^2 = [0, 1]$, since these two vectors span $\mathbb{R}^2$: any $[a, b] = a\mathbf{v}^1 + b\mathbf{v}^2$. Explicitly,
 - (a) $[2, 3] = 2\mathbf{v}^1 + 3\mathbf{v}^2$,
 - (b) $[1, 1] = 1\mathbf{v}^1 + 1\mathbf{v}^2$,
 - (c) $[1, -1] = 1\mathbf{v}^1 - 1\mathbf{v}^2$.

Note (c) is a linear combination but not a non-negative one.

2. Yes. Using the pair $\mathbf{v}^1, \mathbf{v}^2$ with $\lambda = \frac{1}{2}$:

$$\frac{1}{2}[1, 0] + \frac{1}{2}[0, 1] = [\frac{1}{2}, \frac{1}{2}],$$

and the weights are nonnegative and sum to 1. (This is the only pair that works: combinations of $\mathbf{v}^1$ and $\mathbf{v}^3$ have first coordinate 1, and combinations of $\mathbf{v}^2$ and $\mathbf{v}^3$ have second coordinate 1.)

Exercise 6.3 (Linear Independence and Matrix Rank) ★

1. No. The second row $[2, 4]$ equals 2 times the first row $[1, 2]$, so the rows are linearly dependent: $2[1, 2] - [2, 4] = [0, 0]$ is a nontrivial vanishing combination.
2. No. The second column $[2, 4]^\top$ equals 2 times the first column $[1, 2]^\top$, so the columns are dependent as well.
3. $\text{rank}(A) = 1$: the maximum number of linearly independent rows (equivalently columns) is 1. Consistent checks: $\det A = 1 \cdot 4 - 2 \cdot 2 = 0$, so the rank is less than 2, and $A \neq 0$, so the rank is at least 1.

Exercise 6.4 (Convex or Not?) ★

1. *The disk D : convex.* If $\mathbf{x}, \mathbf{y} \in D$ then by the triangle inequality for the Euclidean norm, $\|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\| \leq \lambda \|\mathbf{x}\| + (1 - \lambda) \|\mathbf{y}\| \leq \lambda \cdot 2 + (1 - \lambda) \cdot 2 = 2$, so the segment stays in D . (Balls are convex, as listed in the chapter.)
2. *The ring R : not convex.* Take $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (-1, 0)$; both have norm 1, so both are in R . Their midpoint is $(0, 0)$, with $x_1^2 + x_2^2 = 0 < 1$, which is not in R : the segment passes through the hole.
3. *The halfspace H : convex.* For $\mathbf{x}, \mathbf{y}$ with $x_1 + 2x_2 \leq 3$ and $y_1 + 2y_2 \leq 3$, linearity gives $(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y})_1 + 2(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y})_2 = \lambda(x_1 + 2x_2) + (1 - \lambda)(y_1 + 2y_2) \leq \lambda \cdot 3 + (1 - \lambda) \cdot 3 = 3$. (Halfspaces are convex, as listed in the chapter.)
4. *The set $S = \{\mathbf{x} : |x_1| \geq 1\}$: not convex.* Take $\mathbf{x} = (1, 0) \in S$ and $\mathbf{y} = (-1, 0) \in S$. Their midpoint $(0, 0)$ has $|x_1| = 0 < 1$, so it is not in S : the set is two disjoint halfplanes and the segment crosses the gap between them.

Exercise 6.5 (Writing an LP in Matrix Notation) ★

Order the constraints as: lemon row, water row, then the two rewritten sign constraints $-x_1 \leq 0$ and $-x_2 \leq 0$. Then

$$\mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 4 \\ 0 \\ 0 \end{bmatrix},$$

and the LP is $\max\{\mathbf{c}^\top \mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$.

Verification at $\mathbf{x} = (1.2, 1.6)$:

$$A\mathbf{x} = \begin{bmatrix} 1.2 + 3(1.6) \\ 2(1.2) + 1.6 \\ -1.2 \\ -1.6 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ -1.2 \\ -1.6 \end{bmatrix} \leq \begin{bmatrix} 6 \\ 4 \\ 0 \\ 0 \end{bmatrix},$$

so $\mathbf{x}$ is feasible (verified numerically). The first two constraints hold with equality: they are the two tight constraints at this vertex, which is the optimal solution of the Lemonade Vendor LP from the previous chapter.

Exercise 6.6 (Convex Combinations and Geometry) ★★

The triangle P is the set of convex combinations of $(0, 0)$, $(2, 0)$, $(0, 2)$; equivalently $P = \{\mathbf{x} \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 2\}$. (Matches the book's Selected Solution.)

1. The plot is the closed triangle with the three given vertices: the legs lie on the axes and the hypotenuse is the segment of $x_1 + x_2 = 2$ from $(2, 0)$ to $(0, 2)$.
2. Yes. Take $\lambda_1 = 0$, $\lambda_2 = \lambda_3 = \frac{1}{2}$:

$$0 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

with nonnegative weights summing to 1; so $(1, 1)$ is the midpoint of the hypotenuse.

3. No. Any combination of the vertices hitting $(1.5, 1.5)$ needs $2\lambda_2 = 1.5$ and $2\lambda_3 = 1.5$, forcing $\lambda_2 = \lambda_3 = 0.75$ and $\lambda_2 + \lambda_3 = 1.5 > 1$. Equivalently, $1.5 + 1.5 = 3 > 2$ violates $x_1 + x_2 \leq 2$, so $(1.5, 1.5) \notin P$.

Exercise 6.7 (Halfspace Geometry and Convexity) ★★

1. The boundary is the line $x_1 + 2x_2 = 4$, through $(4, 0)$ and $(0, 2)$. The halfspace H is that line together with everything below it (the side containing the origin, since $0 + 0 \leq 4$).
2. $(2, 1)$: $2 + 2(1) = 4 \leq 4$, so yes, $(2, 1) \in H$; it lies exactly on the boundary. $(1, 2)$: $1 + 2(2) = 5 > 4$, so $(1, 2) \notin H$.
3. The midpoint of $(0, 0)$ and $(2, 1)$ is $(1, \frac{1}{2})$, and $1 + 2 \cdot \frac{1}{2} = 2 \leq 4$, so the midpoint lies in H , consistent with convexity. (Strictly, checking one midpoint illustrates convexity but does not prove it; the general argument is the linearity computation of Theorem “Convexity of Polyhedra” applied to the single row $[1 \ 2]$: for any $\mathbf{x}, \mathbf{y} \in H$ and $\lambda \in [0, 1]$, $(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y})$ satisfies $x_1 + 2x_2 \leq \lambda 4 + (1 - \lambda)4 = 4$.)

Exercise 6.8 (Checking an Extreme Point) ★★

Write the five constraints with rows $A_1 = [1, -4]$, $A_2 = [-1, 1]$, $A_3 = [-1, 2]$, $A_4 = [-1, 0]$, $A_5 = [0, -1]$ and right-hand sides $(0, 1, 4, 0, 0)$.

1. At $(2, 3)$: $A_1\mathbf{x} = 2 - 12 = -10 \leq 0$; $A_2\mathbf{x} = -2 + 3 = 1$ (tight); $A_3\mathbf{x} = -2 + 6 = 4$ (tight); $-x_1 = -2 \leq 0$; $-x_2 = -3 \leq 0$. So $(2, 3) \in P$ and the tight constraints are exactly rows 2 and 3.
3. The matrix of tight rows,

$$A_I = \begin{bmatrix} -1 & 1 \\ -1 & 2 \end{bmatrix}, \quad \det A_I = (-1)(2) - (1)(-1) = -1 \neq 0,$$

has rank 2 (verified numerically), so the two tight rows are linearly independent and $|I| = n = 2$. The system $-x_1 + x_2 = 1$, $-x_1 + 2x_2 = 4$ has the unique solution $(2, 3)$, so $(2, 3)$ is a vertex of P , hence an extreme point by the chapter's equivalence theorem.

2. At $(1, 2)$: $A_1\mathbf{x} = 1 - 8 = -7 \leq 0$; $A_2\mathbf{x} = -1 + 2 = 1$ (tight); $A_3\mathbf{x} = -1 + 4 = 3 < 4$; $-1 < 0$; $-2 < 0$. So $(1, 2) \in P$ and only constraint 2 is tight; one row cannot contain 2 linearly independent rows, so $(1, 2)$ is not a vertex. To exhibit it as a convex combination, slide along the tight line $-x_1 + x_2 = 1$: the points $(0, 1)$ and $(2, 3)$ both lie in P (check $(0, 1)$: $0 - 4 = -4 \leq 0$, $-0 + 1 = 1 \leq 1$, $-0 + 2 = 2 \leq 4$, $0 \geq 0$, $1 \geq 0$), and

$$(1, 2) = \frac{1}{2}(0, 1) + \frac{1}{2}(2, 3),$$

a strict convex combination of two other points of P . Hence $(1, 2)$ is not an extreme point.

Exercise 6.9 (A Convex Combination of Extreme Points) ★★

(Matches the book's Selected Solution.) The first coordinate forces $2\lambda_3 = 1$, so $\lambda_3 = \frac{1}{2}$. The second coordinate then gives $\lambda_2 + \frac{3}{2} = \frac{7}{4}$, so $\lambda_2 = \frac{1}{4}$, and normalization gives $\lambda_1 = 1 - \frac{1}{4} - \frac{1}{2} = \frac{1}{4}$. All multipliers are nonnegative, and indeed

$$\frac{1}{4} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 7/4 \end{bmatrix}$$

(verified numerically). No extreme directions are needed because $(1, \frac{7}{4})$ already lies in the convex hull of the three extreme points; in the Representation Theorem we may take all $\mu_j = 0$. Extreme directions become necessary only for points of P outside that convex hull, farther out in the unbounded part of the region.

Exercise 6.10 (Is the Union of Convex Sets Convex?) ★★

(Matches the book's Selected Solution.)

1. Take C_1 the unit disk centered at $(-2, 0)$ and C_2 the unit disk centered at $(2, 0)$; each disk is convex. The points $(-2, 0) \in C_1$ and $(2, 0) \in C_2$ lie in the union, but their midpoint $(0, 0)$ is at distance $2 > 1$ from both centers, so it lies in neither disk. Hence $C_1 \cup C_2$ is not convex.
2. If one set contains the other, say $C_1 \subseteq C_2$, then $C_1 \cup C_2 = C_2$, which is convex. Nesting is sufficient but not necessary: with $C_1 = \{\mathbf{x} : x_2 \geq 0\}$ and $C_2 = \{\mathbf{x} : x_2 \leq 0\}$, neither halfplane contains the other, yet $C_1 \cup C_2 = \mathbb{R}^2$ is convex.

Exercise 6.11 (Where Boundedness Enters the Vertex Theorem) ★★

1. Boundedness is used exactly once, after the proof establishes $\mathbf{c}^\top \mathbf{d} = 0$: the proof asserts that “since P is bounded, the line $\{\mathbf{x}(t) : t \in \mathbb{R}\}$ cannot be entirely contained in P , so the set $\{t \geq 0 : \mathbf{x}(t) \in P\}$ is a closed, bounded interval $[0, t^+]$,” and at $t = t^+$ a new constraint becomes tight. Without boundedness, the whole line $\mathbf{x}^* + t\mathbf{d}$ could remain feasible for all t ; then no new constraint ever becomes tight, the count of independent tight constraints never grows, and the induction stalls. (Worse, the polyhedron may contain no vertex at all, as in part 2.)
2. Let $P = \{\mathbf{x} \in \mathbb{R}^2 : 0 \leq x_2 \leq 1\}$, a horizontal slab, and maximize $-x_2$.
No vertex. The constraint rows are $[0, 1]$ (for $x_2 \leq 1$) and $[0, -1]$ (for $-x_2 \leq 0$). At any point of P at most one of them can be tight (x_2 cannot equal 0 and 1 simultaneously), and even both together span only the x_2 -axis direction. So the tight constraints at any point have rank

at most $1 < 2 = n$, and no point of P is the unique solution of n linearly independent tight constraints: P has no vertices. (Geometrically, P contains the horizontal line through any of its points, and a set containing a line has no extreme points.)

Optima exist. The objective $-x_2$ is at most 0 on P , and equals 0 exactly when $x_2 = 0$. So the LP has optimal value 0 and the set of optima is the entire boundary line $\{(x_1, 0) : x_1 \in \mathbb{R}\}$, infinitely many optima but none at a vertex.

Which hypothesis fails. P is unbounded (x_1 is free), so the theorem's boundedness hypothesis fails, and the theorem simply does not apply; there is no contradiction. The example shows the hypothesis is not removable: an unbounded feasible region may fail to have any vertex for an optimum to sit at, even when optima exist.

Exercise 6.12 (Intersections of Arbitrary Families of Convex Sets)

★★★

Proof. Let $\mathbf{x}, \mathbf{y} \in C = \bigcap_{i \in I} C_i$ and let $\lambda \in [0, 1]$; set $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. Fix any index $i \in I$. Since $\mathbf{x} \in C$ we have $\mathbf{x} \in C_i$, and likewise $\mathbf{y} \in C_i$. Because C_i is convex, $\mathbf{z} \in C_i$. As $i \in I$ was arbitrary, $\mathbf{z}$ belongs to every C_i , that is, $\mathbf{z} \in C$. Hence C contains all convex combinations of its points and is convex. Nothing in the argument used any bound on $|I|$; it works verbatim for infinite families. If C is empty the statement holds vacuously, since there are no pairs $\mathbf{x}, \mathbf{y}$ to test. ■

Application to polyhedra. A polyhedron $P = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$ with m rows is exactly the intersection of the m halfspaces $H_i = \{\mathbf{x} : A_i\mathbf{x} \leq b_i\}$. Each halfspace is convex (the one-row case of the linearity argument, or the chapter's list of convex sets). By the result just proved, applied to the finite family $\{H_1, \dots, H_m\}$, the intersection P is convex. This reproves Theorem "Convexity of Polyhedra" without redoing the componentwise computation.

7 Chapter 7: Simplex Method

This chapter's seventeen exercises track the chapter's arc: six warm-ups on standard form and the initial dictionary (7.1 to 7.6), seven core simplex solves in dictionary form including two Big-M problems (7.7 to 7.13), three conceptual problems on optimality conditions, geometry of basic feasible solutions, and the ratio test (7.14 to 7.16), and one challenge problem on degeneracy from ratio-test ties (7.17). Every pivot below was verified by exact-fraction basis computations and each optimum was confirmed with an LP solver. Solutions printed in the book's Selected Solutions (7.5, 7.6, 7.7, 7.8, 7.9, 7.14, 7.17) are reproduced here, lightly edited, so that this manual is self-contained.

Exercise 7.1 (Standard form and the first dictionary)

★

(a) Add slack variables $s_1, s_2 \geq 0$:

$$\begin{array}{ll} \max & 4x + 5y \\ \text{s.t.} & x + 2y + s_1 = 12, \\ & 3x + 2y + s_2 = 24, \\ & x, y, s_1, s_2 \geq 0. \end{array}$$

(b) Solving each equation for its slack gives the initial dictionary at the slack basis $\{s_1, s_2\}$:

$$\begin{aligned} \max \quad & z = 0 + 4x + 5y \\ \text{s.t.} \quad & s_1 = 12 - x - 2y, \\ & s_2 = 24 - 3x - 2y, \\ & x, y, s_1, s_2 \geq 0. \end{aligned}$$

(c) Setting the nonbasic variables $x = y = 0$ gives the basic feasible solution $(x, y, s_1, s_2) = (0, 0, 12, 24)$ with objective value $z = 0$.

(d) Under steepest ascent we pick the largest objective coefficient: $5 > 4$, so y enters first.

Exercise 7.2 (Convert to Standard Form) ★

The objective is already a maximization and both variables are already nonnegative. Add a slack variable $s_1 \geq 0$ to the $\leq$ constraint and subtract a surplus variable $s_2 \geq 0$ from the $\geq$ constraint:

$$\begin{aligned} \max \quad & 3x_1 - 5x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 + s_1 = 6, \\ & -x_1 + 3x_2 - s_2 = 2, \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Exercise 7.3 (Convert to Standard Form) ★

Negate the objective to convert min to max: $\max x_1 - 2x_2 - x_3$. Since x_3 is unrestricted, write $x_3 = x_3^+ - x_3^-$ with $x_3^+, x_3^- \geq 0$. The first constraint is already an equality; add a slack $s_1 \geq 0$ to the second:

$$\begin{aligned} \max \quad & x_1 - 2x_2 - x_3^+ + x_3^- \\ \text{s.t.} \quad & x_1 - x_2 + 2x_3^+ - 2x_3^- = 4, \\ & -2x_1 + x_2 + s_1 = 1, \\ & x_1, x_2, x_3^+, x_3^-, s_1 \geq 0. \end{aligned}$$

Exercise 7.4 (Convert to Standard Form) ★

The objective is already a maximization. Handle the variables: x_1 is unrestricted, so set $x_1 = x_1^+ - x_1^-$ with $x_1^+, x_1^- \geq 0$; $x_2, x_3 \leq 0$, so set $x_2 = -x_2'$ and $x_3 = -x_3'$ with $x_2', x_3' \geq 0$. Substituting, the objective becomes $5x_1^+ - 5x_1^- - x_2' + 3x_3'$. Subtract a surplus $s_1 \geq 0$ from the $\geq$ constraint; the equality stays as is:

$$\begin{aligned} \max \quad & 5x_1^+ - 5x_1^- - x_2' + 3x_3' \\ \text{s.t.} \quad & x_1^+ - x_1^- - 2x_2' + x_3' - s_1 = 7, \\ & -x_2' + 4x_3' = 3, \\ & x_1^+, x_1^-, x_2', x_3', s_1 \geq 0. \end{aligned}$$

Exercise 7.5 (Convert to Standard Form) ★

(Book Selected Solution.) Negate the objective: $\max -2x_1 - 4x_2$. Subtract a surplus $s_1 \geq 0$ from the $\geq$ constraint and add a slack $s_2 \geq 0$ to the $\leq$ constraint. Since x_1 is unrestricted, replace

$x_1 = x'_1 - x''_1$ with $x'_1, x''_1 \geq 0$:

$$\begin{array}{ll}\max & -2(x'_1 - x''_1) - 4x_2 \\ \text{s.t.} & (x'_1 - x''_1) + x_2 - s_1 = 3, \\ & 2(x'_1 - x''_1) - x_2 + s_2 = 5, \\ & x'_1, x''_1, x_2, s_1, s_2 \geq 0.\end{array}$$

Exercise 7.6 (Convert to Standard Form) ★

(Book Selected Solution.) Negate the objective: $\max -5y_1 + 3y_2 - y_3$. Add a slack $s_1 \geq 0$ to the $\leq$ constraint; the equality needs no new variable. Since y_3 is unrestricted, replace $y_3 = y'_3 - y''_3$ with $y'_3, y''_3 \geq 0$:

$$\begin{array}{ll}\max & -5y_1 + 3y_2 - (y'_3 - y''_3) \\ \text{s.t.} & 2y_1 - y_2 + 3(y'_3 - y''_3) = 7, \\ & -y_1 + 4y_2 + s_1 = 5, \\ & y_1, y_2, y'_3, y''_3, s_1 \geq 0.\end{array}$$

Exercise 7.7 (Pivoting in a different direction) ★★

(Book Selected Solution, verified.) Entering x first costs an extra pivot compared with the steepest-ascent path: three pivots in all.

Pivot 1: x enters. Ratio test: $9/1 = 9$ (s_1), $16/2 = 8$ (s_2), $14/1 = 14$ (s_3); the minimum is 8, so s_2 leaves. Solving s_2 's row for x and substituting:

$$\begin{array}{ll}\max & z = 16 + 2y - s_2 \\ \text{s.t.} & s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2, \\ & x = 8 - \frac{1}{2}y - \frac{1}{2}s_2, \\ & s_3 = 6 - \frac{3}{2}y + \frac{1}{2}s_2.\end{array}$$

Pivot 2: y enters (only positive coefficient). Ratios: $1/\frac{1}{2} = 2$ (s_1), $8/\frac{1}{2} = 16$ (x), $6/\frac{3}{2} = 4$ (s_3); so s_1 leaves:

$$\begin{array}{ll}\max & z = 20 - 4s_1 + s_2 \\ \text{s.t.} & y = 2 - 2s_1 + s_2, \\ & x = 7 + s_1 - s_2, \\ & s_3 = 3 + 3s_1 - s_2.\end{array}$$

Pivot 3: s_2 enters (coefficient +1). The y row has s_2 with a positive coefficient, so it imposes no limit; ratios $7/1 = 7$ (x) and $3/1 = 3$ (s_3), so s_3 leaves:

$$\begin{array}{ll}\max & z = 23 - s_1 - s_3 \\ \text{s.t.} & x = 4 - 2s_1 + s_3, \\ & s_2 = 3 + 3s_1 - s_3, \\ & y = 5 + s_1 - s_3.\end{array}$$

All objective coefficients are negative, so this dictionary is optimal: $(x, y) = (4, 5)$ with $z^* = 23$, the same optimum the steepest-ascent path reaches in two pivots.

Exercise 7.8 (Simplex Method Practice)

★★

(Book Selected Solution, verified.) Initial dictionary:

$$\begin{array}{ll}\max & z = 5x_1 + 3x_2 \\ \text{s.t.} & s_1 = 12 - x_1 - x_2, \\ & s_2 = 16 - 2x_1 - x_2.\end{array}$$

Pivot 1: x_1 enters (coefficient 5). Ratios: $12/1 = 12$ and $16/2 = 8$, so s_2 leaves:

$$\begin{array}{ll}\max & z = 40 + \frac{1}{2}x_2 - \frac{5}{2}s_2 \\ \text{s.t.} & x_1 = 8 - \frac{1}{2}x_2 - \frac{1}{2}s_2, \\ & s_1 = 4 - \frac{1}{2}x_2 + \frac{1}{2}s_2.\end{array}$$

Pivot 2: x_2 enters. Ratios: $8/\frac{1}{2} = 16$ and $4/\frac{1}{2} = 8$, so s_1 leaves:

$$\begin{array}{ll}\max & z = 44 - s_1 - 2s_2 \\ \text{s.t.} & x_2 = 8 - 2s_1 + s_2, \\ & x_1 = 4 + s_1 - s_2.\end{array}$$

Optimal: $(x_1, x_2) = (4, 8)$ with $z^* = 44$ (both constraints tight).

Exercise 7.9 (Simplex Method Practice)

★★

(Book Selected Solution, verified.) Initial dictionary:

$$\begin{array}{ll}\max & z = 5x_1 + 8x_2 \\ \text{s.t.} & s_1 = 30 - x_1 - 2x_2, \\ & s_2 = 30 - 3x_1 - x_2.\end{array}$$

Pivot 1: x_2 enters ($8 > 5$). Ratios: $30/2 = 15$ and $30/1 = 30$, so s_1 leaves:

$$\begin{array}{ll}\max & z = 120 + x_1 - 4s_1 \\ \text{s.t.} & x_2 = 15 - \frac{1}{2}x_1 - \frac{1}{2}s_1, \\ & s_2 = 15 - \frac{5}{2}x_1 + \frac{1}{2}s_1.\end{array}$$

Pivot 2: x_1 enters. Ratios: $15/\frac{1}{2} = 30$ and $15/\frac{5}{2} = 6$, so s_2 leaves:

$$\begin{array}{ll}\max & z = 126 - \frac{19}{5}s_1 - \frac{2}{5}s_2 \\ \text{s.t.} & x_1 = 6 + \frac{1}{5}s_1 - \frac{2}{5}s_2, \\ & x_2 = 12 - \frac{3}{5}s_1 + \frac{1}{5}s_2.\end{array}$$

Optimal: $(x_1, x_2) = (6, 12)$ with $z^* = 126$.

Exercise 7.10 (Simplex Method Practice)

★★

Add slacks s_1, s_2 :

$$\begin{array}{ll}\max & z = 2x_1 + 3x_2 + x_3 \\ \text{s.t.} & s_1 = 32 - 4x_1 - 2x_2 - 5x_3, \\ & s_2 = 28 - 2x_1 - 4x_2 - 3x_3.\end{array}$$

Pivot 1: x_2 enters (largest coefficient 3). Ratios: $32/2 = 16$ and $28/4 = 7$, so s_2 leaves. With $x_2 = 7 - \frac{1}{2}x_1 - \frac{3}{4}x_3 - \frac{1}{4}s_2$:

$$\begin{aligned} \max \quad & z = 21 + \frac{1}{2}x_1 - \frac{5}{4}x_3 - \frac{3}{4}s_2 \\ \text{s.t.} \quad & s_1 = 18 - 3x_1 - \frac{7}{2}x_3 + \frac{1}{2}s_2, \\ & x_2 = 7 - \frac{1}{2}x_1 - \frac{3}{4}x_3 - \frac{1}{4}s_2. \end{aligned}$$

Pivot 2: x_1 enters (coefficient $\frac{1}{2}$). Ratios: $18/3 = 6$ and $7/\frac{1}{2} = 14$, so s_1 leaves:

$$\begin{aligned} \max \quad & z = 24 - \frac{11}{6}x_3 - \frac{1}{6}s_1 - \frac{2}{3}s_2 \\ \text{s.t.} \quad & x_1 = 6 - \frac{7}{6}x_3 - \frac{1}{3}s_1 + \frac{1}{6}s_2, \\ & x_2 = 4 - \frac{1}{6}x_3 + \frac{1}{6}s_1 - \frac{1}{3}s_2. \end{aligned}$$

All objective coefficients are negative, so this is optimal: $(x_1, x_2, x_3) = (6, 4, 0)$ with $z^* = 24$; both constraints are tight and x_3 never enters.

Exercise 7.11 (Simplex Method Practice) ★★

Add slacks s_1, s_2, s_3 :

$$\begin{aligned} \max \quad & z = 6x_1 + 8x_2 + 5x_3 \\ \text{s.t.} \quad & s_1 = 1800 - 4x_1 - x_2 - x_3, \\ & s_2 = 2000 - 2x_1 - 2x_2 - x_3, \\ & s_3 = 3200 - 4x_1 - 2x_2 - x_3. \end{aligned}$$

Pivot 1: x_2 enters (coefficient 8). Ratios: $1800/1 = 1800$, $2000/2 = 1000$, $3200/2 = 1600$, so s_2 leaves. With $x_2 = 1000 - x_1 - \frac{1}{2}x_3 - \frac{1}{2}s_2$:

$$\begin{aligned} \max \quad & z = 8000 - 2x_1 + x_3 - 4s_2 \\ \text{s.t.} \quad & s_1 = 800 - 3x_1 - \frac{1}{2}x_3 + \frac{1}{2}s_2, \\ & x_2 = 1000 - x_1 - \frac{1}{2}x_3 - \frac{1}{2}s_2, \\ & s_3 = 1200 - 2x_1 + s_2. \end{aligned}$$

Pivot 2: x_3 enters (coefficient $+1$). The s_3 row does not contain x_3 ; ratios $800/\frac{1}{2} = 1600$ (s_1) and $1000/\frac{1}{2} = 2000$ (x_2), so s_1 leaves:

$$\begin{aligned} \max \quad & z = 9600 - 8x_1 - 2s_1 - 3s_2 \\ \text{s.t.} \quad & x_3 = 1600 - 6x_1 - 2s_1 + s_2, \\ & x_2 = 200 + 2x_1 + s_1 - s_2, \\ & s_3 = 1200 - 2x_1 + s_2. \end{aligned}$$

Optimal: $(x_1, x_2, x_3) = (0, 200, 1600)$ with $z^* = 9600$; the first two constraints are tight, the third has slack $s_3 = 1200$, and x_1 stays out of the optimal solution.

Exercise 7.12 (Simplex Method Practice (Big-M)) ★★

Convert the minimization to a maximization, $\max z = -4x_1 - 6x_2 - 7x_3$, subtract surplus variables $e_1, e_2 \geq 0$, and, since the origin is infeasible, add artificial variables $a_1, a_2 \geq 0$ penalized by $-M$ in

the objective. Solving for a_1, a_2 gives the starting dictionary

$$\begin{array}{ll} \max & z = -50M + (2M - 4)x_1 + (3M - 6)x_2 + (3M - 7)x_3 - Me_1 - Me_2 \\ \text{s.t.} & a_1 = 20 - x_1 - x_2 - 2x_3 + e_1, \\ & a_2 = 30 - x_1 - 2x_2 - x_3 + e_2. \end{array}$$

Pivot 1: x_2 enters (largest coefficient $3M - 6$). Ratios: $20/1 = 20$ (a_1) and $30/2 = 15$ (a_2), so a_2 leaves:

$$\begin{array}{ll} \max & z = (-90 - 5M) + \left(\frac{M}{2} - 1\right)x_1 + \left(\frac{3M}{2} - 4\right)x_3 - Me_1 + \left(\frac{M}{2} - 3\right)e_2 + \left(3 - \frac{3M}{2}\right)a_2 \\ \text{s.t.} & a_1 = 5 - \frac{1}{2}x_1 - \frac{3}{2}x_3 + e_1 - \frac{1}{2}e_2 + \frac{1}{2}a_2, \\ & x_2 = 15 - \frac{1}{2}x_1 - \frac{1}{2}x_3 + \frac{1}{2}e_2 - \frac{1}{2}a_2. \end{array}$$

Pivot 2: x_3 enters (coefficient $\frac{3M}{2} - 4$, the largest). Ratios: $5/\frac{3}{2} = \frac{10}{3}$ (a_1) and $15/\frac{1}{2} = 30$ (x_2), so a_1 leaves. Both artificials are now nonbasic and may be dropped:

$$\begin{array}{ll} \max & z = -\frac{310}{3} + \frac{1}{3}x_1 - \frac{8}{3}e_1 - \frac{5}{3}e_2 \\ \text{s.t.} & x_3 = \frac{10}{3} - \frac{1}{3}x_1 + \frac{2}{3}e_1 - \frac{1}{3}e_2, \\ & x_2 = \frac{40}{3} - \frac{1}{3}x_1 - \frac{1}{3}e_1 + \frac{2}{3}e_2. \end{array}$$

Pivot 3: x_1 enters (coefficient $\frac{1}{3}$). Ratios: $\frac{10}{3}/\frac{1}{3} = 10$ (x_3) and $\frac{40}{3}/\frac{1}{3} = 40$ (x_2), so x_3 leaves:

$$\begin{array}{ll} \max & z = -100 - x_3 - 2e_1 - 2e_2 \\ \text{s.t.} & x_1 = 10 - 3x_3 + 2e_1 - e_2, \\ & x_2 = 10 + x_3 - e_1 + e_2. \end{array}$$

Optimal: $(x_1, x_2, x_3) = (10, 10, 0)$ with $\max z = -100$, that is, minimum cost $z^* = 100$; both $\geq$ constraints are tight.

QA flag

Tedium note: with the book's default largest-coefficient entering rule this Big-M solve takes three pivots and passes through fractions in thirds, which is right at the hand-computation limit. A shorter route exists: entering x_1 first (its coefficient $2M - 4$ is also positive) drives a_1 out immediately and reaches the same optimum in only two pivots. An instructor may want to point students to that path, or the book could add a hint such as "entering x_1 first shortens the computation." The exercise itself is correct as stated.

Exercise 7.13 (Simplex Method Practice (Big-M))

★★

Convert to $\max z = -40x_1 - 48x_2 - 30x_3$, subtract surpluses $e_1, e_2 \geq 0$, and add penalized artificials $a_1, a_2 \geq 0$:

$$\begin{array}{ll} \max & z = -55M + (3M - 40)x_1 + (5M - 48)x_2 + (3M - 30)x_3 - Me_1 - Me_2 \\ \text{s.t.} & a_1 = 25 - 2x_1 - 2x_2 - x_3 + e_1, \\ & a_2 = 30 - x_1 - 3x_2 - 2x_3 + e_2. \end{array}$$

Pivot 1: x_2 enters (largest coefficient $5M - 48$). Ratios: $25/2 = 12.5$ (a_1) and $30/3 = 10$ (a_2), so a_2 leaves:

$$\begin{aligned} \max \quad & z = (-480 - 5M) + \left(\frac{4M}{3} - 24\right)x_1 + \left(2 - \frac{M}{3}\right)x_3 - Me_1 + \left(\frac{2M}{3} - 16\right)e_2 + \left(16 - \frac{5M}{3}\right)a_2 \\ \text{s.t.} \quad & a_1 = 5 - \frac{4}{3}x_1 + \frac{1}{3}x_3 + e_1 - \frac{2}{3}e_2 + \frac{2}{3}a_2, \\ & x_2 = 10 - \frac{1}{3}x_1 - \frac{2}{3}x_3 + \frac{1}{3}e_2 - \frac{1}{3}a_2. \end{aligned}$$

Pivot 2: x_1 enters ($\frac{4M}{3} - 24$ beats $\frac{2M}{3} - 16$). Ratios: $5/\frac{4}{3} = \frac{15}{4}$ (a_1) and $10/\frac{1}{3} = 30$ (x_2), so a_1 leaves. Both artificials are now nonbasic; dropping them,

$$\begin{aligned} \max \quad & z = -570 - 4x_3 - 18e_1 - 4e_2 \\ \text{s.t.} \quad & x_1 = \frac{15}{4} + \frac{1}{4}x_3 + \frac{3}{4}e_1 - \frac{1}{2}e_2, \\ & x_2 = \frac{35}{4} - \frac{3}{4}x_3 - \frac{1}{4}e_1 + \frac{1}{2}e_2. \end{aligned}$$

All coefficients are negative, so this is optimal after just two pivots: $(x_1, x_2, x_3) = (\frac{15}{4}, \frac{35}{4}, 0)$ with minimum cost $z^* = 40 \cdot \frac{15}{4} + 48 \cdot \frac{35}{4} = 150 + 420 = 570$. Both $\geq$ constraints are tight.

QA flag

Judgment call: the optimal solution is fractional $(\frac{15}{4}, \frac{35}{4})$ even though all the data are integers. The solve is short (two pivots) and the optimal value 570 is clean, so this is acceptable, but if an integer answer is preferred, changing the first right-hand side from 25 to 24 yields the integral optimum $(3, 9, 0)$ with cost 552 (verified).

Exercise 7.14 (Dictionary Optimality)

★★

(Book Selected Solution, verified.) The basic solution reads off the constant terms: $x_2 = 1$, $x_4 = 2$, $x_3 = b$, with $x_1 = x_5 = x_6 = 0$.

1. **Feasible and optimal:** feasibility requires every constant term nonnegative, so $b \geq 0$; optimality requires no improving direction, so $c_1 \leq 0$, $c_5 \leq 0$, $c_6 \leq 0$. The values a_1, a_2 may be arbitrary.
2. **Not feasible:** $b < 0$ (the constants 1 and 2 are already nonnegative, so only the x_3 row can fail).
3. **Degenerate:** $b = 0$, so that the basic variable x_3 sits at value zero (feasibility of the rest is automatic).
4. **Feasible and unbounded:** we need $b \geq 0$ together with an entering variable whose objective coefficient is positive and whose increase never decreases any basic variable. The variable x_5 appears with fixed coefficient -1 in the x_3 row and x_6 with -1 in both the x_4 and x_3 rows, so the ratio test always stops them. Only x_1 can certify unboundedness: its column coefficients are $+1$, $+1$, and $-a_2$, so the conditions are $c_1 > 0$ and $a_2 \leq 0$ (with $b \geq 0$ for feasibility).

Exercise 7.15 (Basic Feasible Solutions)

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QA flag

Book fix applied: in the source, all but one constraint of this exercise had been commented out, and the one visible constraint ($-x_1 + x_2 \leq 2$) did not match the figure. The constraint system was reconstructed exactly from the figure (all five labeled vertices check out: $A = (2, 1)$, $B = (5, 1)$, $C = (7, 3)$, $D = (0, \frac{16}{3})$, $E = (0, \frac{7}{3})$) and the book statement was repaired to

$$\max 2x_1 + x_2 \quad \text{s.t.} \quad 2x_1 + 3x_2 \geq 7, \quad x_2 \geq 1, \quad x_1 - x_2 \leq 4, \quad x_1 + 3x_2 \leq 16, \quad x_1, x_2 \geq 0,$$

with surplus variables s_1, s_2 on the two $\geq$ constraints and slacks s_3, s_4 on the two $\leq$ constraints. The solution below matches the repaired statement.

(a) The vertex $D = (0, \frac{16}{3})$ is the intersection of the boundaries $x_1 = 0$ and $x_1 + 3x_2 = 16$ (that is, $s_4 = 0$). So the nonbasic variables are $N = \{x_1, s_4\}$ and the basic variables are $B = \{x_2, s_1, s_2, s_3\}$, with values

$$x_2 = \frac{16}{3}, \quad s_1 = 9, \quad s_2 = \frac{13}{3}, \quad s_3 = \frac{28}{3}.$$

(b) Moving from D to $C = (7, 3)$ travels along the edge $s_4 = 0$ while x_1 increases from 0: so x_1 enters. At C the boundary $x_1 - x_2 = 4$ becomes tight, i.e., $s_3 = 0$: so s_3 leaves. This agrees with the ratio test in the dictionary at D , where increasing x_1 (with $s_4 = 0$) drives $s_3 = \frac{28}{3} - \frac{4}{3}x_1$ to zero first, at $x_1 = 7$, before s_2 (at $x_1 = 13$) or x_2 (at $x_1 = 16$).

Exercise 7.16 (What the ratio test protects)

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(a) Pivoting on the s_2 row means solving $s_2 = 9 - \frac{3}{2}x + \frac{1}{2}s_3$ for x , giving $x = 6 - \frac{2}{3}s_2 + \frac{1}{3}s_3$, and substituting everywhere:

$$\begin{aligned} \max \quad & z = 24 - \frac{1}{3}s_2 - \frac{4}{3}s_3 \\ \text{s.t.} \quad & s_1 = -1 + \frac{1}{3}s_2 + \frac{1}{3}s_3, \\ & x = 6 - \frac{2}{3}s_2 + \frac{1}{3}s_3, \\ & y = 4 + \frac{1}{3}s_2 - \frac{2}{3}s_3. \end{aligned}$$

The basic solution is $(x, y, s_1, s_2, s_3) = (6, 4, -1, 0, 0)$: the basic variable s_1 is negative, so the dictionary is infeasible. (Indeed the point $(x, y) = (6, 4)$ violates $x + y \leq 9$.) Note that the “objective value” 24 exceeds the true optimum 23 precisely because the point is outside the feasible region.

(b) As the entering variable x increases, every basic variable with a negative x -coefficient decreases toward zero; the ratio of each row measures how far x can go before that row’s basic variable hits zero. The minimum ratio is the first blocking value. Pivoting on any other row pushes x past that first block, so the basic variable that owned the minimum ratio (here s_1 , blocked at $x = 4$) is driven negative. The minimum-ratio rule protects exactly the nonnegativity of all the basic variables that remain in the basis.

Exercise 7.17 (A tie in the ratio test forces degeneracy)

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(Book Selected Solution, verified, with part (c) restated.)

(a) The initial dictionary is

$$\begin{aligned} \max \quad & z = 2x + y \\ \text{s.t.} \quad & s_1 = 4 - x - y, \\ & s_2 = 8 - 2x - y. \end{aligned}$$

With x entering (and $y = 0$), the ratio test gives $4/1 = 4$ for s_1 and $8/2 = 4$ for s_2 : a tie at $x = 4$.

(b) Let s_1 leave. Then $x = 4 - y - s_1$ and

$$\begin{aligned} \max \quad & z = 8 - y - 2s_1 \\ \text{s.t.} \quad & x = 4 - y - s_1, \\ & s_2 = 0 + y + 2s_1. \end{aligned}$$

The basic variable s_2 has value 0: the dictionary is degenerate. It is also optimal (both reduced costs negative), with $(x, y) = (4, 0)$ and $z^* = 8$.

(c) Let s_2 leave instead. Then $x = 4 - \frac{1}{2}y - \frac{1}{2}s_2$ and

$$\begin{aligned} \max \quad & z = 8 + 0 \cdot y - s_2 \\ \text{s.t.} \quad & x = 4 - \frac{1}{2}y - \frac{1}{2}s_2, \\ & s_1 = 0 - \frac{1}{2}y + \frac{1}{2}s_2. \end{aligned}$$

Again a basic variable (s_1) equals zero: degenerate. The reduced cost of y is exactly 0. A zero reduced cost normally signals alternative optima along the corresponding edge; here the degenerate row $s_1 = 0 - \frac{1}{2}y + \dots$ blocks y at once, so the optimal *solution* $(4, 0)$ is unique, but there are multiple optimal *bases* describing it. This is the typical fingerprint of degeneracy.

(d) If two rows tie at the minimum ratio, the entering variable stops at a value where both corresponding basic variables hit zero simultaneously. Only one of them leaves the basis; the other remains basic at value zero, so the new dictionary necessarily has a basic variable equal to zero, which is the definition of degeneracy. A construction: $\max x + y$ subject to $x \leq 3$, $x + 2y \leq 3$, $x, y \geq 0$; entering x gives ratios $3/1 = 3$ and $3/1 = 3$, a tie.

8 Chapter 8: Simplex - Matrix Calculations

Chapter 8 recasts the simplex method as basis algebra: A_B , $A_B^{-1}\mathbf{b}$, and the reduced costs $\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N$. Its nine exercises split into three warm-ups on computing basic solutions from a basis (8.1 to 8.3), three core computations with reduced costs and the dictionary formula (8.4 to 8.6), two conceptual exercises connecting reduced costs to the dictionary and tableau sign conventions (8.7, 8.8), and one challenge proof on uniqueness of the optimum (8.9). Every matrix inverse, basic solution, and reduced-cost vector below was verified with exact-fraction computations. Book Selected Solutions exist for 8.1, 8.4, and 8.9 and are reproduced here.

Exercise 8.1 (Basis matrix and basic solution) ★

(Book Selected Solution, verified.) The columns of x_2 and x_3 give

$$A_B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A_B^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

The basic solution is

$$\mathbf{x}_B = A_B^{-1}\mathbf{b} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

so $(x_1, x_2, x_3) = (0, 1, 2)$. Both basic variables are nonnegative, so this is a basic feasible solution.

Exercise 8.2 (A basic solution that is not feasible)

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The columns of x_1 and x_3 give

$$A_B = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}, \quad \det A_B = 2 - 3 = -1, \quad A_B^{-1} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}.$$

Setting $x_2 = 0$, the basic solution is

$$\mathbf{x}_B = A_B^{-1} \mathbf{b} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} -3 \\ 7 \end{bmatrix},$$

so $(x_1, x_2, x_3) = (-3, 0, 7)$. It solves $A\mathbf{x} = \mathbf{b}$ (check: $-3 + 2(0) + 7 = 4$ and $3(-3) + 0 + 2(7) = 5$), but the basic variable $x_1 = -3$ violates the nonnegativity constraint $\mathbf{x} \geq \mathbf{0}$. A basic solution only qualifies as a basic *feasible* solution when $\mathbf{x}_B \geq \mathbf{0}$, so this basis gives an infeasible basic solution.

Exercise 8.3 (A fourth basis for the three-bases example)

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The columns of x_2 and s_2 give

$$A_B = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}, \quad A_B^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 1 \end{bmatrix}.$$

The basic solution is

$$\mathbf{x}_B = \begin{bmatrix} x_2 \\ s_2 \end{bmatrix} = A_B^{-1} \mathbf{b} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 1 \end{bmatrix} \begin{bmatrix} 16 \\ 45 \end{bmatrix} = \begin{bmatrix} 8 \\ 21 \end{bmatrix}.$$

Both components are nonnegative, so the basis is *feasible*. Since $x_1 = s_1 = 0$, the corresponding point is $(x_1, x_2) = (0, 8)$: the vertex where the boundary $x_1 + 2x_2 = 16$ meets the x_2 -axis. (This is the basis shown to be optimal in the worked example of the optimality section.)

Exercise 8.4 (Reduced costs at two bases)

★★

(Book Selected Solution, verified, with the intermediate matrices shown.)

Basis $B_1 = \{y, s_1, s_2\}$. Ordering the basic variables (y, s_1, s_2) ,

$$A_{B_1} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}, \quad A_{B_1}^{-1} = \begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \end{bmatrix}, \quad \mathbf{x}_{B_1} = A_{B_1}^{-1} \mathbf{b} = \begin{bmatrix} 7 \\ 2 \\ 9 \end{bmatrix},$$

so $y = 7$, $s_1 = 2$, $s_2 = 9$: a basic feasible solution with objective value $\mathbf{c}_{B_1}^\top \mathbf{x}_{B_1} = 3 \cdot 7 = 21$. With $\mathbf{c}_{B_1}^\top = (3, 0, 0)$, the simplex multipliers are $\mathbf{c}_{B_1}^\top A_{B_1}^{-1} = (0, 0, \frac{3}{2})$, and for the nonbasic variables x and s_3 (columns $(1, 2, 1)^\top$ and $(0, 0, 1)^\top$),

$$\mathbf{c}_N^\top - \mathbf{c}_{B_1}^\top A_{B_1}^{-1} A_N = \begin{bmatrix} 2 & 0 \end{bmatrix} - \begin{bmatrix} \frac{3}{2} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{2} \end{bmatrix}.$$

The reduced cost of x is $\frac{1}{2} > 0$, so B_1 is *not* optimal; this matches the second dictionary of Chapter 7, $z = 21 + \frac{1}{2}x - \frac{3}{2}s_3$.

Basis $B_2 = \{x, y, s_2\}$. Ordering the basic variables (x, y, s_2) ,

$$A_{B_2} = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}, \quad A_{B_2}^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & 1 & 1 \end{bmatrix}, \quad \mathbf{x}_{B_2} = A_{B_2}^{-1} \mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix},$$

a basic feasible solution with value $2 \cdot 4 + 3 \cdot 5 = 23$. With $\mathbf{c}_{B_2}^\top = (2, 3, 0)$ the multipliers are $\mathbf{c}_{B_2}^\top A_{B_2}^{-1} = (1, 0, 1)$, and for the nonbasic variables s_1, s_3 (columns $(1, 0, 0)^\top$ and $(0, 0, 1)^\top$),

$$\mathbf{c}_N^\top - \mathbf{c}_{B_2}^\top A_{B_2}^{-1} A_N = [0 \ 0] - [1 \ 1] = [-1 \ -1] \leq \mathbf{0}^\top,$$

so B_2 is optimal, matching the final dictionary $z = 23 - s_1 - s_3$.

Exercise 8.5 (Which bases are feasible?) ★★

(a) The six column pairs of $A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{bmatrix}$ have determinants

$$\begin{aligned} \det A_{\{x_1, x_2\}} &= -7, & \det A_{\{x_1, s_1\}} &= -5, & \det A_{\{x_1, s_2\}} &= 1, \\ \det A_{\{x_2, s_1\}} &= -3, & \det A_{\{x_2, s_2\}} &= 2, & \det A_{\{s_1, s_2\}} &= 1, \end{aligned}$$

all nonzero, so every pair of columns is linearly independent and each choice is a basis.

(b) Solving $A_B \mathbf{x}_B = \mathbf{b}$ for each basis (nonbasic variables zero):

Basis	Basic solution	Point (x_1, x_2)	Feasible?
$\{x_1, x_2\}$	$x_1 = 6, x_2 = 5$	$(6, 5)$	yes
$\{x_1, s_1\}$	$x_1 = 9, s_1 = 7$	$(9, 0)$	yes
$\{x_1, s_2\}$	$x_1 = 16, s_2 = -35$	$(16, 0)$	no ($s_2 < 0$)
$\{x_2, s_1\}$	$x_2 = 15, s_1 = -14$	$(0, 15)$	no ($s_1 < 0$)
$\{x_2, s_2\}$	$x_2 = 8, s_2 = 21$	$(0, 8)$	yes
$\{s_1, s_2\}$	$s_1 = 16, s_2 = 45$	$(0, 0)$	yes

The four feasible bases correspond to the four vertices of the feasible region, $(0, 0)$, $(9, 0)$, $(6, 5)$, $(0, 8)$; the two infeasible bases are intersections of constraint boundaries lying outside the region.

(c) Take

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix}.$$

The first two columns satisfy $(2, 4)^\top = 2 \cdot (1, 2)^\top$, so the pair $\{1, 2\}$ is linearly dependent ($\det = 0$) and does not form a basis.

Exercise 8.6 (Rebuilding a dictionary from the formula) ★★

With variables ordered (x, y, s_1, s_2) ,

$$A = \begin{bmatrix} 3 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 23 \\ 45 \end{bmatrix}, \quad \mathbf{c}^\top = [2 \ 1 \ 0 \ 0], \quad B = \{x, y\}, \quad N = \{s_1, s_2\}.$$

Then

$$A_B = \begin{bmatrix} 3 & 1 \\ 5 & 3 \end{bmatrix}, \quad A_B^{-1} = \frac{1}{4} \begin{bmatrix} 3 & -1 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 0.75 & -0.25 \\ -1.25 & 0.75 \end{bmatrix}.$$

Since $A_N = I_2$ (the slack columns),

$$A_B^{-1} \mathbf{b} = \frac{1}{4} \begin{bmatrix} 3(23) - 45 \\ -5(23) + 3(45) \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}, \quad A_B^{-1} A_N = A_B^{-1} = \begin{bmatrix} 0.75 & -0.25 \\ -1.25 & 0.75 \end{bmatrix}.$$

The constraint part of the dictionary, $\mathbf{x}_B = A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N$, reads

$$\begin{aligned}x &= 6 - 0.75s_1 + 0.25s_2, \\y &= 5 + 1.25s_1 - 0.75s_2,\end{aligned}$$

exactly the displayed rows. For the objective, $\mathbf{c}_B^\top A_B^{-1}\mathbf{b} = 2(6) + 1(5) = 17$, and

$$\mathbf{c}_B^\top A_B^{-1} = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 0.75 & -0.25 \\ -1.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.25 \end{bmatrix}, \quad \mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N = \begin{bmatrix} -0.25 & -0.25 \end{bmatrix}.$$

So the formula gives $\max 17 - 0.25s_1 - 0.25s_2$, reproducing the displayed dictionary in full. (Both reduced costs are negative, so this basis is optimal.)

Exercise 8.7 (Reduced costs of basic variables) ★★

Applying the reduced-cost formula to the basic columns replaces A_N by A_B :

$$\mathbf{c}_B^\top - \mathbf{c}_B^\top A_B^{-1}A_B = \mathbf{c}_B^\top - \mathbf{c}_B^\top I_m = \mathbf{c}_B^\top - \mathbf{c}_B^\top = \mathbf{0}^\top,$$

since $A_B^{-1}A_B = I_m$ by the definition of the inverse. So every basic variable has reduced cost exactly zero.

Interpretation: the dictionary at basis B expresses z and each basic variable purely in terms of the *nonbasic* variables. Constructing the dictionary means substituting $\mathbf{x}_B = A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N$ into the objective, which eliminates every basic variable from the objective row; the computation above is the algebraic statement that this elimination is exact. A basic variable can never appear in the objective row because, if it did, the row would not be solved for the basic variables, and the “dictionary” property (basic variables and z as functions of nonbasic variables alone) would fail. Equivalently: the value of a basic variable is determined by the nonbasic variables, so it carries no independent rate of change for z .

Exercise 8.8 (Reduced-cost signs and the entering rule) ★★

(a) Rewriting the objective from the viewpoint of basis B ,

$$z = \mathbf{c}_B^\top A_B^{-1}\mathbf{b} + \left(\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N \right) \mathbf{x}_N,$$

which is precisely the objective row of the dictionary: a constant plus a linear function of the nonbasic variables. The coefficient of a single nonbasic variable x_j in this row is the j th component of the reduced-cost vector, namely $c_j - \mathbf{c}_B^\top A_B^{-1}A_{\{j\}}$. So “pick a nonbasic variable with a positive objective-row coefficient” and “pick a nonbasic variable with a positive reduced cost” are literally the same rule, and the stopping conditions (no positive coefficient; all reduced costs ≤ 0) coincide as well.

(b) Increasing a nonbasic variable x_j from zero, while keeping the other nonbasic variables at zero, moves the solution along the ray

$$\mathbf{x}(t) = \bar{\mathbf{x}} + t\mathbf{d}, \quad \mathbf{d}_B = -A_B^{-1}A_{\{j\}}, \quad d_j = 1, \quad d_k = 0 \text{ otherwise.}$$

Geometrically this is a move away from the current vertex along an edge of the feasible region: one tight constraint ($x_j = 0$) is released while the equality constraints stay satisfied. The objective changes at rate equal to the reduced cost, so a positive reduced cost means the edge points uphill. The ratio test stops the move at the neighboring vertex, where some basic variable reaches zero.

(c) The tableau stores the objective as the equation $z - \mathbf{c}^\top \mathbf{x} = 0$, so the entry recorded under a nonbasic variable is the *negative* of its dictionary coefficient, i.e., the negative of its reduced cost. Hunting for a negative z -row entry in the tableau is therefore hunting for a positive reduced cost, and “stop when all z -row entries are nonnegative” is “stop when all reduced costs are ≤ 0 .” The three formulations state one test with two sign conventions; no contradiction arises.

Exercise 8.9 (Strictly negative reduced costs imply a unique optimum) ★★★

(Book Selected Solution, verified.) Write $\bar{\mathbf{c}}_N^\top = \mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N$ and $z^* = \mathbf{c}^\top \bar{\mathbf{x}} = \mathbf{c}_B^\top A_B^{-1} \mathbf{b}$. Every feasible $\mathbf{x}$ satisfies $A_B \mathbf{x}_B + A_N \mathbf{x}_N = \mathbf{b}$, so $\mathbf{x}_B = A_B^{-1} \mathbf{b} - A_B^{-1} A_N \mathbf{x}_N$ and, exactly as in the optimality theorem,

$$\mathbf{c}^\top \mathbf{x} = z^* + \bar{\mathbf{c}}_N^\top \mathbf{x}_N.$$

Take any feasible $\mathbf{x} \neq \bar{\mathbf{x}}$.

Case 1: $\mathbf{x}_N \neq \mathbf{0}$. Then some component $x_j > 0$ with $j \in N$, and since every entry of $\bar{\mathbf{c}}_N$ is strictly negative, $\bar{\mathbf{c}}_N^\top \mathbf{x}_N < 0$. Hence $\mathbf{c}^\top \mathbf{x} < z^*$, so $\mathbf{x}$ is not optimal.

Case 2: $\mathbf{x}_N = \mathbf{0}$. Then $A_B \mathbf{x}_B = \mathbf{b}$, and since A_B is invertible this forces $\mathbf{x}_B = A_B^{-1} \mathbf{b}$, i.e., $\mathbf{x} = \bar{\mathbf{x}}$, contradicting $\mathbf{x} \neq \bar{\mathbf{x}}$.

Therefore every feasible point other than $\bar{\mathbf{x}}$ has objective value strictly less than z^* : the solution $\bar{\mathbf{x}}$ is optimal and no other optimal solution exists.

9 Chapter 9: Simplex Method in Tableau Form

Chapter 9 repackages the simplex method as row operations on a single table. Its twelve exercises cover reading tableaus (9.1 to 9.3), executing pivots and full solves including a Big-M setup (9.4 to 9.7), the sign convention and the tableau-dictionary correspondence (9.8, 9.9), recognizing unboundedness and infeasibility from a tableau (9.10, 9.11), and one challenge problem that turns an unbounded tableau into an explicit certificate ray (9.12). Throughout, recall the chapter’s sign convention: the z -row stores $z - \mathbf{c}^\top \mathbf{x} = 0$, so entering variables have *negative* z -row entries and the tableau is optimal when every z -row entry is nonnegative. All arithmetic below was verified with exact-fraction basis computations, and each optimum with an LP solver. Book Selected Solutions exist for 9.2, 9.3, 9.5, and 9.6 and are reproduced here.

Exercise 9.1 (Reading a mid-solve tableau) ★

(a) The basic variables are the row labels: s_1, s_2, y . The nonbasic variables are x and s_3 .

(b) Setting $x = s_3 = 0$ and reading the RHS column:

$$x = 0, \quad y = 7, \quad s_1 = 2, \quad s_2 = 9, \quad s_3 = 0, \quad z = 21.$$

(c) The z -row entries of the nonbasic variables are $-\frac{1}{2}$ (for x) and $\frac{3}{2}$ (for s_3). In the chapter’s convention these are the recorded reduced costs; because of the stored sign flip, increasing x by one unit *raises* z by $\frac{1}{2}$, while increasing s_3 would lower it by $\frac{3}{2}$.

(d) Not optimal: the z -row has the negative entry $-\frac{1}{2}$ under x , so x enters next. (The subsequent ratio test gives $2/\frac{1}{2} = 4$, $9/\frac{3}{2} = 6$, $7/\frac{1}{2} = 14$, so s_1 would leave, leading to the optimal tableau with $z = 23$.)

Exercise 9.2 (Reading a tableau)

★

(Book Selected Solution, verified.) The basic variables are the row labels x_1 and x_2 ; the nonbasic variables are s_1 and s_2 . Setting the nonbasic variables to zero and reading the RHS column gives the basic feasible solution

$$x_1 = 6, \quad x_2 = 4, \quad s_1 = s_2 = 0, \quad z = 40.$$

The objective-row entries of the nonbasic variables are 3 and 1, both nonnegative. In this convention a positive entry means that increasing the corresponding nonbasic variable would *decrease* z , so no pivot can improve the objective and the tableau is optimal.

Exercise 9.3 (Entering variable, ratio test, and pivot element)

★

(Book Selected Solution, verified.) The most negative objective-row entry is -4 , so x_1 enters. The ratio test divides each RHS by the positive entries of the x_1 column:

$$\frac{6}{2} = 3, \quad \frac{8}{4} = 2, \quad \frac{12}{3} = 4.$$

The minimum ratio 2 occurs in the s_2 row, so s_2 leaves and the pivot element is the 4 in the s_2 row of the x_1 column.

Exercise 9.4 (Perform one pivot)

★★

Pivot on the 4 in the s_2 row, x_1 column. Scale the pivot row by $\frac{1}{4}$ to get the new x_1 -row $[1, \frac{1}{4}, 0, \frac{1}{4}, 0 \mid 2]$, then clear the x_1 column: s_1 -row $\leftarrow s_1$ -row $- 2(x_1$ -row), s_3 -row $\leftarrow s_3$ -row $- 3(x_1$ -row), z -row $\leftarrow z$ -row $+ 4(x_1$ -row). The result:

Basis	x_1	x_2	s_1	s_2	s_3	RHS
z	0	-2	0	1	0	8
s_1	0	$\frac{5}{2}$	1	$-\frac{1}{2}$	0	2
x_1	1	$\frac{1}{4}$	0	$\frac{1}{4}$	0	2
s_3	0	$\frac{13}{4}$	0	$-\frac{3}{4}$	1	6

The x_1 column is now the unit column $(0; 0, 1, 0)$, and the objective value in the z -row RHS has increased from 0 to 8 (the new BFS is $x_1 = 2$, $x_2 = 0$ with $z = 4 \cdot 2 = 8$). The tableau is not yet optimal (-2 under x_2), but the exercise asks only for this one pivot.

Exercise 9.5 (Solve a linear program in tableau form)

★★

(Book Selected Solution, verified, with the starting tableau displayed.) Adding slacks s_1, s_2 , the initial tableau is

Basis	x	y	s_1	s_2	RHS	ratio
z	-1	-2	0	0	0	
s_1	1	1	1	0	4	$4/1 = 4$
s_2	1	3	0	1	6	$6/3 = 2$ (min)

Pivot 1: y enters (-2 most negative); ratios 4 and 2, so s_2 leaves; pivot on the boxed 3. Dividing the pivot row by 3 and clearing the y column gives

Basis	x	y	s_1	s_2	RHS	ratio
z	$-\frac{1}{3}$	0	0	$\frac{2}{3}$	4	
s_1	$\frac{2}{3}$	0	1	$-\frac{1}{3}$	2	$2/\frac{2}{3} = 3$ (min)
y	$\frac{1}{3}$	1	0	$\frac{1}{3}$	2	$2/\frac{1}{3} = 6$

Pivot 2: x enters ($-\frac{1}{3}$); ratios 3 and 6, so s_1 leaves; pivot on the boxed $\frac{2}{3}$:

Basis	x	y	s_1	s_2	RHS
z	0	0	$\frac{1}{2}$	$\frac{1}{2}$	5
x	1	0	$\frac{3}{2}$	$-\frac{1}{2}$	3
y	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	1

Every z -row entry is nonnegative, so the tableau is optimal: $(x, y) = (3, 1)$ with $z^* = 5$ (both constraints tight).

Exercise 9.6 (A full two-pivot solve)

★★

(Book Selected Solution, verified.) Adding slacks s_1, s_2 , the starting tableau is

Basis	x	y	s_1	s_2	RHS	ratio
z	-3	-2	0	0	0	
s_1	$\frac{2}{3}$	1	1	0	10	$10/2 = 5$ (min)
s_2	1	3	0	1	15	$15/1 = 15$

Pivot 1: x enters (-3 most negative); s_1 leaves; pivot on the boxed 2. Scaling the pivot row by $\frac{1}{2}$ and clearing the x column:

Basis	x	y	s_1	s_2	RHS	ratio
z	0	$-\frac{1}{2}$	$\frac{3}{2}$	0	15	
x	1	$\frac{1}{2}$	$\frac{1}{2}$	0	5	$5/\frac{1}{2} = 10$
s_2	0	$\frac{5}{2}$	$-\frac{1}{2}$	1	10	$10/\frac{5}{2} = 4$ (min)

Pivot 2: y enters ($-\frac{1}{2}$); ratios 10 and 4, so s_2 leaves; pivot on the boxed $\frac{5}{2}$:

Basis	x	y	s_1	s_2	RHS
z	0	0	$\frac{7}{5}$	$\frac{1}{5}$	17
x	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	3
y	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	4

Every z -row entry is nonnegative, so the tableau is optimal: $(x, y) = (3, 4)$ with $z^* = 17$ and $s_1 = s_2 = 0$ (both constraints are tight at the optimum).

Exercise 9.7 (Setting up a Big-M tableau)

★★

The augmented constraints are

$$x + y - e_1 + a_1 = 4, \quad x + 2y + s_2 = 10, \quad x, y, e_1, s_2, a_1 \geq 0,$$

with objective $z - 4x - 3y + Ma_1 = 0$. Written naively, the tableau is

Basis	x	y	e_1	s_2	a_1	RHS
z	-4	-3	0	0	M	0
a_1	1	1	-1	0	1	4
s_2	1	2	0	1	0	10

This is not yet a proper tableau: the *basic* variable a_1 has a nonzero entry (M) in the z -row. Pricing out ($z\text{-row} \leftarrow z\text{-row} - M \cdot a_1\text{-row}$) gives

Basis	x	y	e_1	s_2	a_1	RHS	ratio
z	-4 - M	-3 - M	M	0	0	$-4M$	
a_1	1	1	-1	0	1	4	$4/1 = 4$ (min)
s_2	1	2	0	1	0	10	$10/1 = 10$

Now a_1 's objective entry is zero, as required. The most negative z -row entry is $-4 - M$ (for large M it beats $-3 - M$), so x *enters first*. The ratio test gives $4/1 = 4$ in the a_1 row versus $10/1 = 10$ in the s_2 row, so *yes: the artificial variable a_1 leaves on the very first pivot*, and the method continues with an ordinary feasible tableau (the pivot lands at $(x, y) = (4, 0)$ with $z = 16$).

Exercise 9.8 (The z -row sign convention)

★★

(a) A dictionary displays $z = \bar{z} + \sum_j \bar{c}_j x_j$ over the nonbasic variables. The tableau instead stores this equation with everything on the left: $z - \sum_j \bar{c}_j x_j = \bar{z}$. Each coefficient $\bar{c}_j$ crosses the equals sign and gets negated, so a variable that is attractive in the dictionary (positive $\bar{c}_j$) shows up in the tableau with a negative entry $-\bar{c}_j$. The entering rules “most positive dictionary coefficient” and “most negative z -row entry” therefore select the same variable.

(b) From $z = 12 + 3x_1 - 2x_2$, move the variable terms left: $z - 3x_1 + 2x_2 = 12$. The z -row is

	x_1	x_2	RHS
z	-3	2	12

(entries 0 under any slack or basic columns not shown).

(c) The stored equation is $z - \sum_j \bar{c}_j x_j = \bar{z}$, and it only involves *nonbasic* variables besides z . At the current basic solution every nonbasic variable equals zero, so the equation collapses to $z = \bar{z}$: the RHS entry is the current objective value. The constant term never crosses the equals sign (it was already on the right), so it is the one number in the row that is not negated.

Exercise 9.9 (From tableau to dictionary)

★★

Each tableau row is an equation. The x_1 row says $x_1 + 2s_1 - s_2 = 6$ and the x_2 row says $x_2 - s_1 + s_2 = 4$; the z -row says $z + 3s_1 + s_2 = 40$. Solving each for its basic variable (and z):

$$\begin{aligned} \max \quad & z = 40 - 3s_1 - s_2 \\ \text{s.t.} \quad & x_1 = 6 - 2s_1 + s_2, \\ & x_2 = 4 + s_1 - s_2, \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Setting the nonbasic variables $s_1 = s_2 = 0$ gives $x_1 = 6$, $x_2 = 4$, $z = 40$: the same basic feasible solution as the tableau. (Both dictionary coefficients in the objective are negative, matching the optimality of the tableau.)

Exercise 9.10 (Recognizing an unbounded problem)

★★

The only negative z -row entry is -2 under x_1 , so x_1 should enter. The ratio test looks for *positive* entries in the x_1 column, but they are -1 (in the x_2 row) and -3 (in the s_1 row): there are none, so the ratio test fails; no row limits the increase of x_1 . Increasing $x_1 = t$ while keeping $s_2 = 0$ gives $x_2 = 3 + t$ and $s_1 = 5 + 3t$, which stay nonnegative for all $t \geq 0$, while $z = 10 + 2t \rightarrow \infty$. The linear program is **unbounded**: it has feasible solutions of arbitrarily large objective value and no optimal solution.

Exercise 9.11 (Reading off infeasibility)

★★

Yes, the tableau is optimal for the penalized (Big-M) problem: for large M every z -row entry $(0, 0, M, 2, 0)$ is nonnegative, so no pivot can improve the penalized objective. But the artificial variable is still basic at the positive value $a_1 = 3$ (read from its RHS). A feasible solution of the original problem would allow all artificial variables to be set to zero, so if the best the penalized problem can do still keeps $a_1 = 3 > 0$, no feasible solution of the original program exists: the original linear program is **infeasible**.

The objective value $8 - 3M$ tells the same story: it contains the unavoidable penalty $-Ma_1 = -3M$. Since M stands for an arbitrarily large constant, $8 - 3M$ is arbitrarily bad; the penalized problem cannot escape the penalty, which is exactly the algebraic signature of infeasibility.

Exercise 9.12 (Certifying unboundedness from a tableau)

★★★

(a) The only negative z -row entry is -2 under x_2 , so x_2 enters. Its column entries are -1 (in the x_1 row) and 0 (in the s_2 row); the ratio test requires a *positive* entry and finds none, so no row limits the increase of x_2 : the problem is unbounded.

(b) Set $x_2 = t \geq 0$ and keep the other nonbasic variable $s_1 = 0$. The x_1 row ($x_1 - x_2 + s_1 = 1$) gives $x_1 = 1 + t$, and the s_2 row ($s_1 + s_2 = 3$) gives $s_2 = 3$. So

$$\mathbf{x}(t) = (x_1, x_2, s_1, s_2) = (1 + t, t, 0, 3), \quad t \geq 0.$$

Direct substitution into the original constraints:

$$x_1 - x_2 = (1 + t) - t = 1 \leq 1 \quad \checkmark \quad -x_1 + x_2 = -(1 + t) + t = -1 \leq 2 \quad \checkmark$$

and $x_1 = 1 + t \geq 0$, $x_2 = t \geq 0$, $s_1 = 0 \geq 0$, $s_2 = 2 - (-1) = 3 \geq 0$. So $\mathbf{x}(t)$ is feasible for every $t \geq 0$.

(c) Along the ray, $z(t) = x_1 + x_2 = 1 + 2t \rightarrow \infty$ as $t \rightarrow \infty$. Geometrically, the feasible region is the strip between the parallel-ish lines $x_1 - x_2 = 1$ and $-x_1 + x_2 = 2$ in the first quadrant; it is unbounded in the direction $\mathbf{d} = (1, 1)$. The certificate ray starts at the vertex $(1, 0)$ and points along $(1, 1)$, hugging the boundary $x_1 - x_2 = 1$; since $\mathbf{c}^\top \mathbf{d} = 1 + 1 = 2 > 0$, the objective grows without bound along it.

10 Chapter 10: Sensitivity Analysis

This chapter has 12 exercises. The two warm-ups (10.1, 10.2) read shadow prices and a right-hand-side range straight off a given final dictionary. The core problems (10.3 to 10.7) carry out cost ranging, a full sensitivity workup, basic versus nonbasic coefficient ranging, ranging from the basis inverse, and the reading of a solver sensitivity report. The concept problems (10.8 to 10.11) interpret shadow prices, explain why slack constraints have price zero, why allowable ranges exist, and what degeneracy does to shadow prices. The challenge problem (10.12) constructs an instance where the shadow price prediction fails outside its range. Exercises 10.4, 10.6, 10.8, and 10.12 have Selected Solutions in the book; those are reproduced here and expanded. Every range below was verified numerically at both endpoints and just outside them.

Exercise 10.1 (Shadow Prices from the Final Dictionary) ★

The final dictionary of the running example ($\max 2x + 3y$, constraints hours $x + y \leq 9$, flour $2x + y \leq 16$, sugar $x + 2y \leq 14$) is

$$z = 23 - s_1 - s_3, \quad x = 4 - 2s_1 + s_3, \quad y = 5 + s_1 - s_3, \quad s_2 = 3 + 3s_1 - s_3.$$

1. The shadow price of constraint i is the negative of the coefficient of s_i in the objective row (raising b_i by Δ is equivalent to setting $s_i = -\Delta$). From $z = 23 - s_1 - 0 \cdot s_2 - s_3$:

$$y_1^* = 1 \text{ (hours)}, \quad y_2^* = 0 \text{ (flour)}, \quad y_3^* = 1 \text{ (sugar)}.$$

2. Flour is not binding. In the dictionary this shows up in two equivalent ways: the slack s_2 is *basic and positive* ($s_2 = 3$ at the optimum, so three units of flour are unused), and s_2 is absent from the objective row, so its shadow price is 0. A resource with leftovers is worth nothing at the margin.
3. The shadow price of sugar is \$1 per unit, so one extra unit of sugar raises profit by \$1 > \$0.75: accept the offer and net \$0.25. The most the bakery should pay is \$1 per unit, and (from the ranging in the chapter, $11 \leq b_3 \leq 18$) that price is valid for up to 4 additional units of sugar.

Verification. `linprog` returns $z^* = 23$ with duals $(1, 0, 1)$; with $b_3 = 15$ the optimal value is 24.

Exercise 10.2 (RHS Sensitivity Range) ★

The LP is $\max 4x_1 + 3x_2$ subject to $x_1 + x_2 \leq 10$, $2x_1 + x_2 \leq 16$, $x_1, x_2 \geq 0$, with final dictionary

$$z = 36 - 2s_1 - s_2, \quad x_1 = 6 + s_1 - s_2, \quad x_2 = 4 - 2s_1 + s_2.$$

Perturb $b_1 = 10 + \Delta$. The slack of constraint 1 absorbs the change: setting $s'_1 = s_1 + \Delta$ restores the original right-hand side, so we substitute $s_1 = s'_1 - \Delta$ into the dictionary:

$$\begin{aligned} z &= 36 - 2(s'_1 - \Delta) - s_2 = (36 + 2\Delta) - 2s'_1 - s_2, \\ x_1 &= 6 + (s'_1 - \Delta) - s_2 = (6 - \Delta) + s'_1 - s_2, \\ x_2 &= 4 - 2(s'_1 - \Delta) + s_2 = (4 + 2\Delta) - 2s'_1 + s_2. \end{aligned}$$

Feasibility of the basic solution ($s'_1 = s_2 = 0$) requires

$$x_1 = 6 - \Delta \geq 0 \Rightarrow \Delta \leq 6, \quad x_2 = 4 + 2\Delta \geq 0 \Rightarrow \Delta \geq -2.$$

Hence

$$-2 \leq \Delta \leq 6 \quad \Longleftrightarrow \quad \boxed{8 \leq b_1 \leq 16},$$

and inside the range $z = 36 + 2\Delta$ (the shadow price of constraint 1 is 2, matching the coefficient of s_1 in the objective row).

The same range follows from the basis inverse: with basis $B = \{x_1, x_2\}$, $A_B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, $A_B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$, and

$$\mathbf{x}_B(b_1) = A_B^{-1} \begin{bmatrix} b_1 \\ 16 \end{bmatrix} = \begin{bmatrix} 16 - b_1 \\ 2b_1 - 16 \end{bmatrix} \geq \mathbf{0} \quad \Longleftrightarrow \quad 8 \leq b_1 \leq 16.$$

Verification. At $b_1 = 8$: optimum $(8, 0)$, $z = 32 = 36 + 2(-2)$. At $b_1 = 16$: optimum $(0, 16)$, $z = 48 = 36 + 2(6)$. At $b_1 = 7.9$ the duals jump to $(4, 0)$ and at $b_1 = 16.1$ to $(0, 3)$, confirming the basis change at both endpoints.

Exercise 10.3 (Objective Coefficient Sensitivity) ★★

Same LP and dictionary as Exercise 10.2. Let $c_1 = 4 + \delta$. Since x_1 is basic, express δx_1 through the dictionary row of x_1 and add it to the objective row:

$$z = 36 - 2s_1 - s_2 + \delta(6 + s_1 - s_2) = (36 + 6\delta) + (-2 + \delta)s_1 + (-1 - \delta)s_2.$$

Optimality (all reduced costs ≤ 0 in a maximization) requires

$$-2 + \delta \leq 0 \Rightarrow \delta \leq 2, \quad -1 - \delta \leq 0 \Rightarrow \delta \geq -1.$$

Hence

$$-1 \leq \delta \leq 2 \quad \Longleftrightarrow \quad \boxed{3 \leq c_1 \leq 6}.$$

Inside the range the optimal solution stays at $(x_1, x_2) = (6, 4)$ and the objective value moves linearly:

$$z^* = 36 + 6\delta = 12 + 6c_1$$

(the slope is $x_1^* = 6$, the value of the variable whose coefficient is changing).

Verification. At $c_1 = 3$: $z^* = 30$ (attained by both $(6, 4)$ and $(0, 10)$, alternative optima at the endpoint). At $c_1 = 6$: $z^* = 48$ at $(6, 4)$. At $c_1 = 2.99$ the optimum jumps to $(0, 10)$ and at $c_1 = 6.01$ to $(8, 0)$.

Exercise 10.4 (Full Sensitivity Workup)

★★

LP: $\max 3x_1 + 5x_2$ subject to $x_1 + 2x_2 \leq 14$ (assembly), $3x_1 + 2x_2 \leq 18$ (machine), $x_1, x_2 \geq 0$. This exercise has a Selected Solution in the book; it is reproduced and annotated here.

- Both constraints are binding at the optimum: solving $x_1 + 2x_2 = 14$ and $3x_1 + 2x_2 = 18$ gives $(x_1, x_2) = (2, 6)$ with $z^* = 3(2) + 5(6) = 36$.
- The basis is $B = \{x_1, x_2\}$ with

$$A_B = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}, \quad A_B^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & -\frac{1}{4} \end{bmatrix},$$

so the shadow prices are $\mathbf{y}^\top = \mathbf{c}_B^\top A_B^{-1} = [3, 5] A_B^{-1} = [\frac{9}{4}, \frac{1}{4}]$: an assembly hour is worth \$2.25 and a machine hour \$0.25.

- With $\mathbf{b} = (b_1, 18)$, feasibility of $\mathbf{x}_B = A_B^{-1}\mathbf{b} = \left(-\frac{b_1}{2} + 9, \frac{3b_1}{4} - \frac{9}{2}\right) \geq \mathbf{0}$ gives

$$\boxed{6 \leq b_1 \leq 18}, \quad z = 36 + \frac{9}{4}(b_1 - 14) \text{ inside the range.}$$

With $\mathbf{b} = (14, b_2)$, feasibility of $\left(\frac{b_2}{2} - 7, \frac{21}{2} - \frac{b_2}{4}\right) \geq \mathbf{0}$ gives

$$\boxed{14 \leq b_2 \leq 42}, \quad z = 36 + \frac{1}{4}(b_2 - 18).$$

- With $\mathbf{c}_B = (c_1, 5)$ the dual prices are $\left(-\frac{c_1}{2} + \frac{15}{4}, \frac{c_1}{2} - \frac{5}{4}\right)$; both are nonnegative (equivalently, both slack reduced costs stay ≤ 0) exactly when

$$\boxed{\frac{5}{2} \leq c_1 \leq \frac{15}{2}}.$$

With $\mathbf{c}_B = (3, c_2)$ the dual prices are $\left(\frac{3c_2}{4} - \frac{3}{2}, \frac{3}{2} - \frac{c_2}{4}\right)$, giving

$$\boxed{2 \leq c_2 \leq 6}.$$

- Software confirms every claim: at $b_1 = 6$ the solution is $(6, 0)$ with $z = 18 = 36 + \frac{9}{4}(-8)$; at $b_1 = 18$ it is $(0, 9)$ with $z = 45 = 36 + \frac{9}{4}(4)$; at $b_2 = 14$: $(0, 7)$, $z = 35$; at $b_2 = 42$: $(14, 0)$, $z = 42$; at $c_1 = 2.5$ and $c_1 = 7.5$ the value is 35 and 45 with the basis still optimal, while at $c_1 = 2.49$ and 7.51 the optimal vertex moves to $(0, 7)$ and $(6, 0)$ respectively; similarly for $c_2 = 2$ and 6 .

Exercise 10.5 (Ranging a Basic and a Nonbasic Coefficient)

★★

LP: $\max 5x_1 + 2x_2$ subject to $2x_1 + x_2 \leq 10$, $x_1 + 3x_2 \leq 15$, $x_1, x_2 \geq 0$, with optimum $(5, 0)$, $z^* = 25$, basis $B = \{x_1, s_2\}$.

- The basis matrix (columns of x_1 and s_2) and its inverse are

$$A_B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \quad A_B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}, \quad \mathbf{y}^\top = \mathbf{c}_B^\top A_B^{-1} = [5, 0] A_B^{-1} = \left[\frac{5}{2}, 0\right].$$

Reduced costs of the nonbasic variables:

$$\bar{c}_{x_2} = 2 - \mathbf{y}^\top \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 2 - \frac{5}{2} = -\frac{1}{2} \leq 0, \quad \bar{c}_{s_1} = 0 - y_1 = -\frac{5}{2} \leq 0.$$

Both are nonpositive, so $(5, 0)$ is optimal (the objective row is $z = 25 - \frac{1}{2}x_2 - \frac{5}{2}s_1$). As a check, $\mathbf{x}_B = A_B^{-1}\mathbf{b} = (5, 10)$, so $x_1 = 5$, $s_2 = 10 \geq 0$.

2. x_2 is nonbasic, so c_2 appears only in its own reduced cost: $\bar{c}_{x_2}(c_2) = c_2 - \frac{5}{2} \leq 0$ if and only if

$$\boxed{c_2 \leq \frac{5}{2}} \quad (\text{allowable range } -\infty < c_2 \leq \frac{5}{2}).$$

Inside this range the solution and the value do not move at all: $z^* = 25$. At $c_2 = \frac{5}{2}$ the vertex $(3, 4)$ becomes an alternative optimum, and for $c_2 > \frac{5}{2}$ the optimum jumps there.

3. x_1 is basic, so c_1 enters $\mathbf{c}_B$ and moves *every* reduced cost. With $\mathbf{c}_B = (c_1, 0)$, $\mathbf{y}(c_1)^\top = [c_1, 0] A_B^{-1} = [\frac{c_1}{2}, 0]$, so

$$\bar{c}_{x_2} = 2 - \frac{c_1}{2} \leq 0 \Rightarrow c_1 \geq 4, \quad \bar{c}_{s_1} = -\frac{c_1}{2} \leq 0 \Rightarrow c_1 \geq 0.$$

Intersecting,

$$\boxed{c_1 \geq 4} \quad (\text{allowable range } 4 \leq c_1 < \infty),$$

and inside it $z^* = 5c_1$ (slope $x_1^* = 5$), so $z^* = 25 + 5\delta$ for $c_1 = 5 + \delta$, $\delta \geq -1$.

4. A nonbasic coefficient touches only its own reduced cost, so the range comes from a single inequality and is one-sided: lowering the profit of a product you are not making can never make you start making it, so there is no finite lower endpoint. A basic coefficient sits inside $\mathbf{c}_B$, hence inside $\mathbf{y}^\top = \mathbf{c}_B^\top A_B^{-1}$, and therefore shifts the reduced cost of *every* nonbasic variable at once; each nonbasic variable contributes one inequality, and the range is the intersection of all of them.

Verification. At $c_2 = 2.49$ the optimum is $(5, 0)$; at $c_2 = 2.51$ it is $(3, 4)$ with $z = 25.04$. At $c_1 = 4.01$ the optimum is $(5, 0)$ with $z = 20.05$; at $c_1 = 3.99$ it is $(3, 4)$; at $c_1 = 4$ both vertices give $z = 20$.

Exercise 10.6 (Sensitivity from the Basis Inverse) ★★

LP: $\max 5x_1 + 4x_2$ subject to $x_1 + x_2 + s_1 = 6$, $3x_1 + 2x_2 + s_2 = 15$, all variables ≥ 0 , with basis $B = \{x_1, x_2\}$,

$$A_B = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}, \quad A_B^{-1} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix}.$$

This exercise has a Selected Solution in the book; it is reproduced here.

- 1.

$$\mathbf{x}_B = A_B^{-1}\mathbf{b} = \begin{bmatrix} -2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 6 \\ 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix},$$

so $(x_1, x_2) = (3, 3) \geq \mathbf{0}$ is a feasible basic solution with objective value $z = 5(3) + 4(3) = 27$.

- Replacing b_1 by a variable value,

$$\mathbf{x}_B(b_1) = A_B^{-1} \begin{bmatrix} b_1 \\ 15 \end{bmatrix} = \begin{bmatrix} -2b_1 + 15 \\ 3b_1 - 15 \end{bmatrix} \geq \mathbf{0} \iff \boxed{5 \leq b_1 \leq 7.5}.$$

Inside the range $z = \mathbf{c}_B^\top \mathbf{x}_B(b_1) = 5(15 - 2b_1) + 4(3b_1 - 15) = 15 + 2b_1$, so the shadow price of constraint 1 is 2.

- The dual prices are $\mathbf{y}^\top = \mathbf{c}_B^\top A_B^{-1} = [5, 4] A_B^{-1} = [2, 1]$, so the reduced costs of the nonbasic slacks are $\bar{c}_{s_1} = -y_1 = -2$ and $\bar{c}_{s_2} = -y_2 = -1$; both are nonpositive, confirming optimality (equivalently, $z = 27 - 2s_1 - s_2$). With c_1 variable, $\mathbf{y}(c_1)^\top = [c_1, 4] A_B^{-1} = [-2c_1 + 12, c_1 - 4]$, and the basis stays optimal while both components are nonnegative:

$$-2c_1 + 12 \geq 0 \text{ and } c_1 - 4 \geq 0 \iff \boxed{4 \leq c_1 \leq 6}.$$

Verification. Duals (2, 1) confirmed by `linprog`; at $b_1 = 5$: (5, 0), $z = 25$; at $b_1 = 7.5$: (0, 7.5), $z = 30$; at $b_1 = 4.99$ the duals jump to (5, 0) and at 7.51 to (0, 2). At $c_1 = 3.99$ the optimum moves to (0, 6) and at 6.01 to (5, 0).

Exercise 10.7 (Reading a Sensitivity Report) ★★

- The optimal value is read from the variable table:

$$z^* = 6(8) + 5(4) = 48 + 20 = 68.$$

- The change is $\Delta c_1 = 7.5 - 6 = 1.5$, which is within the allowable increase of 2. The optimal *basis* (and the solution (8, 4)) does not change; only the objective value moves, to $z = 7.5(8) + 5(4) = 80$.
- Constraint 2 is non-binding. Two independent tells in the report: its shadow price is 0, and its allowable increase is ∞ (adding more of a resource you are not fully using changes nothing, no matter how much you add). The allowable decrease of 5 is exactly the current slack.
- The increase of 2 is within the allowable increase of 4 for constraint 1, so the shadow price 2.5 applies to the full change:

$$z_{\text{new}} \approx 68 + 2 \times 2.5 = 73.$$

Exercise 10.8 (Shadow Price Interpretation) ★★

LP: $\max 50x_1 + 30x_2$ subject to $4x_1 + 2x_2 \leq 40$ (wood), $2x_1 + 3x_2 \leq 30$ (labor), $x_1, x_2 \geq 0$; optimum (7.5, 5), $z^* = 525$, duals (11.25, 2.50). This exercise has a Selected Solution in the book; it is reproduced here.

At the optimal solution $(x_1, x_2) = (7.5, 5)$ both constraints are binding: $4(7.5) + 2(5) = 40$ and $2(7.5) + 3(5) = 30$.

- The shadow price of wood is $y_1 = 11.25$: one additional board-foot of wood increases the optimal profit by \$11.25 (as long as the basis does not change). The shadow price of labor is $y_2 = 2.50$: one additional hour of labor is worth \$2.50 at the margin.

- Two additional board-feet increase profit by approximately $2 \times 11.25 = \$22.50$, from \$525 to \$547.50. (This is exact here: the wood RHS can range from 20 to 60 before the basis changes, and 42 is well inside that range.)
- Wood is more valuable at the margin, since $11.25 > 2.50$: a unit of wood buys more additional profit than a unit of labor.

Verification. `linprog` gives duals (11.25, 2.5) and $z^*(b_1 = 42) = 547.5$ exactly. The wood range [20, 60] checks out: at $b_1 = 19.9$ the duals become (15, 0) and at $b_1 = 60.1$ they become (0, 25).

Exercise 10.9 (Why a Slack Constraint Has Price Zero)

★★

- Geometric and economic argument.* If constraint i is not binding, the optimal point sits strictly inside the half-space of that constraint; the constraint's boundary does not touch the optimal vertex. Nudging b_i up moves that boundary even further away from the optimum, so the feasible region near the optimum, and hence the optimal solution and value, do not change at all. Economically: you already have leftovers of resource i . More of something you are throwing away is worth exactly nothing, so the most you would pay for an extra unit, which is what the shadow price measures, is 0.
- Dictionary argument.* If constraint i has slack, its slack variable s_i is basic with $s_i > 0$ in the final dictionary, and therefore s_i does not appear in the objective row (the objective row contains only nonbasic variables). Perturbing b_i to $b_i + \Delta$ and substituting $s'_i = s_i + \Delta$ changes only the constant in the row of s_i : it becomes $s_i = (\text{old value} + \Delta) + \dots$, and for small $|\Delta|$ this stays nonnegative, so feasibility persists. The objective row is untouched, so z^* is unchanged: shadow price 0. This is exactly the perturbation of b_2 worked in the book, where $s_2 = 3 + \Delta \geq 0$ holds for all $\Delta \geq -3$ and $z = 23$ throughout.

Exercise 10.10 (Why Ranges Exist at All)

★★

- The fixed object is the optimal *basis* B (equivalently, the matrix A_B^{-1} , i.e., the final dictionary's structure). For the basis to remain the answer, it must keep satisfying two families of conditions: *feasibility*, $\mathbf{x}_B = A_B^{-1}\mathbf{b} \geq \mathbf{0}$, which is what an RHS change can break, and *optimality*, all reduced costs $\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1} A_N \leq \mathbf{0}$ (for a max problem), which is what a cost change can break. Each condition is a linear inequality in the perturbation, and the allowable range is the intersection of these inequalities: a closed interval (possibly unbounded on one side).
- As long as the basis is fixed, the optimal value has the closed form $z^* = \mathbf{c}_B^\top A_B^{-1} \mathbf{b}$. This formula is linear in each b_i (with slope $y_i = (\mathbf{c}_B^\top A_B^{-1})_i$, the shadow price) and linear in each c_i for $i \in B$ (with slope x_i^* , the current value of that variable). Linearity of the formula in the data, with the basis frozen, is the whole story.
- At an endpoint of the range, one of the inequalities becomes tight: either a basic variable reaches 0 (RHS ranging) or a reduced cost reaches 0 (cost ranging). One step beyond, the dictionary is either infeasible or no longer optimal, and the simplex method must pivot to a different basis (dual simplex pivot in the first case, primal pivot in the second). After the pivot, z^* is still piecewise linear in the parameter, but with a new slope; the sensitivity report's numbers all describe only the current piece.

Exercise 10.11 (When the Shadow Price Prediction Breaks)

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LP: $\max x_1 + x_2$ subject to $x_1 \leq 1$, $x_2 \leq 1$, $x_1 + x_2 \leq b_3$ with $b_3 = 2$, $x_1, x_2 \geq 0$. The optimum $(1, 1)$ is degenerate: all three constraints are tight but only two are needed to define the vertex.

1. *Increase* b_3 to 3: the constraints $x_1 \leq 1$ and $x_2 \leq 1$ alone already force $z = x_1 + x_2 \leq 2$, and $(1, 1)$ is still feasible ($1 + 1 = 2 \leq 3$), so $z^* = 2$: no change. *Decrease* b_3 to 1: now $z = x_1 + x_2 \leq b_3 = 1$, and $z = 1$ is attained (e.g., at $(1, 0)$), so $z^* = 1$: a decrease of exactly 1. Verified with `linprog`: $z^*(3) = 2$, $z^*(1) = 1$.
2. The value function $z^*(b_3)$ has a *kink* at $b_3 = 2$: its right derivative is 0 and its left derivative is 1. A shadow price is supposed to be the derivative dz^*/db_3 , and at a kink that derivative does not exist. Behind the kink is dual degeneracy: the dual problem, $\min y_1 + y_2 + 2y_3$ subject to $y_1 + y_3 \geq 1$, $y_2 + y_3 \geq 1$, $\mathbf{y} \geq \mathbf{0}$, has a whole segment of optimal solutions

$$\mathbf{y}(t) = (1 - t, 1 - t, t), \quad t \in [0, 1],$$

all with value 2. Any $y_3 \in [0, 1]$ is a legitimate “shadow price” of constraint 3; the right derivative of z^* is the minimum of y_3 over the optimal dual set (0) and the left derivative is the maximum (1). No single number describes both directions.

3. A solver reports the dual solution at whichever optimal dual vertex its algorithm happens to terminate (HiGHS reports $\mathbf{y} = (1, 1, 0)$ here, i.e., price 0 for constraint 3; another solver or another pivot rule could report $(0, 0, 1)$). With a degenerate optimum, the reported shadow price may be valid only for increases, only for decreases, or may have allowable range 0 in one direction, so it should be checked by re-solving with the perturbed RHS before money changes hands.

Exercise 10.12 (Pushing Past the Range)

★★★

This exercise has a Selected Solution in the book; it is reproduced here with the verification spelled out.

The running example of this chapter works. Its hours constraint $x + y \leq 9$ has shadow price $p = 1$, valid for $7 \leq b_1 \leq 10$ (allowable increase 1). Increasing b_1 by 1 to 10 raises the optimal value from 23 to 24, exactly p . Increasing b_1 by 5 to 14 raises it only to 24, not to $23 + 5 = 28$: solving the LP with $b_1 = 14$ gives the optimum $(x, y) = (6, 4)$ determined by the flour and sugar constraints, and $z^* = 2(6) + 3(4) = 24$. At the range boundary $b_1 = 10$ the basic variable s_2 hits 0 and the basis changes; beyond it the hours constraint is no longer binding, so its shadow price drops to 0 and further increases in b_1 buy nothing. In general, the optimal value is a piecewise-linear concave function of b_1 , and the shadow price is only its slope on the current piece.

Verification. `linprog` gives $z^*(b_1 = 9) = 23$ at $(4, 5)$, $z^*(b_1 = 10) = 24$ at $(6, 4)$, and $z^*(b_1 = 14) = 24$ at $(6, 4)$, so the one-unit prediction is exact and the five-unit prediction over-promises by 4.

11 Chapter 11: Duality

This chapter has 17 exercises drawn from two book files that share one exercise counter: the duality sections contribute 11.1 to 11.9 (warm-ups on writing duals and weak duality, core problems on mixed constraint forms and shadow prices, concept problems on certificates and infeasibility, and the dual-of-the-dual proof), and the complementary slackness section contributes 11.10 to 11.17

(condition checking, recovering one side of a primal-dual pair from the other, disproving claimed optima, and two larger modeling problems). Duals are written out in full and every complementary slackness condition is checked term by term. Exercises 11.2, 11.4, 11.9, 11.11, 11.12, and 11.15 have Selected Solutions in the book; those are reproduced here, expanded where useful. All numeric claims were verified with `scipy`.

QA flag

Book fixes made during this QA pass (all verified numerically, none affecting exercise data or Selected Solutions):

- `duality.tex`, “Case 2” box: the primal read $\min \mathbf{b}^\top \mathbf{x}$ s.t. $A\mathbf{x} \leq \mathbf{c}$ with dual $\max \mathbf{c}^\top \mathbf{y}$ s.t. $A^\top \mathbf{y} \leq \mathbf{b}$, $\mathbf{y} \geq 0$, which violates weak duality (test instance: $\min x$ s.t. $x \leq 3$, $x \geq 0$ has value 0 but the stated dual has value 3). Fixed to the intended mirror of Case 1: minimization with $\geq$ constraints.
- `duality.tex`, “Complicated Dual” example (both derivations): the sign restrictions on y_2 and y_3 were swapped. For a maximization primal, the $\geq$ constraint must take $y_2 \leq 0$ and the $\leq$ constraint $y_3 \geq 0$. Decisive check: that primal is infeasible, yet the dual as previously printed has finite optimal value $11/3$, which strong duality forbids; with the corrected signs the dual is unbounded, as it must be. A remark about the (harmless) infeasibility of that example primal was also added. The aggregation “second look” additionally treated the free variable x_2 as nonnegative, deriving $y_1 - y_2 - y_3 \geq 1$ where the first derivation has $= 1$; made consistent (x_2 free, equality).
- `complimentary-slackness.tex`, Exercise 11.13: “a dual solution $\mathbf{y}^* = (1, 1)$ ” now reads “a candidate dual solution,” since $(1, 1)$ is intentionally not dual feasible (that discovery is the point of parts (b) and (c)).

Exercise 11.1 (The Dual of a Small Bakery) ★

Primal: $\max 5x_1 + 4x_2$ subject to $2x_1 + 3x_2 \leq 12$ (flour), $x_1 + x_2 \leq 5$ (sugar), $x_1, x_2 \geq 0$.

1. One dual variable per resource: y_1 for flour, y_2 for sugar. The dual is

$$\begin{aligned} \min \quad & 12y_1 + 5y_2 \\ \text{s.t.} \quad & 2y_1 + y_2 \geq 5 \quad (\text{loaves}) \\ & 3y_1 + y_2 \geq 4 \quad (\text{rolls}) \\ & y_1, y_2 \geq 0. \end{aligned}$$

2. Read y_1 as a price per unit of flour and y_2 as a price per unit of sugar. The first dual constraint says the rival’s offer for the ingredients in one loaf (2 flour, 1 sugar) must be at least the \$5 the bakery could earn by baking the loaf itself; the second says the ingredients in one roll (3 flour, 1 sugar) must fetch at least \$4. The dual objective is the rival shopping for the cheapest total payment $12y_1 + 5y_2$ that still convinces the baker to sell.
3. The primal right-hand sides (12, 5) (resource amounts) become the dual objective coefficients, and the primal objective coefficients (5, 4) (unit profits) become the dual right-hand sides. The constraint matrix is transposed.

Verification. Primal optimum $(5, 0)$ with $z^* = 25$ (flour has slack 2); dual optimum $\mathbf{y}^* = (0, 5)$ with value 25, consistent with complementary slackness ($y_1 = 0$ on the slack flour constraint).

Exercise 11.2 (Formulate the Dual) ★

This exercise has a Selected Solution in the book; it is reproduced here. There are two constraints, so the dual has two variables $y_1, y_2 \geq 0$, one per constraint. Each of the three primal variables produces a dual constraint whose right-hand side is that variable's objective coefficient:

$$\begin{aligned} \min \quad & 10y_1 + 12y_2 \\ \text{s.t.} \quad & y_1 + 3y_2 \geq 4 \\ & 2y_1 + y_2 \geq 5 \\ & y_1 + 2y_2 \geq 3 \\ & y_1, y_2 \geq 0. \end{aligned}$$

Exercise 11.3 (Verifying Weak Duality) ★

Primal: $\max 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 5$, $x_1, x_2 \geq 0$; dual: $\min 4y_1 + 5y_2$ subject to $y_1 + 2y_2 \geq 3$, $y_1 + y_2 \geq 2$, $y_1, y_2 \geq 0$.

1. $\mathbf{x} = (1, 2)$: $1 + 2 = 3 \leq 4$ and $2(1) + 2 = 4 \leq 5$, with $x_1, x_2 \geq 0$, so $\mathbf{x}$ is primal feasible. Its value is $\mathbf{c}^\top \mathbf{x} = 3(1) + 2(2) = 7$.
2. $\mathbf{y} = (2, 1)$: $2 + 2(1) = 4 \geq 3$ and $2 + 1 = 3 \geq 2$, with $y_1, y_2 \geq 0$, so $\mathbf{y}$ is dual feasible. Its value is $\mathbf{b}^\top \mathbf{y} = 4(2) + 5(1) = 13$.
3. Weak duality gives $\mathbf{c}^\top \mathbf{x} \leq z^* \leq \mathbf{b}^\top \mathbf{y}$ for this pair, i.e., the sandwich

$$7 \leq z^* \leq 13.$$

The true optimum $\mathbf{x}^* = (1, 3)$ with $z^* = 9$ indeed satisfies $7 \leq 9 \leq 13$. (Verified: `linprog` returns $z^* = 9$ at $(1, 3)$ with duals $(1, 1)$.)

Exercise 11.4 (Weak Duality Bound) ★★

This exercise has a Selected Solution in the book; it is reproduced here.

The dual is $\min 4y_1 + 6y_2$ subject to $y_1 + 2y_2 \geq 3$, $y_1 + y_2 \geq 2$, $y_1, y_2 \geq 0$. For $\mathbf{y} = (1, 1)$: $1 + 2 = 3 \geq 3$ and $1 + 1 = 2 \geq 2$, so $\mathbf{y}$ is dual feasible with objective $4(1) + 6(1) = 10$. For $\mathbf{x} = (2, 2)$: $2 + 2 = 4 \leq 4$ and $4 + 2 = 6 \leq 6$, so $\mathbf{x}$ is primal feasible with objective $3(2) + 2(2) = 10$. By weak duality, every primal value is at most every dual value, so the optimal value z^* satisfies $10 \leq z^* \leq 10$. Since the two bounds coincide, both solutions are in fact optimal and $z^* = 10$. (Verified: `linprog` returns $z^* = 10$ at $(2, 2)$ with duals $(1, 1)$.)

Exercise 11.5 (Dual with Mixed Constraints) ★★

Primal: $\max 2x_1 + 3x_2$ subject to $x_1 + x_2 = 5$, $x_1 - x_2 \leq 2$, $x_1 \geq 0$, x_2 free.

Assign y_1 to the equality and y_2 to the inequality. For a maximization primal: an equality constraint imposes no sign restriction (y_1 free), and a $\leq$ constraint takes a nonnegative multiplier

($y_2 \geq 0$). The dual constraints come from the primal variables: $x_1 \geq 0$ gives an inequality ($\geq$), x_2 free forces an equality. The dual is

$$\begin{aligned} \min \quad & 5y_1 + 2y_2 \\ \text{s.t.} \quad & y_1 + y_2 \geq 2 \quad (\text{from } x_1 \geq 0) \\ & y_1 - y_2 = 3 \quad (\text{from } x_2 \text{ free}) \\ & y_1 \text{ free, } y_2 \geq 0. \end{aligned}$$

Summary of the mappings: equality constraint $\rightarrow$ free dual variable; $\leq$ constraint $\rightarrow y_2 \geq 0$; nonnegative primal variable $\rightarrow \geq$ dual constraint; free primal variable $\rightarrow$ dual equality.

Verification. Primal optimum: from $x_2 = 5 - x_1$ the objective is $15 - x_1$, maximized at $x_1 = 0$, so $\mathbf{x}^* = (0, 5)$, $z^* = 15$. Dual: substituting $y_1 = 3 + y_2$ gives objective $15 + 7y_2$, minimized at $y_2 = 0$, so $\mathbf{y}^* = (3, 0)$ with value $15 = z^*$; `linprog` confirms both.

Exercise 11.6 (Shadow Prices and Dual Variables) ★★

Primal: $\max 5x_1 + 4x_2$ subject to $2x_1 + x_2 \leq 20$ (flour), $x_1 + 2x_2 \leq 18$ (sugar), $x_1 + x_2 \leq 13$ (oven); $\mathbf{x}^* = (7\frac{1}{3}, 5\frac{1}{3})$, $z^* = 58$; $\mathbf{y}^* = (2, 1, 0)$.

1. $\mathbf{b}^\top \mathbf{y}^* = 20(2) + 18(1) + 13(0) = 40 + 18 + 0 = 58 = z^*$, so strong duality holds.

2. Substituting $\mathbf{x}^* = (\frac{22}{3}, \frac{16}{3})$:

$$\begin{aligned} 2\left(\frac{22}{3}\right) + \frac{16}{3} &= \frac{60}{3} = 20 && (\text{flour binding}), \\ \frac{22}{3} + 2\left(\frac{16}{3}\right) &= \frac{54}{3} = 18 && (\text{sugar binding}), \\ \frac{22}{3} + \frac{16}{3} &= \frac{38}{3} \approx 12.67 < 13 && (\text{oven non-binding}). \end{aligned}$$

3. Complementary slackness, condition by condition:

$$\begin{aligned} y_1^* (2x_1^* + x_2^* - 20) &= 2 \cdot 0 = 0 \checkmark & y_2^* (x_1^* + 2x_2^* - 18) &= 1 \cdot 0 = 0 \checkmark \\ y_3^* (x_1^* + x_2^* - 13) &= 0 \cdot \left(-\frac{1}{3}\right) = 0 \checkmark & x_1^* (5 - 2y_1^* - y_2^* - y_3^*) &= \frac{22}{3} (5 - 5) = 0 \checkmark \\ x_2^* (4 - y_1^* - 2y_2^* - y_3^*) &= \frac{16}{3} (4 - 4) = 0 \checkmark \end{aligned}$$

Both dual constraints are tight (as they must be, since $x_1^*, x_2^* > 0$), and the only slack primal constraint (oven) carries dual price 0. All conditions hold, so the pair is optimal by the Primal Dual Optimality theorem of the book (feasibility on both sides plus complementary slackness).

4. The shadow price of flour is $y_1^* = 2$: one more kg of flour raises profit by about \$2. Sugar is worth \$1 per kg at the margin, and oven capacity is worth \$0 (it is not fully used, so more of it buys nothing). One additional kg of flour: profit increases by approximately \$2, from 58 to 60. (Verified: $z^*(b_1 = 21) = 60$ exactly.)

Exercise 11.7 (Certificates in Your Own Words) ★★

1. A dual feasible $\mathbf{y}$ certifies an *upper bound* on every feasible primal value: it is a recipe of multipliers that combines the primal constraints into the single valid inequality $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$. To convince a skeptical friend that no plan earns more than a given amount M , you do not need to enumerate plans; you hand over the numbers $y_1, \dots, y_m \geq 0$, let the friend check

the n dominance inequalities $A^\top \mathbf{y} \geq \mathbf{c}$ (one per product), and let them compute $\mathbf{b}^\top \mathbf{y} \leq M$. Multiplying out constraints and adding is arithmetic the friend can verify in minutes, with no optimization and no trust required.

2. Multiplying an inequality by a negative number flips its direction. The primal constraints face $\leq$, and the certificate needs the combined constraint to still face $\leq$ so that it caps the objective from above; that is only guaranteed when every multiplier is nonnegative. (Equality constraints can be combined with multipliers of either sign, which is exactly why their dual variables come out free.)
3. The set of valid certificates is cut out by linear conditions on $\mathbf{y}$: the sign conditions $\mathbf{y} \geq \mathbf{0}$ and the dominance conditions $A^\top \mathbf{y} \geq \mathbf{c}$. Minimizing the linear bound $\mathbf{b}^\top \mathbf{y}$ over this polyhedral set is a linear program, the dual. Weak duality says every certificate is an over-estimate or exact; strong duality says the hunt is perfectly successful: the best certificate is not merely close but *exactly* equal to the optimal primal value, so an optimal plan and an optimal certificate together prove each other optimal.

Exercise 11.8 (Strong Duality and Infeasibility)

★★

Primal: $\max x_1 + x_2$ subject to $x_1 - x_2 \leq 1$, $-x_1 + x_2 \leq -3$, $x_1, x_2 \geq 0$.

1. Two constraints give two dual variables $y_1, y_2 \geq 0$; the dual is

$$\begin{aligned} \min \quad & y_1 - 3y_2 \\ \text{s.t.} \quad & y_1 - y_2 \geq 1 \quad (\text{from } x_1) \\ & -y_1 + y_2 \geq 1 \quad (\text{from } x_2) \\ & y_1, y_2 \geq 0. \end{aligned}$$

2. Add the two primal constraints: $(x_1 - x_2) + (-x_1 + x_2) = 0$, while the right-hand sides add to $1 + (-3) = -2$. So every feasible point would need $0 \leq -2$, impossible. The primal is infeasible (no $(x_1, x_2) \geq 0$, or indeed any (x_1, x_2) at all, satisfies both).
3. The dual is *infeasible* as well: adding the two dual constraints gives $(y_1 - y_2) + (-y_1 + y_2) = 0 \geq 2$, impossible. How this squares with the strong duality theorem: since the primal is infeasible, the theorem rules out the dual having an optimal solution (case 3 needs both feasible) and, read contrapositively, an infeasible primal leaves the dual only two options, unbounded or infeasible. Here a direct check shows it is infeasible. This pair is the standard example that *both* problems of a primal-dual pair can be infeasible simultaneously; that fourth possibility is consistent with the theorem even though it is not one of the three listed cases. (Verified: HiGHS reports both problems infeasible.)

QA flag

Judgment call, no edit made: The Strong Duality theorem in `duality.tex` lists three cases and never mentions that primal and dual can both be infeasible, yet this exercise's part (c) hinges on exactly that fourth possibility, and part (c)'s instruction "explain using the strong duality theorem" may push students toward the wrong answer "unbounded." Suggest adding a one-sentence remark after the theorem (e.g., "It is also possible for both (P) and (D) to be infeasible; see the exercises.").

Exercise 11.9 (The Dual of the Dual is the Primal)

★★★

This exercise has a Selected Solution in the book; it is reproduced here.

Rewrite (D) in symmetric maximization form. Minimizing $\mathbf{b}^\top \mathbf{y}$ is the same as maximizing $(-\mathbf{b})^\top \mathbf{y}$, and the constraints $A^\top \mathbf{y} \geq \mathbf{c}$ are equivalent to $(-A^\top) \mathbf{y} \leq -\mathbf{c}$. So (D) is

$$\max (-\mathbf{b})^\top \mathbf{y} \quad \text{s.t.} \quad (-A^\top) \mathbf{y} \leq -\mathbf{c}, \quad \mathbf{y} \geq 0,$$

which is the symmetric form with objective vector $\tilde{\mathbf{c}} = -\mathbf{b}$, matrix $\tilde{A} = -A^\top$, and right-hand side $\tilde{\mathbf{b}} = -\mathbf{c}$. Its dual, by definition, is

$$\min \tilde{\mathbf{b}}^\top \mathbf{z} \quad \text{s.t.} \quad \tilde{A}^\top \mathbf{z} \geq \tilde{\mathbf{c}}, \quad \mathbf{z} \geq 0,$$

that is, $\min (-\mathbf{c})^\top \mathbf{z}$ subject to $(-A) \mathbf{z} \geq -\mathbf{b}$, $\mathbf{z} \geq 0$. Multiplying the objective and the constraints by -1 turns this into

$$\max \mathbf{c}^\top \mathbf{z} \quad \text{s.t.} \quad A \mathbf{z} \leq \mathbf{b}, \quad \mathbf{z} \geq 0,$$

which is exactly (P) with $\mathbf{z}$ in place of $\mathbf{x}$. Hence the dual of the dual is the primal.

Exercise 11.10 (Checking the Conditions)

★

Pair: $\max 4x_1 + 3x_2$ s.t. $x_1 + x_2 \leq 6$, $2x_1 + x_2 \leq 10$, $\mathbf{x} \geq \mathbf{0}$; dual $\min 6y_1 + 10y_2$ s.t. $y_1 + 2y_2 \geq 4$, $y_1 + y_2 \geq 3$, $\mathbf{y} \geq \mathbf{0}$. Candidates $\mathbf{x}^* = (4, 2)$, $\mathbf{y}^* = (2, 1)$.

1. Primal feasibility: $4 + 2 = 6 \leq 6$ and $2(4) + 2 = 10 \leq 10$, with $\mathbf{x}^* \geq \mathbf{0}$. Dual feasibility: $2 + 2(1) = 4 \geq 4$ and $2 + 1 = 3 \geq 3$, with $\mathbf{y}^* \geq \mathbf{0}$. Both feasible (in fact all four constraints are tight).
2. The four conditions:

$$\begin{aligned} y_1^* (x_1^* + x_2^* - 6) &= 2(6 - 6) = 0 \checkmark & y_2^* (2x_1^* + x_2^* - 10) &= 1(10 - 10) = 0 \checkmark \\ x_1^* (4 - y_1^* - 2y_2^*) &= 4(4 - 4) = 0 \checkmark & x_2^* (3 - y_1^* - y_2^*) &= 2(3 - 3) = 0 \checkmark \end{aligned}$$

3. Feasibility on both sides plus complementary slackness is exactly the optimality certificate of the book's Primal Dual Optimality theorem, so both are optimal. Objective check: $4(4) + 3(2) = 22$ and $6(2) + 10(1) = 22$; equal values reconfirm optimality by strong duality. (Verified with `linprog`: $z^* = 22$, duals $(2, 1)$.)

Exercise 11.11 (Bakery Production and Duality Interpretation)

★★

This exercise has a Selected Solution in the book; it is reproduced here.

1. **Feasibility:** Substituting $(x_1, x_2, x_3) = (2, 0, 8)$: flour $8(2) + 6(0) + 8 = 24 \leq 48$, sugar $4(2) + 2(0) + 1.5(8) = 20 \leq 20$, baking time $2(2) + 1.5(0) + 0.5(8) = 8 \leq 8$, so the primal solution is feasible. Substituting $(y_1, y_2, y_3) = (0, 10, 10)$: the dual constraints evaluate to $60 \geq 60$, $35 \geq 30$, and $20 \geq 20$, so the dual solution is feasible.
2. **Complementary slackness:** In the primal, sugar and baking time are tight while flour has slack ($24 < 48$); accordingly $y_1 = 0$ while $y_2, y_3 > 0$. In the dual, constraints 1 and 3 are tight while constraint 2 has slack ($35 > 30$); accordingly $x_2 = 0$ while $x_1, x_3 > 0$. Every condition $y_i(\mathbf{a}_i^\top \mathbf{x} - b_i) = 0$ and $x_j(c_j - (A^\top \mathbf{y})_j) = 0$ is satisfied.

3. **Objective values:** Primal: $60(2) + 30(0) + 20(8) = 280$. Dual: $48(0) + 20(10) + 8(10) = 280$. They match, confirming optimality by strong duality.
4. **Shadow prices:** Flour has shadow price $y_1 = 0$: with 24 kg left over, more flour is worthless. Sugar has shadow price $y_2 = 10$: one more kg of sugar would increase profit by \$10. Baking time has shadow price $y_3 = 10$: one more oven hour is worth \$10.
5. **Sensitivity:** (a) One more kg of flour: no change (\$0). (b) One more kg of sugar: profit increases by \$10. (c) One more hour of baking time: profit increases by \$10. These estimates are valid for small changes in the right-hand sides (as long as the optimal basis does not change).

QA flag

Judgment call, no edit made: this exercise is nearly a verbatim restatement of the worked “Bakery 2.0” example and of Examples “Verifying Optimality with Complementary Slackness” and “Recovering Dual Variables via Complementary Slackness” in the same chapter; every requested computation appears, with the same numbers, in the chapter body, and the book also prints a Selected Solution. As a graded exercise it is fully copyable. Consider changing the data (e.g., new profits or resource levels) or dropping the printed dual optimum so students must recover it themselves.

Exercise 11.12 (Box Constraints and a Dual Certificate)

★★

This exercise has a Selected Solution in the book; it is reproduced here.

(a) The dual is

$$\begin{aligned} \min \quad & y_1 + y_2 + y_3 + y_4 + 8y_5 \\ \text{s.t.} \quad & y_1 + y_5 \geq 1, \quad y_2 + 2y_5 \geq 4, \quad y_3 + 3y_5 \geq 9, \quad y_4 + 4y_5 \geq 16, \\ & y_1, \dots, y_5 \geq 0. \end{aligned}$$

(b) By weak duality, every dual feasible $\mathbf{y}$ gives an upper bound $\mathbf{b}^\top \mathbf{y}$ on the primal objective. If we exhibit a primal feasible $\mathbf{x}$ and a dual feasible $\mathbf{y}$ whose objective values are equal, both must be optimal.

(c) At $\mathbf{x}^* = (0, \frac{1}{2}, 1, 1)$, constraints 1 and 2 have slack ($0 < 1$ and $\frac{1}{2} < 1$), so $y_1 = y_2 = 0$. Since $x_2, x_3, x_4 > 0$, dual constraints 2, 3, 4 must be tight: $2y_5 = 4$ gives $y_5 = 2$; then $y_3 = 9 - 3(2) = 3$ and $y_4 = 16 - 4(2) = 8$. The remaining dual constraint holds with slack: $y_1 + y_5 = 2 \geq 1$, consistent with $x_1 = 0$. So $\mathbf{y}^* = (0, 0, 3, 8, 2)$ is dual feasible with objective $3 + 8 + 8(2) = 27$, matching the primal value $4(\frac{1}{2}) + 9 + 16 = 27$. By part (b), both solutions are optimal.

Verification. `linprog` returns $z^* = 27$ at $(0, 0.5, 1, 1)$ with duals $(0, 0, 3, 8, 2)$, exactly the certificate above. Feasibility of $\mathbf{x}^*$: constraint 5 gives $0 + 2(\frac{1}{2}) + 3(1) + 4(1) = 8 \leq 8$, tight, consistent with $y_5 = 2 > 0$.

Exercise 11.13 (Deducing the Primal from a Dual Guess)

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Pair: $\max 5x_1 + 4x_2$ s.t. $2x_1 + x_2 \leq 8$, $x_1 + 3x_2 \leq 9$, $\mathbf{x} \geq \mathbf{0}$; dual $\min 8y_1 + 9y_2$ s.t. $2y_1 + y_2 \geq 5$, $y_1 + 3y_2 \geq 4$, $\mathbf{y} \geq \mathbf{0}$. Candidate: $\mathbf{y}^* = (1, 1)$.

- (a) Since $y_1^* = 1 > 0$ and $y_2^* = 1 > 0$, complementary slackness would force both primal constraints to be tight:

$$2x_1 + x_2 = 8, \quad x_1 + 3x_2 = 9 \quad \implies \quad x_1^* = 3, \quad x_2^* = 2.$$

- (b) Yes: $(3, 2) \geq \mathbf{0}$ satisfies $2(3) + 2 = 8 \leq 8$ and $3 + 3(2) = 9 \leq 9$, so it is primal feasible, with value $5(3) + 4(2) = 23$.
- (c) No, this is not a valid optimal pair, and the cleanest witness is weak duality: the “dual value” of $\mathbf{y}^* = (1, 1)$ is $8(1) + 9(1) = 17 < 23$, which no genuine dual feasible point can achieve. Indeed $\mathbf{y}^*$ is *not dual feasible*: the first dual constraint fails, $2(1) + 1 = 3 < 5$. Complementary slackness also fails as a diagnostic: $x_1^* = 3 > 0$ requires the first dual constraint to be tight, but $3 \neq 5$. The lesson: complementary slackness is only a certificate when *both* points are feasible (the book’s Primal Dual Optimality theorem requires primal feasibility, dual feasibility, and the CS conditions together).

Interestingly, the primal point $(3, 2)$ recovered from the bogus guess happens to be the true primal optimum: re-solving the tight-constraint system for the *correct* dual, with $x_1^*, x_2^* > 0$ forcing both dual constraints tight, gives $2y_1 + y_2 = 5$, $y_1 + 3y_2 = 4$, so $\mathbf{y}^{\text{opt}} = (\frac{11}{5}, \frac{3}{5})$ with dual value $8(\frac{11}{5}) + 9(\frac{3}{5}) = \frac{115}{5} = 23$, matching the primal value. (Verified: `linprog` gives $z^* = 23$ at $(3, 2)$ with duals $(2.2, 0.6)$.)

Exercise 11.14 (A Positive Price on a Slack Constraint)

★★

- By the book’s Primal Dual Optimality theorem, if $\mathbf{x}^*$ and $\mathbf{y}^*$ are both optimal, then, both being feasible, they must satisfy complementary slackness: $y_2^* (\mathbf{a}_2^\top \mathbf{x}^* - b_2) = 0$. Here $\mathbf{a}_2^\top \mathbf{x}^* < b_2$ (slack) and $y_2^* = 3 > 0$, so the product is strictly negative, not zero. The claim is therefore false: $\mathbf{x}^*$ and $\mathbf{y}^*$ cannot both be optimal.
- Be careful about what is ruled out: only the *joint* optimality of this particular pair. The argument does not identify the culprit. It is possible that $\mathbf{x}^*$ is optimal and $\mathbf{y}^*$ is a merely feasible, suboptimal dual point; possible that $\mathbf{y}^*$ is optimal and $\mathbf{x}^*$ is suboptimal; and possible that neither is optimal. (If both were optimal, any optimal primal paired with any optimal dual satisfies CS, so the observed violation convicts at least one of them, without saying which.) A quick way to find out more: compare $\mathbf{c}^\top \mathbf{x}^*$ with $\mathbf{b}^\top \mathbf{y}^*$; by strong duality the gap between them bounds how far each is from optimal.
- Economically, y_2^* is the price the rival (or the market) puts on one unit of resource 2. Paying a strictly positive price for additional units of a resource of which you already have unused stock is absurd: an extra unit would join the leftovers and generate no additional profit, so its marginal value is 0. Equivalently, at economic equilibrium a good in excess supply has price zero; a positive price on a slack resource signals that the reported prices and the reported plan cannot describe the same equilibrium.

Exercise 11.15 (Disproving a Claimed Optimum)

★★

This exercise has a Selected Solution in the book; it is reproduced here.

- (a) The conditions are $y_1(x_1 + x_2 - 4) = 0$, $y_2(2x_1 + x_2 - 5) = 0$, $x_1(y_1 + 2y_2 - 3) = 0$, and $x_2(y_1 + y_2 - 2) = 0$.

(b) At $(x_1, x_2) = (2, 0)$: the first primal constraint gives $2 < 4$ (slack), forcing $y_1 = 0$; the second gives $4 < 5$ (slack), forcing $y_2 = 0$. But $x_1 = 2 > 0$ requires the first dual constraint to be tight: $y_1 + 2y_2 = 3$.

(c) With $y_1 = y_2 = 0$ we get $0 = 3$, a contradiction: no dual solution is compatible, so $(2, 0)$ is not optimal. (It is feasible, but both constraints have slack, so we can do better.) The actual optimum is $(x_1, x_2) = (1, 3)$ with value 9: both primal constraints are tight, and solving $y_1 + 2y_2 = 3$, $y_1 + y_2 = 2$ gives the dual optimum $\mathbf{y}^* = (1, 1)$ with matching value $4(1) + 5(1) = 9$. (Verified with `linprog`.)

Exercise 11.16 (Meal Kit Optimization and Duality)

★ ★ ★

Primal: $\max 3x_1 + 2x_2 + 4x_3 + x_4 + 5x_5 + 2x_6$ subject to kitchen $x_1 + 2x_2 + x_3 + x_5 \leq 10$, packaging $2x_1 + x_4 + 3x_5 + x_6 \leq 12$, $\mathbf{x} \geq \mathbf{0}$.

- Two resources give two dual variables: y_1 = the marginal value of one kitchen hour, y_2 = the marginal value of one packaging hour. Each kit gives one dual constraint (its resource bundle, priced at y , must cover its profit):

$$\begin{aligned} \min \quad & 10y_1 + 12y_2 \\ \text{s.t.} \quad & y_1 + 2y_2 \geq 3 \quad (x_1 \text{ Classic Chicken}) \\ & 2y_1 \geq 2 \quad (x_2 \text{ Vegan Delight}) \\ & y_1 \geq 4 \quad (x_3 \text{ Protein Boost}) \\ & y_2 \geq 1 \quad (x_4 \text{ Quick \& Light}) \\ & y_1 + 3y_2 \geq 5 \quad (x_5 \text{ Gourmet Chef Special}) \\ & y_2 \geq 2 \quad (x_6 \text{ Breakfast Box}) \\ & y_1, y_2 \geq 0. \end{aligned}$$

- In the (y_1, y_2) plane the six constraints reduce to two: $y_1 \geq 4$ (Protein Boost; it dominates $2y_1 \geq 2$) and $y_2 \geq 2$ (Breakfast Box; it dominates $y_2 \geq 1$). The two coupling constraints are then automatic: $y_1 + 2y_2 \geq 4 + 4 = 8 \geq 3$ and $y_1 + 3y_2 \geq 4 + 6 = 10 \geq 5$. The feasible region is the “northeast quadrant” $\{y_1 \geq 4, y_2 \geq 2\}$, and the level lines of $10y_1 + 12y_2$ decrease toward the southwest, so the optimum is the corner

$$\mathbf{y}^* = (4, 2), \quad \text{dual value } 10(4) + 12(2) = 64.$$

- A kitchen hour is worth \$4 and a packaging hour \$2 at the margin: with one more kitchen hour the startup could earn about \$4 more (by making one more Protein Boost), and one more packaging hour is worth \$2 (one more Breakfast Box). The whole resource endowment, priced this way, is worth \$64, and no kit beats its own resource bill at these prices.
- Since $y_1^* = 4 > 0$ and $y_2^* = 2 > 0$, complementary slackness forces *both* primal constraints to be tight. On the variable side, the dual constraints with slack at $\mathbf{y}^*$ force those variables to zero:

$$\begin{aligned} x_1 : \quad & 4 + 2(2) = 8 > 3 \quad \Rightarrow x_1 = 0, & x_2 : \quad & 2(4) = 8 > 2 \quad \Rightarrow x_2 = 0, \\ x_4 : \quad & 2 > 1 \quad \Rightarrow x_4 = 0, & x_5 : \quad & 4 + 3(2) = 10 > 5 \quad \Rightarrow x_5 = 0, \end{aligned}$$

while the constraints for x_3 ($y_1 \geq 4$, tight) and x_6 ($y_2 \geq 2$, tight) allow $x_3, x_6 > 0$. The basic variables are x_3 and x_6 .

5. With x_3, x_6 the only nonzeros, the two tight resource constraints become $x_3 = 10$ and $x_6 = 12$, so

$$\mathbf{x}^* = (0, 0, 10, 0, 0, 12), \quad z^* = 4(10) + 2(12) = 64,$$

matching the dual value, so both solutions are optimal by strong duality. (Verified: `linprog` returns $z^* = 64$ at exactly this $\mathbf{x}^*$ with duals $(4, 2)$.)

Exercise 11.17 (Duality with Mixed Constraint Types: Eco-Friendly Packaging Startup) ★★★

Primal: $\max 4x_1 + 6x_2 + 5x_3$ subject to labor $2x_1 + 4x_2 + 3x_3 \leq 140$, material $4x_1 + 6x_2 + 4x_3 = 240$, retail $6x_1 + 9x_2 + 8x_3 \geq 80$, $\mathbf{x} \geq \mathbf{0}$.

- Dual variables, with signs dictated by the constraint types of this *maximization* primal: $y_1 \geq 0$ for the $\leq$ labor constraint (price per labor hour), y_2 *free* for the material equality (price per unit of material; it can be negative, since being forced to consume exactly 240 units could in principle hurt), and $y_3 \leq 0$ for the $\geq$ retail commitment (a $\geq$ constraint in a max problem takes a nonpositive multiplier; its “price” can only penalize, reflecting that the commitment restricts the startup). Each nonnegative primal variable gives a $\geq$ dual constraint:

$$\begin{aligned} \min \quad & 140y_1 + 240y_2 + 80y_3 \\ \text{s.t.} \quad & 2y_1 + 4y_2 + 6y_3 \geq 4 \quad (x_1 \text{ FreshBox Standard}) \\ & 4y_1 + 6y_2 + 9y_3 \geq 6 \quad (x_2 \text{ FreshBox Premium}) \\ & 3y_1 + 4y_2 + 8y_3 \geq 5 \quad (x_3 \text{ Market Crate}) \\ & y_1 \geq 0, \quad y_2 \text{ free}, \quad y_3 \leq 0. \end{aligned}$$

- The dual is three-dimensional, so solve either problem with software. `linprog` on the primal returns

$$\mathbf{x}^* = (40, 0, 20), \quad z^* = 4(40) + 5(20) = 260,$$

with labor tight ($2(40) + 3(20) = 140$), material exact ($4(40) + 4(20) = 240$), and retail slack ($6(40) + 8(20) = 400 > 80$). Solving the dual directly gives $\mathbf{y}^* = (1, \frac{1}{2}, 0)$ with value $140(1) + 240(\frac{1}{2}) + 80(0) = 260$. (Alternatively: guessing $y_3 = 0$ from the retail slack reduces the dual to two variables, and the graphical optimum of $\min 140y_1 + 240y_2$ over the three constraints in the (y_1, y_2) plane is the intersection of the x_1 and x_3 constraints, $2y_1 + 4y_2 = 4$ and $3y_1 + 4y_2 = 5$, i.e., $(1, \frac{1}{2})$.)

- Complementary slackness, both directions:

- Retail has slack ($400 > 80$) $\Rightarrow y_3^* = 0$. Consistent. Labor is tight, so $y_1^* = 1 > 0$ is allowed; the material equality is always “tight,” so the free $y_2^* = \frac{1}{2}$ carries no condition.
- At $\mathbf{y}^* = (1, \frac{1}{2}, 0)$: the x_1 constraint is tight ($2 + 2 + 0 = 4$), the x_3 constraint is tight ($3 + 2 + 0 = 5$), and the x_2 constraint has slack ($4 + 3 + 0 = 7 > 6$) $\Rightarrow x_2^* = 0$. So the candidate basic variables are x_1 and x_3 .

- With $x_2 = 0$, the two tight resource constraints give the square system

$$2x_1 + 3x_3 = 140, \quad 4x_1 + 4x_3 = 240 \quad \Longrightarrow \quad x_1 = 40, \quad x_3 = 20,$$

(from the second equation $x_1 + x_3 = 60$; substituting $x_1 = 60 - x_3$ into the first gives $120 + x_3 = 140$). Total profit: $z^* = 4(40) + 6(0) + 5(20) = 260$, equal to the dual value, so

the pair is optimal by strong duality. Interpretation: a labor hour is worth \$1, a unit of the material quota is worth \$0.50, and the retail commitment costs nothing since it is comfortably over-fulfilled.

Note. The dual variable signs above follow the corrected sign conventions in `duality.tex` (see the QA box at the top of this chapter); before the fix, the book’s “Complicated Dual” example would have suggested $y_3 \geq 0$ for the retail constraint, which is wrong for a maximization primal (here it is harmless only because $y_3^* = 0$).

12 Chapter 12: Software – Python

This chapter has nine exercises, all built around the two running PuLP examples of the chapter’s PuLP section (Section 12.12): the product mix LP and the complete transportation LP. The warm-ups (12.1–12.3) modify data in the worked examples; the core problems (12.4–12.6) build a model from scratch, extract duals and slacks, and run an ε -constraint sweep; the concept problems (12.7–12.8) diagnose the two classic PuLP modeling bugs (a missing constraint and a wrong optimization sense); and the challenge problem (12.9) traces optimal profit as a function of a right-hand side. Every solution below was produced by actually running the code shown (PuLP 3.3.2 with the bundled CBC solver); printed output is reproduced verbatim. The book’s Selected Solutions for 12.2, 12.5, and 12.9 were re-executed and confirmed; they are reused and expanded here.

Exercise 12.1 (A New Profit Margin) ★

Only the objective line changes: the profit coefficient of X_1 becomes 5.

```
from pulp import *

prob = LpProblem("Product_Mix", LpMaximize)
x1 = LpVariable("X1", lowBound=0)
x2 = LpVariable("X2", lowBound=0)

prob += 5*x1 + 2*x2, "Total_Profit"      # was 3*x1 + 2*x2
prob += 10*x1 + 5*x2 <= 300, "Material"
prob += 4*x1 + 4*x2 <= 160, "Labor"
prob += 2*x1 + 6*x2 <= 180, "Machine"

prob.solve(PULP_CBC_CMD(msg=0))
print(f"Status: {LpStatus[prob.status]}")
print(f"Maximum Profit: ${value(prob.objective)}")
print(f"X1 = {x1.varValue}")
print(f"X2 = {x2.varValue}")
for name, c in prob.constraints.items():
    print(name, "slack =", c.slack)
```

Actual output:

```
Status: Optimal
Maximum Profit: $150.0
```

```

X1 = 30.0
X2 = 0.0
Material slack = -0.0
Labor slack = 40.0
Machine slack = 120.0

```

The new plan is $(X_1, X_2) = (30, 0)$ with profit \$150. Product 1 is now so profitable per unit of the scarce material (profit 5 per 10 units of material, versus 2 per 5 for Product 2, and material is the first resource to run out) that the company specializes completely. Only the **Material** constraint is binding ($10 \cdot 30 + 5 \cdot 0 = 300$); Labor has slack 40 and Machine has slack 120 (the nonnegativity bound $X_2 \geq 0$ is also active). As promised, the answer is no longer $(20, 20)$.

Exercise 12.2 (From Math to PuLP) ★

The book's Selected Solution is correct as printed; the code below was re-executed to confirm. Following the six-step pattern:

```

from pulp import *

prob = LpProblem("Three_Var", LpMaximize)           # Step 1: problem
x1 = LpVariable("X1", lowBound=0)                  # Step 2: variables
x2 = LpVariable("X2", lowBound=0)
x3 = LpVariable("X3", lowBound=0)

prob += 4*x1 + 6*x2 + 5*x3, "Objective"            # Step 3: objective
prob += 2*x1 + x2 + x3 <= 9, "C1"                  # Step 4: constraints
prob += x1 + 3*x2 + 2*x3 <= 14, "C2"
prob += x1 + x2 + 2*x3 <= 10, "C3"

prob.solve(PULP_CBC_CMD(msg=0))                    # Step 5: solve
print(LpStatus[prob.status], value(prob.objective))
for v in prob.variables():                          # Step 6: results
    print(v.name, "=", v.varValue)

```

Actual output:

```

Optimal 35.0
X1 = 2.0
X2 = 2.0
X3 = 3.0

```

The optimal objective value is 35 at $(x_1, x_2, x_3) = (2, 2, 3)$. All three constraints are binding ($4 + 2 + 3 = 9$, $2 + 6 + 6 = 14$, $2 + 2 + 6 = 10$);

QA flag

Resolved 2026-07-10: the original data for this exercise matched a well-known copyrighted textbook example coefficient for coefficient. The data was replaced with an original instance (verified optimum $(2, 2, 3)$, value 35) in both the book and this manual.

Exercise 12.3 (New Demand Data)

★

Only the demand dictionary changes.

```
from pulp import *

plants = ['Chicago', 'Denver']
markets = ['NYC', 'LA', 'Houston']

supply = {'Chicago': 100, 'Denver': 150}
demand = {'NYC': 100, 'LA': 80, 'Houston': 70}    # updated demand

cost = {
    ('Chicago', 'NYC'): 3, ('Chicago', 'LA'): 5, ('Chicago', 'Houston'): 2,
    ('Denver', 'NYC'): 4, ('Denver', 'LA'): 2, ('Denver', 'Houston'): 3
}

prob = LpProblem("Transportation", LpMinimize)
routes = [(p, m) for p in plants for m in markets]
x = LpVariable.dicts("Ship", routes, lowBound=0)
prob += lpSum([cost[r] * x[r] for r in routes]), "Total_Cost"
for p in plants:
    prob += lpSum([x[p,m] for m in markets]) <= supply[p], f"Supply_{p}"
for m in markets:
    prob += lpSum([x[p,m] for p in plants]) >= demand[m], f"Demand_{m}"

prob.solve(PULP_CBC_CMD(msg=0))
print(f"Status: {LpStatus[prob.status]}")
print(f"Total Cost: ${value(prob.objective)}")
for p in plants:
    for m in markets:
        if x[p,m].varValue > 0:
            print(f"  {p} -> {m}: {x[p,m].varValue}")
```

Actual output:

```
Status: Optimal
Total Cost: $670.0
  Chicago -> NYC: 100.0
  Denver -> LA: 80.0
  Denver -> Houston: 70.0
```

The new total cost is \$670 (up from \$610). CBC's plan sends all of Chicago's supply to NYC and covers LA and Houston from Denver. (There are alternative optima with the same cost, e.g., Denver serving LA and NYC while Chicago covers Houston and part of NYC; students may report a different plan at cost 670.)

One-sentence explanation: total demand rose from 240 to 250 units and shifted from LA (which Denver serves at the system's cheapest rate, \$2) toward NYC (cheapest inbound rate \$3), so the network must ship more units and ship them on more expensive routes. Note also that demand

now exactly equals total supply ($250 = 250$), so neither plant has spare capacity left to exploit cheaper routings.

Exercise 12.4 (A Three-Plant Transportation Model)

★★

```
from pulp import *

# === DATA ===
plants = ['A', 'B', 'C']
markets = ['M1', 'M2', 'M3']
supply = {'A': 60, 'B': 70, 'C': 80}
demand = {'M1': 50, 'M2': 60, 'M3': 70}
cost = {
    ('A','M1'): 4, ('A','M2'): 6, ('A','M3'): 8,
    ('B','M1'): 5, ('B','M2'): 4, ('B','M3'): 3,
    ('C','M1'): 6, ('C','M2'): 7, ('C','M3'): 5
}

# === MODEL ===
prob = LpProblem("Three_Plant_Transport", LpMinimize)
routes = [(p, m) for p in plants for m in markets]
x = LpVariable.dicts("Ship", routes, lowBound=0)

prob += lpSum([cost[r] * x[r] for r in routes]), "Total_Cost"
for p in plants:
    prob += lpSum([x[p,m] for m in markets]) <= supply[p], f"Supply_{p}"
for m in markets:
    prob += lpSum([x[p,m] for p in plants]) >= demand[m], f"Demand_{m}"

# === SOLVE / RESULTS ===
prob.solve(PULP_CBC_CMD(msg=0))
print(f"Status: {LpStatus[prob.status]}")
print(f"Total Cost: {value(prob.objective)}")
for p in plants:
    shipped = sum(x[p,m].varValue for m in markets)
    print(f"Plant {p}: shipped {shipped} of {supply[p]}"
          f" (unused {supply[p]-shipped})")
for p in plants:
    for m in markets:
        if x[p,m].varValue > 0:
            print(f"  {p} -> {m}: {x[p,m].varValue}")
```

Actual output:

```
Status: Optimal
Total Cost: 770.0
Plant A: shipped 50.0 of 60 (unused 10.0)
Plant B: shipped 70.0 of 70 (unused 0.0)
```

Plant C: shipped 60.0 of 80 (unused 20.0)

A -> M1: 50.0

B -> M2: 60.0

B -> M3: 10.0

C -> M3: 60.0

Minimum total cost: \$770. Plan: A supplies all of M1 (50 units at cost 4), B supplies all of M2 (60 at 4) plus 10 units of M3 (at 3), and C supplies the remaining 60 units of M3 (at 5). Plants A and C have unused capacity (10 and 20 units respectively); B, whose routes to M2 and M3 are the cheapest in their columns, is used to capacity. Sanity check by hand: the greedy assignment $M1 \leftarrow A$, $M2 \leftarrow B$, $M3 \leftarrow B$ then C gives $200 + 240 + 30 + 300 = 770$, and the LP confirms no cheaper plan exists.

Exercise 12.5 (Duals and Slacks from a Solved Model)

★★

Complete script (original product mix data, profit $3X_1 + 2X_2$):

```
from pulp import *

prob = LpProblem("Product_Mix", LpMaximize)
x1 = LpVariable("X1", lowBound=0)
x2 = LpVariable("X2", lowBound=0)
prob += 3*x1 + 2*x2, "Total_Profit"
prob += 10*x1 + 5*x2 <= 300, "Material"
prob += 4*x1 + 4*x2 <= 160, "Labor"
prob += 2*x1 + 6*x2 <= 180, "Machine"
prob.solve(PULP_CBC_CMD(msg=0))

print(LpStatus[prob.status], value(prob.objective), x1.varValue, x2.varValue)
for name, c in prob.constraints.items():
    print(name, "dual =", c.pi, "slack =", c.slack)
```

Actual output:

```
Optimal 100.0 20.0 20.0
Material dual = 0.2 slack = -0.0
Labor dual = 0.25 slack = -0.0
Machine dual = -0.0 slack = 20.0
```

1. Material: dual 0.2, slack 0. Labor: dual 0.25, slack 0. Machine: dual 0, slack 20. (CBC prints -0.0 ; that is a signed floating point zero, i.e., 0.) Hand check of the duals: the binding rows give $10y_1 + 4y_2 = 3$ and $5y_1 + 4y_2 = 2$, so $y_1 = 0.2$, $y_2 = 0.25$.
2. The Machine constraint has dual value 0 and slack 20: at the optimum (20, 20), machine usage is $2(20) + 6(20) = 160 < 180$. A resource with leftover capacity cannot be worth anything at the margin, so its shadow price is zero. This is complementary slackness: for each constraint, the dual value or the slack (or both) must be zero.
3. The Labor dual is 0.25, so one extra hour of labor raises the optimal profit by \$0.25, from \$100 to \$100.25 (valid as long as the current basis stays optimal; Exercise 12.9 shows this rate holds up to $b = 168$).

Exercise 12.6 (An Epsilon-Constraint Sweep)

★★

Following the template from the multi-objective chapter (Section 13.3), we re-solve the product mix LP with the extra cap $2X_1 + 6X_2 \leq \varepsilon$.

```
from pulp import *

print(f"{'eps':>4} {'X1':>6} {'X2':>6} {'profit':>7}")
for eps in [60, 90, 120, 150, 180]:
    prob = LpProblem("Product_Mix_Eps", LpMaximize)
    x1 = LpVariable("X1", lowBound=0)
    x2 = LpVariable("X2", lowBound=0)
    prob += 3*x1 + 2*x2
    prob += 10*x1 + 5*x2 <= 300, "Material"
    prob += 4*x1 + 4*x2 <= 160, "Labor"
    prob += 2*x1 + 6*x2 <= 180, "Machine"
    prob += 2*x1 + 6*x2 <= eps, "Machine_cap"
    prob.solve(PULP_CBC_CMD(msg=0))
    print(f"{'eps':>4} {'x1.varValue':>6.1f} {'x2.varValue':>6.1f}"
          f" {'value(prob.objective)':>7.1f}")
```

Actual output:

eps	X1	X2	profit
60	30.0	0.0	90.0
90	27.0	6.0	93.0
120	24.0	12.0	96.0
150	21.0	18.0	99.0
180	20.0	20.0	100.0

Trade-off: tightening the machine cap costs profit, but cheaply. From $\varepsilon = 160$ (where the unconstrained optimum $(20, 20)$ uses exactly 160 machine hours) down to $\varepsilon = 60$, the cap is binding and each machine hour given up costs exactly \$0.10 of profit (the cap's shadow price): profit falls linearly $100 \rightarrow 99 \rightarrow 96 \rightarrow 93 \rightarrow 90$ as ε falls $160 \rightarrow 150 \rightarrow 120 \rightarrow 90 \rightarrow 60$. For $\varepsilon \geq 160$ (including the printed $\varepsilon = 180$) the cap is slack and the solution is the original optimum. Along the sweep the plan shifts steadily from the machine-heavy mix toward Product 1 only: the company can cut machine usage by nearly two thirds (from 160 to 60 hours) while losing only 10% of profit. Each row is a point on the (profit, machine usage) Pareto frontier.

Exercise 12.7 (The Constraint That Wasn't There)

★★

1. Prediction and experiment agree: PuLP neither raises an error nor prints a warning; it happily solves the model you gave it. PuLP cannot know that a constraint is missing; the two-constraint LP is a perfectly valid problem. The run returns status `Optimal`.

```
from pulp import *

prob = LpProblem("Product_Mix_NoLabor", LpMaximize)
x1 = LpVariable("X1", lowBound=0)
```



```

x2 = LpVariable("X2", lowBound=0)
prob += 3*x1 + 2*x2, "Total_Profit"
prob += 10*x1 + 5*x2 <= 300, "Material"
prob += 2*x1 + 6*x2 <= 180, "Machine"      # Labor line is missing
prob.solve(PULP_CBC_CMD(msg=0))
print(LpStatus[prob.status], value(prob.objective), x1.varValue, x2.varValue)
print("Labor used:", 4*x1.varValue + 4*x2.varValue)

```

Actual output:

```

Optimal 102.0 18.0 24.0
Labor used: 168.0

```

2. The returned solution is (18, 24) with profit \$102: the intersection of the Material and Machine lines ($10x_1 + 5x_2 = 300$, $2x_1 + 6x_2 = 180$). The profit exceeds \$100 because dropping a constraint enlarges the feasible region, and the true optimum (20, 20) had Labor binding, so relaxing Labor strictly helps. The plan is *not* implementable: it needs $4(18) + 4(24) = 168$ labor hours but only 160 are available.
3. With `prob += 4*x1 + 4*x2` (no `<= 160`), PuLP treats the expression as a new *objective* and replaces $3x_1 + 2x_2$ with $4x_1 + 4x_2$. Recent PuLP versions print

`UserWarning: Overwriting previously set objective.`

but the model still solves and reports `Optimal` (our run returns (18, 24) with objective value 168, which is the value of $4x_1 + 4x_2$, not profit). This bug is more dangerous than an omitted line because it does two bad things at once, both invisibly: the labor constraint is still missing *and* the objective is now the wrong function, yet every status check passes. An omitted line at least leaves the objective intact. Defensive habits: print `prob` (or `prob.objective`) before solving, and treat any `UserWarning` from PuLP as an error.

QA flag

Book fix applied (sync note): the warn box in the product mix example (Subsection 12.12.3, Step 4) said PuLP *silently* overwrites the objective. Verified with PuLP 3.3.2 that a `UserWarning` is printed. The book text was updated to say a warning is printed but the model still solves. Exercise wording itself needed no change; part (1) of this exercise refers to the omitted-line case, which is truly silent.

Exercise 12.8 (Minimize or Maximize?)

★★

1. Minimizing $3x_1 + 2x_2$ over the feasible region returns $(x_1, x_2) = (0, 0)$ with objective 0, status `Optimal`. Verified:

```

from pulp import *
prob = LpProblem("Product_Mix", LpMinimize)    # wrong sense
x1 = LpVariable("X1", lowBound=0)

```

```

x2 = LpVariable("X2", lowBound=0)
prob += 3*x1 + 2*x2, "Total_Profit"
prob += 10*x1 + 5*x2 <= 300, "Material"
prob += 4*x1 + 4*x2 <= 160, "Labor"
prob += 2*x1 + 6*x2 <= 180, "Machine"
prob.solve(PULP_CBC_CMD(msg=0))
print(LpStatus[prob.status], value(prob.objective), x1.varValue, x2.varValue)

```

Output: `Optimal 0.0 0.0 0.0`. The model is not infeasible because the constraints are all $\leq$ constraints with nonnegative data: the origin satisfies every one of them. Minimizing a nonnegative objective over a region containing the origin gives the useless but perfectly legal answer “produce nothing.”

2. Two one-line fixes, both verified to return the plan (20,20):

- Change the sense: `prob = LpProblem("Product_Mix", LpMaximize)`.
Output: `Optimal 100.0 20.0 20.0`.
- Change the objective (keep `LpMinimize`): minimize the negated profit, `prob += -3*x1 - 2*x2`.
Output: `Optimal -100.0 20.0 20.0`; the optimal plan is the same (20,20) and the profit is the negative of the reported objective, \$100.

3. `LpStatus[prob.status]` only reports whether the solver finished successfully on *the model you built*; the wrong-sense model is a valid LP with a legitimate optimum, so the status is `Optimal` in both the buggy and the correct runs. Useful sanity checks: assert that the objective value is strictly positive (a profit of \$0 from a company trying to make money should raise eyebrows); assert the solution is not identically zero; compare the objective against any known feasible plan (e.g., (10,10) earns \$50, so the optimum must be at least \$50); or check that at least one resource constraint is binding, since a profit maximum with every resource slack is suspicious.

Exercise 12.9 (Profit as a Function of Labor)

★★★

The book’s Selected Solution is correct as printed; here is the full script and its actual output (plans added for part 2’s discussion).

```

from pulp import *

def solve_mix(b):
    prob = LpProblem("Product_Mix", LpMaximize)
    x1 = LpVariable("X1", lowBound=0)
    x2 = LpVariable("X2", lowBound=0)
    prob += 3*x1 + 2*x2
    prob += 10*x1 + 5*x2 <= 300, "Material"
    prob += 4*x1 + 4*x2 <= b, "Labor"
    prob += 2*x1 + 6*x2 <= 180, "Machine"
    prob.solve(PULP_CBC_CMD(msg=0))
    return value(prob.objective), prob.constraints["Labor"].pi, \
           x1.varValue, x2.varValue

```

```

for b in range(100, 200, 10):
    profit, dual, v1, v2 = solve_mix(b)
    print(f"b = {b}: profit = {profit:<6} labor dual = {dual:<5}"
          f" plan = ({v1}, {v2})")

```

Actual output:

```

b = 100: profit = 75.0    labor dual = 0.75    plan = (25.0, 0.0)
b = 110: profit = 82.5    labor dual = 0.75    plan = (27.5, 0.0)
b = 120: profit = 90.0    labor dual = 0.75    plan = (30.0, 0.0)
b = 130: profit = 92.5    labor dual = 0.25    plan = (27.5, 5.0)
b = 140: profit = 95.0    labor dual = 0.25    plan = (25.0, 10.0)
b = 150: profit = 97.5    labor dual = 0.25    plan = (22.5, 15.0)
b = 160: profit = 100.0   labor dual = 0.25    plan = (20.0, 20.0)
b = 170: profit = 102.0   labor dual = -0.0    plan = (18.0, 24.0)
b = 180: profit = 102.0   labor dual = -0.0    plan = (18.0, 24.0)
b = 190: profit = 102.0   labor dual = -0.0    plan = (18.0, 24.0)

```

1. The ten profits are 75, 82.5, 90, 92.5, 95, 97.5, 100, 102, 102, 102. The curve is piecewise linear and concave with slope changes at $b = 120$ and at $b = 168$ (the latter lies between the printed points 160 and 170; it is where the plan (18, 24), the intersection of Material and Machine, uses exactly $4(18) + 4(24) = 168$ labor hours).
2. The slope of the profit curve at each b equals the printed shadow price `c.pi` of the Labor constraint. For $b \leq 120$ only Labor limits production, the plan is $(b/4, 0)$, and each hour is worth 0.75 (profit rises 7.5 per 10 hours: $75 \rightarrow 82.5 \rightarrow 90$). For $120 \leq b \leq 168$, Material and Labor are both binding, the plan slides along the Material line, and an extra hour is worth only 0.25 (profit rises 2.5 per 10 hours). The strictly decreasing sequence of shadow prices $0.75 \rightarrow 0.25 \rightarrow 0$ is exactly the concavity of the curve.
3. For $b \geq 168$ the optimum is pinned at (18, 24), where Material and Machine are binding and Labor is slack; extra labor has shadow price 0 and profit stays flat at \$102. Buying more of a non-binding resource is worthless. This is the right-hand-side sensitivity analysis of Chapter 10 computed by brute force: the loop reproduces the shadow price and its allowable range (the dual stays 0.25 precisely for $120 \leq b \leq 168$) without any tableau work.

13 Chapter 13: Multi-Objective Optimization

This chapter has ten exercises. The warm-ups (13.1–13.3) cover the definition of Pareto optimality, dominance checking in a small table, and weighted-sum scoring. The core problems (13.4–13.8) work the chapter's main methods on concrete data: frontier identification, weighted sums, the ε -constraint method (on a table, on two small LPs, and by hand), and trade-off analysis. The concept problems (13.9–13.10, with 13.10 as the challenge) probe the known blind spot of weighted sums on non-convex frontiers and compute the exact weight intervals that select each vertex in the chapter's furniture example. Every numerical claim below was verified by direct computation; the LP-based exercises (13.6 and the furniture sweeps) were additionally re-solved with PuLP/CBC, and outputs shown are actual. The book's Selected Solutions for 13.1, 13.4, 13.7, and 13.10 were checked and found correct; they are reused and expanded here.

Exercise 13.1 (Defining Pareto Optimality) ★

For a problem minimizing f_1 and f_2 over a feasible set P , a point $\mathbf{x} \in P$ is *Pareto optimal* if there is no $\bar{\mathbf{x}} \in P$ with $f_i(\bar{\mathbf{x}}) \leq f_i(\mathbf{x})$ for both $i = 1, 2$ and $f_i(\bar{\mathbf{x}}) < f_i(\mathbf{x})$ for at least one i . In words: no other feasible point is at least as good in both objectives and strictly better in one. (Equivalently, adapt the book's maximization definition by flipping the inequalities.) The Pareto frontier is the set of all such points.

Exercise 13.2 (Spotting Dominated Designs) ★

The data (both objectives minimized): $G = (2, 9)$, $H = (4, 7)$, $I = (5, 8)$, $J = (7, 3)$, $K = (6, 5)$.

1. Only contract I is dominated: $H = (4, 7)$ beats $I = (5, 8)$ in both cost and delivery time. Checking every other pair, one contract is always cheaper while the other is faster, so no other domination occurs (e.g., $K = (6, 5)$ vs. $J = (7, 3)$: K is cheaper, J is faster).
2. In order of increasing f_1 , the Pareto frontier is

$$G = (2, 9), \quad H = (4, 7), \quad K = (6, 5), \quad J = (7, 3).$$

Note the f_2 values strictly decrease along this list, as they must along a frontier of a two-objective minimization problem.

3. True. If G is the *unique* minimizer of f_1 , every other contract has strictly larger f_1 , so no contract can be at least as good as G in both objectives; hence nothing dominates G . (Uniqueness matters: if two contracts tie for the minimum of f_1 , the one with larger f_2 is dominated by the other.)

Exercise 13.3 (Computing Weighted-Sum Scores) ★

1. With $(w_1, w_2) = (0.6, 0.4)$, $S = 0.6f_1 + 0.4f_2$:

Contract	G	H	I	J	K
S	4.8	5.2	6.2	5.4	5.6

For example, $S_G = 0.6(2) + 0.4(9) = 1.2 + 3.6 = 4.8$. The winner is contract G .

2. With $(w_1, w_2) = (0.2, 0.8)$:

Contract	G	H	I	J	K
S	7.6	6.4	7.4	3.8	5.2

The winner is now contract J . The winner changes because the weights encode the decision maker's exchange rate between the objectives: with most of the weight on delivery time ($w_2 = 0.8$), the contract with by far the best delivery time (J , at 3 days) overcomes its high cost, whereas with the weight mostly on cost, the cheapest contract (G) wins. Geometrically, the two weight vectors correspond to supporting lines of different slopes, which touch the frontier at different points.

3. $S_I - S_H = w_1(5 - 4) + w_2(8 - 7) = w_1 + w_2 > 0$ whenever $w_1, w_2 > 0$. So I scores strictly worse than H for every positive weight pair, and in general a dominated point loses to any point dominating it: the score difference is a nonnegative combination of the coordinatewise gaps, at least one of which is strictly positive.

Exercise 13.4 (Pareto Frontier, Weighted Sum, and ε -Constraint)

★★

The candidates (both objectives minimized): $A = (6, 5)$, $B = (4, 8)$, $C = (7, 4)$, $D = (3, 6)$, $E = (5, 6)$, $F = (10, 3)$.

1. Compare pairs for domination. $D = (3, 6)$ dominates $B = (4, 8)$, and $E = (5, 6)$ is dominated by D ($3 \leq 5$, $6 \leq 6$, with the first strict). A , C , F are not dominated by anything. The Pareto frontier is $\{D, A, C, F\}$ with objective vectors $(3, 6)$, $(6, 5)$, $(7, 4)$, $(10, 3)$.
2. A design is a weighted-sum optimum for some α exactly when it lies on the lower-left convex hull of the frontier points. Plotting $(3, 6)$, $(6, 5)$, $(7, 4)$, $(10, 3)$: the hull is formed by D , C , and F , while A lies above it. Indeed, D (best f_1) and F (best f_2) are always obtainable at extreme weights; C lies on the hull between them (the segment $D \rightarrow F$ has slope $-3/7$, and at $f_1 = 7$ it sits at $6 - 4 \cdot \frac{3}{7} \approx 4.29 > 4$, so C is below that segment, hence on the hull); but $A = (6, 5)$ sits above the segment from D to C (that segment at $f_1 = 6$ has value $6 - \frac{2}{4}(6 - 3) = 4.5 < 5$). So D , C , F are obtainable by weighted sums and A is not.
3. Choose $\varepsilon = 5$: minimizing f_1 subject to $f_2 \leq 5$ leaves candidates $A = (6, 5)$, $C = (7, 4)$, $F = (10, 3)$, and selects $A = (6, 5)$, exactly the frontier point unreachable by any weighted sum. (Any $\varepsilon \in [5, 6)$ works.)
4. Sketch description: plot the six points in the (f_1, f_2) plane. Mark B and E as dominated (B is up and to the right of D ; E is directly right of D). The frontier is the “staircase” of points D, A, C, F running from upper-left to lower-right; improvement arrows point left (lower cost) and down (lower emissions), toward the empty lower-left region. A correct sketch shows that no frontier point has another point in its lower-left quadrant.

Exercise 13.5 (Weighted Sum Method)

★★

In minimization form: $P1 = (f_1, f_2) = (2, -7)$, $P2 = (5, -9)$, $P3 = (3, -6)$.

1. $P1$ and $P2$ are Pareto optimal; $P3$ is not. $P1$ dominates $P3$: it is cheaper ($2 < 3$) and has the higher accessibility score ($7 > 6$). Between $P1$ and $P2$ there is a genuine trade-off ($P1$ cheaper, $P2$ more accessible), so neither dominates the other.
2. Scores $S = 0.5f_1 + 0.5f_2$:

$$S_{P1} = 0.5(2) + 0.5(-7) = -2.5,$$

$$S_{P2} = 0.5(5) + 0.5(-9) = -2.0,$$

$$S_{P3} = 0.5(3) + 0.5(-6) = -1.5.$$

The lowest score wins: project $P1$ is selected.

3. No. Since $P1$ dominates $P3$, $S_{P3} - S_{P1} = w_1(3 - 2) + w_2(-6 + 7) = w_1 + w_2 > 0$ for all $w_1, w_2 > 0$, so $P3$ always scores strictly worse than $P1$. Even the degenerate boundary weights fail to select $P3$: with $w = (1, 0)$ (cost only) $P1$ wins, and with $w = (0, 1)$ (accessibility only) $P2$ wins. A dominated alternative can never be chosen by the weighted-sum method with positive weights.

Exercise 13.6 (Epsilon-Constraint Method)

★★

1. Keeping f_1 as the objective and bounding f_2 :

$$\begin{aligned} \min \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \leq \varepsilon, \\ & x_1 + x_2 \geq 4, \\ & 0 \leq x_1 \leq 5, \quad 0 \leq x_2 \leq 5. \end{aligned}$$

2. For $\varepsilon = 10$: the optimum is $(x_1, x_2) = (3, 1)$ with $f_1 = 5$ and $f_2 = 10$. For $\varepsilon = 6$: the optimum is $(1, 3)$ with $f_1 = 7$ and $f_2 = 6$. In both cases the optimum is the intersection of $3x_1 + x_2 = \varepsilon$ with $x_1 + x_2 = 4$ (along $x_1 + x_2 = 4$ we have $f_1 = 8 - x_1$, so the LP pushes x_1 as high as the ε bound allows: $2x_1 = \varepsilon - 4$). Both solutions are Pareto optimal: each is the unique minimizer of f_1 given $f_2 \leq \varepsilon$, and the ε bound is active, so no feasible point improves either objective without worsening the other. Verified with PuLP:

```
import pulp

for eps in [10, 6]:
    prob = pulp.LpProblem("biobj", pulp.LpMinimize)
    x1 = pulp.LpVariable("x1", lowBound=0, upBound=5)
    x2 = pulp.LpVariable("x2", lowBound=0, upBound=5)
    prob += x1 + 2*x2                                # f1 (primary objective)
    prob += x1 + x2 >= 4
    prob += 3*x1 + x2 <= eps                          # f2 <= epsilon
    prob.solve(pulp.PULP_CBC_CMD(msg=0))
    print(f"eps={eps}: x=({x1.value()}, {x2.value()}), "
          f"f1={x1.value() + 2*x2.value()}, "
          f"f2={3*x1.value() + x2.value()}")
```

Actual output:

```
eps=10: x=(3.0, 1.0), f1=5.0, f2=10.0
eps=6: x=(1.0, 3.0), f1=7.0, f2=6.0
```

3. Minimizing f_2 alone gives $f_2^* = 4$ at $(0, 4)$; minimizing f_1 alone gives $f_1 = 4$ at $(4, 0)$, where $f_2 = 12$. So: for $\varepsilon \geq 12$ the bound is inactive and the solution stays at $(4, 0)$ with $(f_1, f_2) = (4, 12)$. As ε decreases from 12 toward $f_2^* = 4$, the optimum slides along the edge $x_1 + x_2 = 4$ from $(4, 0)$ to $(0, 4)$, with $x_1 = (\varepsilon - 4)/2$ and $f_1 = 8 - (\varepsilon - 4)/2$ increasing linearly from 4 to 8: each unit of tightening in f_2 costs exactly $\frac{1}{2}$ unit of f_1 . The sweep traces the entire Pareto frontier, the segment from $(4, 12)$ to $(8, 4)$ in objective space. For $\varepsilon < 4$ the LP is infeasible, so the sweep stops at f_2^* .

Exercise 13.7 (Epsilon-Constraint by Hand)

★★

1. At $(0, 0)$: $(f_1, f_2) = (0, 0)$. At $(6, 0)$: $(18, 6)$. At $(0, 6)$: $(6, 18)$. So $(6, 0)$ maximizes f_1 alone and $(0, 6)$ maximizes f_2 alone.

2. For $\varepsilon > 6$ the constraint $x_1 + 3x_2 \geq \varepsilon$ is binding at the optimum, and the solution lies where it meets the edge $x_1 + x_2 = 6$: solving the two equations gives $x_2 = (\varepsilon - 6)/2$ and $x_1 = (18 - \varepsilon)/2$. (For $\varepsilon = 6$ the vertex $(6, 0)$ itself is optimal.) Tabulating:

ε	x_1	x_2	f_1	f_2
6	6	0	18	6
10	4	2	14	10
14	2	4	10	14
18	0	6	6	18

Each row is easily confirmed graphically: the sliced region is the triangle cut by the line $x_1 + 3x_2 = \varepsilon$, and the objective $3x_1 + x_2$ increases toward the lower-right, so the optimum sits at the corner of the slice on the hypotenuse.

3. Each tabulated solution has $f_2 = \varepsilon$ exactly and the largest possible f_1 given that bound, so each is Pareto optimal. The Pareto frontier in decision space is the edge $x_1 + x_2 = 6$ from $(6, 0)$ to $(0, 6)$: parametrizing by $x_2 = t$ gives $f_1 = 18 - 2t$ and $f_2 = 6 + 2t$, so the trade-off rate is one for one ($\Delta f_1 / \Delta f_2 = -1$): each unit gained in f_2 costs exactly one unit of f_1 .
4. The maximum of f_2 over X is 18 (at $(0, 6)$), so for $\varepsilon > 18$ the constrained LP is *infeasible*; the sweep should stop at $\varepsilon = 18$. (A practical lesson for implementations: first compute each objective's individual optimum to set the sweep range.)

Exercise 13.8 (Trade-Off Analysis)

★★

1. Plot description: the four points $(40, 20)$, $(45, 14)$, $(55, 10)$, $(70, 8)$ in the (cost, emissions) plane form a decreasing chain; connecting them from upper-left to lower-right gives the (discrete, convex) Pareto frontier. No point lies below-left of another, so all four are on the frontier; improvement is toward the lower-left.
2. Marginal rates of substitution (emissions reduced per \$1000 of extra cost):

Step	Δf_1	Δf_2	tons per \$1000
S1 $\rightarrow$ S2	+5	-6	$6/5 = 1.20$
S2 $\rightarrow$ S3	+10	-4	$4/10 = 0.40$
S3 $\rightarrow$ S4	+15	-2	$2/15 \approx 0.13$

The step S1 $\rightarrow$ S2 gives by far the largest emissions reduction per dollar (1.2 tons per \$1000). The steadily deteriorating ratios ($1.20 \rightarrow 0.40 \rightarrow 0.13$) mean S2 is the natural elbow (knee point) of this frontier.

3. With a budget of at most \$50,000, the feasible Pareto solutions are S1 (\$40k) and S2 (\$45k); the firm should choose **S2**: it fits the budget and has lower emissions (14 vs. 20 tons) than S1, and S3 (\$55k) is unaffordable. (Consistent with part (2), S2 also happens to be the best value step on the frontier.)

Exercise 13.9 (What Weighted Sums Cannot See)

★★

The points, both objectives minimized: $P = (0, 4)$, $Q = (3, 3)$, $R = (4, 0)$.

1. Pairwise: P vs. Q : P is better in f_1 ($0 < 3$), Q better in f_2 ($3 < 4$). Q vs. R : Q better in f_1 , R better in f_2 . P vs. R : P better in f_1 , R better in f_2 . No domination in any pair, so all three are Pareto optimal.
2. The weighted scores with weights $(\alpha, 1 - \alpha)$:

$$S_P(\alpha) = 4(1 - \alpha) = 4 - 4\alpha, \quad S_Q(\alpha) = 3\alpha + 3(1 - \alpha) = 3, \quad S_R(\alpha) = 4\alpha.$$

For any $\alpha \in [0, 1]$, $\min\{S_P, S_R\} = \min\{4 - 4\alpha, 4\alpha\} \leq 2$, since the two values sum to 4 (their minimum is maximized at $\alpha = \frac{1}{2}$, where both equal 2). Hence $\min\{S_P, S_R\} \leq 2 < 3 = S_Q$ for every α : either P or R always scores strictly better than Q , so Q is never a weighted-sum optimum, not even tied.

3. Geometrically, the weighted-sum method minimizes a linear function over the points in objective space, so its optima are exactly the points on the lower-left convex hull, where some supporting line with normal $(\alpha, 1 - \alpha) \geq 0$ touches. Here the hull's lower boundary is the segment from $P = (0, 4)$ to $R = (4, 0)$; its midpoint is $(2, 2)$, and $Q = (3, 3)$ lies strictly above the segment (the line through P and R is $f_1 + f_2 = 4$, while Q has $f_1 + f_2 = 6$). Every supporting line touches only P , R , or the whole segment, never Q . This is precisely the “Cons” entry for the weighted-sum method in Table 13.1: it may miss Pareto optimal solutions in non-convex regions of the frontier, and this three-point frontier is non-convex at Q .
4. The ε -constraint method recovers Q : minimize f_1 subject to $f_2 \leq \varepsilon$ with $\varepsilon = 3$. The feasible points are then Q (with $f_2 = 3$) and R (with $f_2 = 0$), and Q wins with $f_1 = 3 < 4$. (Goal programming with goals $(3, 3)$ and any positive weights also selects Q , whose total deviation is 0.)

Exercise 13.10 (Weights That Select Each Vertex)

★★★

The book's Selected Solution is correct as printed and is expanded here; all breakpoints below were re-verified both algebraically and by a PuLP sweep over α .

1. The weighted objective $\alpha(8x + 2y) - (1 - \alpha)(10x + 2y)$ at each vertex of P :

$$(0, 0) : 0, \quad (0, 20) : 80\alpha - 40, \quad (10, 8) : 212\alpha - 116, \quad (12, 0) : 216\alpha - 120.$$

(For example, $(10, 8)$ has $f_1 = 96$ and $f_2 = 116$, giving $96\alpha - 116(1 - \alpha) = 212\alpha - 116$.)

2. The optimal value is the upper envelope of these four linear functions of α . Comparing pairs: $(0, 20)$ beats $(0, 0)$ iff $80\alpha - 40 \geq 0$, i.e., $\alpha \geq \frac{1}{2}$; $(10, 8)$ beats $(0, 20)$ iff $212\alpha - 116 \geq 80\alpha - 40$, i.e., $132\alpha \geq 76$, i.e., $\alpha \geq \frac{19}{33} \approx 0.576$; and $(10, 8)$ beats $(12, 0)$ iff $4 \geq 4\alpha$, which holds for all $\alpha \leq 1$. Hence

$$(0, 0) \text{ optimal for } \alpha \in [0, \frac{1}{2}], \quad (0, 20) \text{ for } \alpha \in [\frac{1}{2}, \frac{19}{33}], \quad (10, 8) \text{ for } \alpha \in [\frac{19}{33}, 1],$$

with breakpoints $\alpha = \frac{1}{2}$ and $\alpha = \frac{19}{33}$. A numerical sweep confirms the switches (actual PuLP/CBC output of the weighted LP): $\alpha = 0.4999$ returns $(0, 0)$; $\alpha = 0.5001$ and $\alpha = 0.5757$ return $(0, 20)$; $\alpha = 0.5758$ returns $(10, 8)$.

3. At $\alpha = \frac{1}{2}$ the entire edge $x = 0$ is optimal (every plan $(0, y)$, $0 \leq y \leq 20$, scores 0); at $\alpha = \frac{19}{33}$ the whole rosewood edge from $(0, 20)$ to $(10, 8)$ is optimal. At $\alpha = 1$ waste is ignored, so all revenue-optimal plans are optimal: the entire edge from $(10, 8)$ to $(12, 0)$, *including the dominated vertex* $(12, 0)$, which earns the same \$96,000 as $(10, 8)$ but wastes $120 > 116$ bdf. Lesson: a weighted sum with a zero weight can return points that are not Pareto optimal; any strictly positive weight on waste breaks the tie in favor of $(10, 8)$.
4. With revenue in raw dollars, $f_1 = 8000x + 2000y$, the vertex values become 0, $40040\alpha - 40$, $96116\alpha - 116$, and $96120\alpha - 120$, and the same comparisons give breakpoints

$$\alpha = \frac{40}{40040} = \frac{1}{1001} \approx 0.001 \quad \text{and} \quad \alpha = \frac{76}{56076} = \frac{19}{14019} \approx 0.00136 :$$

virtually every weight in $[0, 1]$ now selects $(10, 8)$, and the entire trade-off structure is squeezed into the first 0.14% of the weight range. When objectives live on wildly different scales, the weights lose their intended meaning as relative importances, which is why objectives should be normalized before weighting (Section 13.2).

14 Chapter 14: Graph Algorithms

This chapter's exercises come in two layers. Four in-chapter exercises (numbered 14.1 through 14.4 in order of appearance: Kruskal's algorithm, NetworkX, Prim's algorithm, and a shortest path problem) sit next to the material they drill, and the book answers three of them in its Exercise Answers block. The end-of-chapter Exercises section is arranged in three bands: **Skills** items 1 to 14 (graph drawing, degrees, connectivity, Dijkstra, spanning trees), **Concepts** items 15 to 17 (handshake lemma, correctness of Kruskal and Dijkstra), and **Explorations** items 18 and 19 (social networks and Levenshtein distance), for which we give guidance and rubric notes rather than a single answer. Every shortest path, spanning tree, and degree count below was re-verified computationally with NetworkX, and the book's Exercise Answers block and Selected Solutions all check out. One data typo was fixed in the book (the Bern to Frankfurt travel time read "3.55" instead of "3:55" in the Skills 9 graph). Much of this chapter, including many exercises, is adapted from the Graph Theory chapter of *Math in Society* by David Lippman (CC BY-SA 3.0); that attribution was missing and has now been added to the chapter source header and to the book's Sources and Attribution front matter.

Learning checkpoint (five cities airfare graph)

The book's Exercise Answers block already answers this; we confirm: (a) 5 vertices and 10 edges (every pair of cities is joined, so the graph is K_5); (b) yes, it is connected, since K_5 contains a path between every pair; (c) LA has degree 4, one edge to each other city; (d) Seattle to Dallas to Atlanta is a path (it does not return to its start); (e) LA to Chicago to Dallas to LA is a circuit (it starts and ends at LA).

Exercise 14.1: Min Cost Spanning Tree (Kruskal)

Sort the fifteen edges by weight and scan, rejecting any edge that closes a circuit:

AB	11	add	
BG	13	add	
AE	14	add	
AG	15	reject	closes circuit A–B–G–A
EF	16	add	
CE	17	add	tree complete (5 edges, 6 vertices)

The minimum cost spanning tree is $\{AB, BG, AE, EF, CE\}$ with total weight $11+13+14+16+17 = 71$. This matches the book’s Exercise Answers block (which lists the same edges; the total 71 is worth adding on the board). Verified with NetworkX.

Exercise 14.2: NetworkX

After running the provided loading code, two lines solve the problem:

```
T = nx.minimum_spanning_tree(G)
print(sorted(T.edges(data='weight')), T.size(weight='weight'))
```

Output: edges ($A, B, 11$), ($A, E, 14$), ($B, G, 13$), ($C, E, 17$), ($E, F, 16$) and total weight 71.0, agreeing with the hand computation in Exercise 14.1. Instructors may also ask for `nx.dijkstra_path` on the same graph as a follow-up.

Exercise 14.3: Prim’s Algorithm

Starting from vertex A and always adding the cheapest edge leaving the current tree:

Tree $\{A\}$	cheapest leaving edge AB (11)	add AB
Tree $\{A, B\}$	candidates: BG 13, AE 14, AG 15, BF 23, ...	add BG
Tree $\{A, B, G\}$	candidates: AE 14, AG rejected (both ends in tree), FG 19, ...	add AE
Tree $\{A, B, G, E\}$	candidates: EF 16, EC 17, ...	add EF
Tree $\{A, B, G, E, F\}$	candidates: EC 17, BC 25, ...	add EC

Selection order: AB (11), BG (13), AE (14), EF (16), EC (17). Prim’s algorithm produces the same tree as Kruskal’s, total 71, as expected since the edge weights are distinct (the MST is unique). Verified with NetworkX’s Prim iterator.

QA flag

The floating figure captioned “Prim’s algorithm” that appears right after Exercise 14.3 shows a *different* seven-vertex graph (vertices a through g) with a highlighted solution, and nothing in the text refers to it. Students may mistake it for the answer to Exercise 14.3. Suggest either adding a sentence introducing it as a second worked illustration, or captioning it with its own data, or removing it.

Exercise 14.4: Shortest Path Problem (A to G)

Dijkstra from A. Permanent labels in order: A [0], B [1], C [4], D [4] (via B, since $1 + 3 = 4$ beats $4 + 2 = 6$), E [6] (via D), F [8] (tie: via D $4 + 4$ or via E $6 + 2$), G [13] (via E, since $6 + 7 = 13$ beats $8 + 6 = 14$). The shortest path is

$$A \rightarrow B \rightarrow D \rightarrow E \rightarrow G, \quad 1 + 3 + 2 + 7 = 13,$$

and it is unique. This matches the book’s Exercise Answers entry (“ABDEG, with length 13”). Verified with NetworkX.

Skills 1: Postal carrier graph (drawing)

Rubric item; there is one intended construction. Place a vertex at each street intersection: the six blocks sit in a 2 by 3 arrangement, so the streets form a grid with $3 \times 4 = 12$ intersections. Draw an edge along every street segment that has at least one house (black square) on it, and omit segments with no houses, since the carrier only needs streets “with houses.” Grade on three points: (i) vertices at intersections only, not at houses; (ii) one edge per street segment even when houses sit on both sides of the street (the carrier walks the street once); (iii) segments bordering only the empty block and carrying no houses are omitted. A follow-up worth asking: does the resulting graph have an Euler path (count odd-degree vertices)?

Skills 2: Seven bridges graph (drawing)

The construction mirrors the Königsberg example: shrink each land mass to a single vertex and draw one edge per bridge, giving a multigraph with 7 edges (parallel edges are expected where two bridges join the same pair of banks). Reading the map, the river splits the town into four land masses, so the answer is a 4-vertex, 7-edge multigraph; the degrees must sum to $2 \times 7 = 14$. Grade on: vertices are land masses (not bridges), exactly seven edges, and parallel edges drawn where appropriate. The natural follow-up, which the item’s stem points to, is that an all-bridges walk exists exactly when the number of odd-degree vertices is 0 or 2.

QA flag

Skills 1 and 2 depend on low-resolution raster images (`GraphExercise1.png`, `GraphExercise2.png`), and in Skills 2 the exact assignment of bridges to land-mass pairs is hard to read from the picture, so student answers may legitimately differ in the multigraph details. Consider redrawing both as TikZ figures with unambiguous house and bridge placement, or labeling the land masses in the image.

Skills 3: Dallas driving-time graph (drawing)

Five vertices (Fort Worth, Plano, Mesquite, Arlington, Denton). The table gives all ten pairwise times, so the answer is a complete graph K_5 with edge weights, in minutes: FW–Plano 54, FW–Mesquite 52, FW–Arlington 19, FW–Denton 42, Plano–Mesquite 38, Plano–Arlington 53, Plano–Denton 41, Mesquite–Arlington 43, Mesquite–Denton 56, Arlington–Denton 50. Any planar arrangement is acceptable; grade on all ten weighted edges being present and correctly labeled.

Skills 4: Airfare graph (drawing)

Again a weighted K_5 , on Seattle, Honolulu, London, Moscow, Cairo, with fares: Seattle–Honolulu \$159, Seattle–London \$370, Seattle–Moscow \$654, Seattle–Cairo \$684, Honolulu–London \$830, Honolulu–Moscow \$854, Honolulu–Cairo \$801, London–Moscow \$245, London–Cairo \$323, Moscow–Cairo \$329. This graph is reused in Skills 14, so it is worth having students keep their drawing.

Skills 5: Degrees in the first graph

Label the vertices A (0, 0), B (1, 0), C (0, 1), D (1, 1), E (2, 1). The six edges are AB, BC, BD, BE, CD, DE. Degrees:

$$\deg A = 1, \quad \deg B = 4, \quad \deg C = 2, \quad \deg D = 3, \quad \deg E = 2.$$

Check: the degrees sum to $12 = 2 \times 6$ edges. Verified with NetworkX.

Skills 6: Degrees in the second graph

Label the vertices A (0, 0) at the bottom, B and C the two middle vertices (left and right), D and E the two top vertices (left and right). The five edges are AB, AC, BC, BD, CE. Degrees:

$$\deg A = 2, \quad \deg B = 3, \quad \deg C = 3, \quad \deg D = 1, \quad \deg E = 1,$$

summing to $10 = 2 \times 5$. Verified with NetworkX.

Skills 7: Which graphs are connected (five vertices)

Label each graph's vertices a (0, 0), b (1, 0), c (0, 1), d (1, 1), e (2, 1).

- First graph (edges ab, bd, cd, be): connected; from b one reaches a, d, e directly and c through d.
- Second graph (edges bd, be, de, ac): *not* connected; the triangle $\{b, d, e\}$ and the edge $\{a, c\}$ form two separate pieces.
- Third graph (edges ab, bd, cd, be, de, ac): connected; it contains the first graph plus two more edges.

So the first and third graphs are connected. Verified with NetworkX.

Skills 8: Which graphs are connected (eight vertices)

Label the vertices a (0, 0), b (1, 0), c (2, 0) across the bottom, d (0.5, 1), e (1.5, 1) in the middle, and f (0, 2), g (1, 2), h (2, 2) across the top.

- First graph (edges ab, bc, de, fg, gh): *not* connected; three components (bottom path a–b–c, middle edge d–e, top path f–g–h).
- Second graph (edges ab, be, de, df, gh, eh, ce): connected; e touches b, d, h, c, and from those a, f, g are reached.
- Third graph (edges ab, be, ed, df, fa, gh, hc): *not* connected; the 5-cycle a–b–e–d–f–a and the path g–h–c are separate components.

Only the middle graph is connected. Verified with NetworkX.

Skills 9: Eurail, Bern to Berlin (Dijkstra)

Run Dijkstra from Bern (equivalently from Berlin; the book’s Selected Solution runs it from Bern and is correct). Working in hours:minutes, the labels become permanent in the order

Bern	0:00	
Lyon	3:50	via Bern
Frankfurt	3:55	via Bern
Paris	5:45	via Lyon ($3:50 + 1:55$)
München	7:05	via Frankfurt ($3:55 + 3:10$)
Amsterdam	7:10	via Paris ($5:45 + 1:25$; beats Frankfurt’s $3:55 + 4:00 = 7:55$)
Berlin	12:50	via München ($7:05 + 5:45$; beats Amsterdam’s $7:10 + 6:10 = 13:20$)

Shortest route: Bern $\rightarrow$ Frankfurt $\rightarrow$ München $\rightarrow$ Berlin, total $3:55 + 3:10 + 5:45 = 12:50$. Verified with NetworkX (770 minutes). Note: the Bern–Frankfurt edge label in the book read “3.55”; this has been corrected to “3:55” in the chapter source, matching the value the book’s own Selected Solution uses.

Skills 10: Eurail, Paris to München

Same graph. Dijkstra from München gives permanent labels München 0:00, Frankfurt 3:10, Berlin 5:45, Bern 7:05, Amsterdam 7:10 ($3:10 + 4:00$), Paris 8:35 (via Amsterdam, $7:10 + 1:25$), Lyon 10:30 (via Paris). The shortest route is

$$\text{Paris} \rightarrow \text{Amsterdam} \rightarrow \text{Frankfurt} \rightarrow \text{München}, \quad 1:25 + 4:00 + 3:10 = 8:35.$$

The southern alternative through Lyon and Bern takes $1:55 + 3:50 + 3:55 + 3:10 = 12:50$, nearly four and a half hours longer. Verified with NetworkX (515 minutes).

Skills 11: MST of the Dallas graph

Kruskal on the graph from Skills 3. Sorted scan:

FW–Arlington	19	add
Plano–Mesquite	38	add
Plano–Denton	41	add
FW–Denton	42	add; tree complete (4 edges, 5 vertices)

No edge is rejected before the tree completes. The MST is {FW–Arlington, Plano–Mesquite, Plano–Denton, FW–Denton} with total $19 + 38 + 41 + 42 = 140$ minutes. Verified with NetworkX.

Skills 12: MST for the five buildings

Kruskal, in thousands of dollars: CE 4.0 add, BD 4.3 add, AE 4.4 add, AD 4.7 add, and the tree is complete with no rejections. Total $4.0 + 4.3 + 4.4 + 4.7 = 17.4$, that is \$17,400 per the cost units. This matches the book’s Selected Solution, including its remark that a tree offers no backup path if a cable is cut. Verified with NetworkX.

Skills 13: Virginia rail, Bristol to Washington (Dijkstra)

Dijkstra from Washington. Permanent labels in order: Washington [0], Richmond [115], Charlottesville [140], Lynchburg [210] (via Charlottesville, $140+70$, beating Richmond's $115+125 = 240$), Roanoke [260] (via Charlottesville, $140 + 120$, beating Lynchburg's $210 + 55 = 265$), Bristol [405] (via Roanoke). Shortest route:

Bristol $\rightarrow$ Roanoke $\rightarrow$ Charlottesville $\rightarrow$ Washington, $145 + 120 + 140 = 405$ minutes = 6h45.

Matches the book's Selected Solution. Verified with NetworkX.

Skills 14: Kruskal on the airfare graph

Sorted scan of the ten fares from Skills 4:

Seattle–Honolulu	\$159	add
London–Moscow	\$245	add
London–Cairo	\$323	add
Moscow–Cairo	\$329	reject; closes circuit London–Moscow–Cairo
Seattle–London	\$370	add; tree complete

MST total: $159 + 245 + 323 + 370 = \1097 . An airline alliance could read this tree as the cheapest set of routes that still connects all five cities, for example as a minimal backbone of through-fares, with the caveat that trips between cities in different branches require connections. Matches the book's Selected Solution; verified with NetworkX.

Concepts 15: Odd-degree vertices

No. Each edge contributes exactly 2 to the sum of all vertex degrees, so $\sum_v \deg(v) = 2|E|$ is even (the handshake lemma). If exactly one vertex had odd degree, the sum would be odd, a contradiction. The same argument rules out any *odd number* of odd-degree vertices: 1, 3, 5, and so on are impossible, while every even count (0, 2, 4, ...) can occur. This fact is what makes the Euler path test (at most two odd-degree vertices) well posed.

Concepts 16: Why greedy works for spanning trees

Let T be Kruskal's tree and suppose some spanning tree T^* is cheaper. Pick an edge $e \in T^* \setminus T$. Kruskal's algorithm examined e and rejected it only because e closed a circuit with already-accepted edges; since edges are scanned from cheapest to most expensive, every edge of that circuit costs at most $c(e)$. The circuit cannot lie entirely inside T^* (trees have no circuits), so it contains an edge $f \notin T^*$ with $c(f) \leq c(e)$. Then $T^* - e + f$ is again a spanning tree (removing e splits T^* into two pieces, and the circuit provides the edge f reconnecting them), costs no more than T^* , and agrees with T on one more edge. Repeating the exchange transforms T^* into T without ever increasing cost, so $c(T) \leq c(T^*)$: the greedy tree is optimal.

Negative costs change nothing: the argument compares costs only through the *order* in which Kruskal's algorithm scans edges, never through their signs. This is a genuine contrast with Dijkstra's algorithm (Concepts 17), which does need nonnegativity. The book's Selected Solution for this item is correct.

Concepts 17: Why Dijkstra needs nonnegative weights

Justification: when the algorithm makes vertex u permanent, u carries the smallest tentative distance among all unvisited vertices. Any other route from the source to u must first leave the visited set through some unvisited vertex w , whose tentative label is already $\geq \text{label}(u)$; with nonnegative edge weights the rest of that route can only add length. So no shorter route to u can exist, and finalizing u is safe.

Counterexample with a negative weight (directed, three vertices): edges $A \rightarrow B$ with weight 2, $A \rightarrow C$ with weight 3, and $C \rightarrow B$ with weight -2 . The true shortest path from A to B is $A \rightarrow C \rightarrow B$ with length $3 + (-2) = 1$. Dijkstra from A gives B the tentative label 2, makes B permanent first (since $2 < 3$), and never revisits it, reporting 2 instead of 1. Verified against Bellman–Ford in NetworkX. Any three-vertex example with the same structure earns full credit; students who use an undirected negative edge should note that undirected negative edges make “shortest walk” degenerate, so the directed version is the clean answer.

Explorations 18: Social network (guidance and rubric)

(a) The friendship graph has vertices A through I and the fifteen edges read from the upper triangle of the table: AB, AC, AF, AG, BC, BE, CE, DE, DI, EG, EI, FH, FI, GH, HI. Check: every X above the diagonal is used once; row I is omitted from the table because all of I’s friendships already appear in earlier rows.

(b) The shortest path from A to D has length 3, so A and D are three degrees of separation apart. There are four shortest paths: A–B–E–D, A–C–E–D, A–G–E–D, and A–F–I–D (any one suffices). No length-2 path exists because A’s friends are $\{B, C, F, G\}$ and D’s friends are $\{E, I\}$, which do not intersect. Verified with NetworkX.

(c) Rubric for the movie extension (group activity, no single answer): completeness of the actor graph for the chosen movies (vertices are actors, one edge per shared movie, labeled); correctness of at least one multi-step path between two actors in different movies; quality of the quiz interaction. A natural assessment question is to have each group compute a “degrees of separation” number between two named actors and explain the path. Instructors may connect this to the Six Degrees of Kevin Bacon game and to breadth-first search as the unweighted special case of Dijkstra.

Explorations 19: Levenshtein distance (guidance and rubric)

(a) Any correct star of dictionary words at distance 1 from “moke” earns credit. A convenient list of more than ten: substitutions give make, mike, mode, mole, more, mote, move (position changes inside the word) and coke, joke, poke, woke, yoke, toke (first letter); the insertion smoke also works. Students should then connect pairs of *those* words at distance 1 from each other (for example make–mike, poke–joke) to get a genuine graph rather than only a star. A spell checker uses the graph by suggesting the dictionary neighbors of the misspelled word, that is, the vertices adjacent to “moke.”

(b) Rubric: (i) the weighting scheme is explicit and defensible (for example keyboard adjacency: substituting a letter by a neighboring key is cheap; common phonetic errors cheap; rare substitutions expensive); (ii) Dijkstra is run from “moke” and produces a distance label for each candidate word; (iii) the interpretation is stated: the spell checker ranks suggestions by increasing shortest-path distance, so likelier intended words appear first. Any consistent weights are acceptable; grade the reasoning, not the specific numbers.

QA flag

Minor maintenance notes for the Exercises section. (1) Skills 11 and Skills 14 refer to “problem #3” and “problem #4” by hand-typed number; if items are ever inserted or reordered these references will silently break. Consider labeling the items and using `\ref`. (2) The standalone Exercise 14.4 (Shortest Path Problem) sits between the band introduction and the Skills list, outside all three bands; moving it into Skills (it is a mechanics drill) would make the band structure exact.

15 Chapter 15: Introduction to Integer Programming Formulations

The sixteen exercises of this chapter are grouped in four bands: Warm-ups (15.1 to 15.4: knapsacks, set covering, and a big- M drill), Core problems (15.5 to 15.11: capital budgeting, covering, facility location, coloring, and logical constraints), Concepts and connections (15.12 to 15.14: LP rounding, big- M tightness, and reading the kidney exchange model), and Challenge problems (15.15 and 15.16: piecewise-linear cost and a small flow shop). Every model below was implemented and solved with PuLP (CBC), and every claimed optimal value in the exercise statements checked out, including the hinted values $52/3$ in 15.12, profit \$40 in 15.15, and makespan 12 in 15.16. The book’s six Selected Solutions (15.2, 15.3, 15.8, 15.9, 15.10, 15.11) were all verified; the solution to 15.3 was extended in the book to record the third optimal placement $\{1, 6\}$ alongside $\{2, 5\}$ and $\{3, 4\}$. The exercises appear to be original to this book (no external-source attribution issues arose in this chapter).

Exercise 15.1 (Knapsack Problem) ★

(a) Let $x_i \in \{0, 1\}$ equal 1 if item i is packed, for $i \in \{T(\text{ent}), S(\text{leeping bag}), V(\text{stove}), F(\text{ood}), C(\text{amera})\}$:

$$\begin{aligned} \max \quad & 8x_T + 6x_S + 5x_V + 10x_F + 4x_C \\ \text{s.t.} \quad & 5x_T + 3x_S + 4x_V + 7x_F + 2x_C \leq 15 \\ & x \in \{0, 1\}^5. \end{aligned}$$

The optimum packs the tent, sleeping bag, and food: weight $5 + 3 + 7 = 15$ (the capacity exactly), value $8 + 6 + 10 = 24$. Runner-up loads such as $\{\text{tent, bag, stove, camera}\}$ (weight 14, value 23) fall short.

(b) “If the stove is packed then the food is packed” is the implication trick from the modeling tricks section:

$$x_V \leq x_F.$$

The optimal solution from (a) does not pack the stove, so it remains optimal with value 24. Both versions verified with PuLP.

Exercise 15.2 (Courier Van Knapsack) ★

(a) With $x_i = 1$ if package i is loaded,

$$\max \quad 5x_1 + 3x_2 + 3x_3 + 7x_4 + 4x_5 \quad \text{s.t.} \quad 11x_1 + 3x_2 + 2x_3 + 5x_4 + 4x_5 \leq 15, \quad x \in \{0, 1\}^5.$$

(b) Optimal load: packages 2, 3, 4, 5, with weight $3+2+5+4 = 14 \leq 15$ and payout $3+3+7+4 = \$17$. Taking the heavy package 1 leaves only 4kg for the rest, capping the payout at $5+3 = \$8$. Matches the book's Selected Solution; verified with PuLP.

Exercise 15.3 (Covering a Small Town) ★

(a) With the district itself included (as in the fire station example):

$$V_1 = \{1, 2, 4\}, V_2 = \{1, 2, 3, 5\}, V_3 = \{2, 3, 6\}, V_4 = \{1, 4, 5\}, V_5 = \{2, 4, 5, 6\}, V_6 = \{3, 5, 6\}.$$

(b) With $x_i = 1$ if a station is placed in district i :

$$\min \sum_{i=1}^6 x_i \quad \text{s.t.} \quad \sum_{j \in V_i} x_j \geq 1 \quad \text{for } i = 1, \dots, 6, \quad x \in \{0, 1\}^6.$$

(c) The optimum is 2 stations. One station cannot suffice since no V_i contains all six districts (the largest have four elements). Enumerating all pairs shows exactly three optimal placements: $\{2, 5\}$, $\{3, 4\}$, and $\{1, 6\}$; for example $\{1, 6\}$ works because $1 \in V_1 \cap V_2 \cap V_4$ and $6 \in V_3 \cap V_5 \cap V_6$. Verified with PuLP plus complete enumeration. (The book's Selected Solution originally listed only $\{2, 5\}$ and $\{3, 4\}$; the third placement has been added there.)

Exercise 15.4 (Big-M Activation Drill) ★

(a) The policy “if $\delta = 0$ then $x_1 + x_2 + x_3 \leq 40$ ” becomes the single constraint

$$x_1 + x_2 + x_3 \leq 40 + M\delta.$$

When $\delta = 0$ the cap of 40 is enforced; when $\delta = 1$ the constraint must be inert.

(b) With $0 \leq x_i \leq 30$, the left side is at most 90. Inert means $40 + M \geq 90$, so the smallest valid constant is

$$M = 90 - 40 = 50.$$

Any smaller M would wrongly cut off legal solutions such as $x_1 = x_2 = x_3 = 30$ with a second guide hired.

Exercise 15.5 (Locating Distribution Terminals) ★★

The three conditions translate directly: (1) is a cardinality constraint, (2) says $y_1 = 0 \Rightarrow y_2 = 1$, i.e. $y_1 + y_2 \geq 1$, and (3) is a covering constraint:

$$\begin{aligned} \min \quad & 10y_1 + 12y_2 + 20y_3 + 18y_4 + 25y_5 \\ \text{s.t.} \quad & y_1 + y_2 + y_3 + y_4 + y_5 \leq 2 \\ & y_1 + y_2 \geq 1 \\ & y_3 + y_4 + y_5 \geq 1 \\ & y \in \{0, 1\}^5. \end{aligned}$$

The exercise only asks for the model, but the instance is worth solving in class: constraints (2) and (3) force one Maritimes terminal and one central terminal, the budget of two makes it exactly one of each, and the cheapest pair is Halifax plus Ottawa, $y_1 = y_4 = 1$, cost \$28 million. Verified with PuLP.

Exercise 15.6 (Warehouse Location)

★★

(a) Let $y_i = 1$ if warehouse $i \in \{A, B, C, D\}$ opens. Each region needs an open warehouse that serves it:

$$\begin{aligned}
 \min \quad & 150y_A + 200y_B + 120y_C + 180y_D \\
 \text{s.t.} \quad & y_A + y_C \geq 1 && (\text{region 1}) \\
 & y_A + y_B \geq 1 && (\text{region 2}) \\
 & y_A + y_B + y_D \geq 1 && (\text{region 3}) \\
 & y_B + y_C + y_D \geq 1 && (\text{region 4}) \\
 & y_B + y_D \geq 1 && (\text{region 5}) \\
 & y \in \{0, 1\}^4.
 \end{aligned}$$

The optimum opens B and C for $200 + 120 = \$320\text{K}$: C covers regions 1 and 4, B covers 2, 3, 4, 5. For comparison, A with D costs 330 and A with B costs 350; C with D misses region 2. Verified with PuLP.

(b) This is a (weighted) *set covering* problem: warehouses are the sets, regions are the elements to cover, and fixed costs replace the unit objective.

Exercise 15.7 (Capital Budgeting with Dependencies)

★★

(a) With $x_j = 1$ if project j is funded:

$$\begin{aligned}
 \max \quad & 20x_1 + 40x_2 + 20x_3 + 15x_4 + 30x_5 \\
 \text{s.t.} \quad & 5x_1 + 4x_2 + 3x_3 + 7x_4 + 8x_5 \leq 20 && (\text{year 1}) \\
 & x_1 + 7x_2 + 9x_3 + 4x_4 + 6x_5 \leq 20 && (\text{year 2}) \\
 & x_3 \leq x_1 && (3 \text{ only with } 1) \\
 & x_2 + x_5 \leq 1 && (2 \text{ and } 5 \text{ conflict}) \\
 & x \in \{0, 1\}^5.
 \end{aligned}$$

(b) Optimum: fund projects 1, 2, and 3 for NPV $20 + 40 + 20 = 80$ (\$80,000). Outlays: year 1 uses $5 + 4 + 3 = 12 \leq 20$; year 2 uses $1 + 7 + 9 = 17 \leq 20$. Adding project 4 would push year 2 to 21 and is infeasible; swapping 2 for 5 drops the NPV to 70. Verified with PuLP.

Exercise 15.8 (Facility Location: Two Distribution Centers)

★★

(a) Total demand is $10 + 12 + 15 = 37$ against total capacity $40 + 25 = 65$, so the instance is feasible. Center 1 alone ($u_1 = 40 \geq 37$) can serve everything; center 2 alone cannot ($25 < 37$).

(b) Following the capacitated facility location model, with $y_i \in \{0, 1\}$ and $x_{ij} \geq 0$ units of store j 's demand served by center i :

$$\begin{aligned}
 \min \quad & 80y_1 + 60y_2 + \sum_{i,j} c_{ij}x_{ij} \\
 \text{s.t.} \quad & x_{1j} + x_{2j} = d_j && j = 1, 2, 3 \\
 & x_{11} + x_{12} + x_{13} \leq 40y_1 \\
 & x_{21} + x_{22} + x_{23} \leq 25y_2 \\
 & x_{ij} \geq 0, y_i \in \{0, 1\},
 \end{aligned}$$

with c and d as tabulated.

(c) Optimum: open center 1 only and serve all demand from it, $x_{11} = 10$, $x_{12} = 12$, $x_{13} = 15$:

$$\text{cost} = 80 + 2(10) + 3(12) + 1(15) = 80 + 71 = \$151.$$

Opening center 2 as well would add its \$60 fixed cost and save at most $2 \cdot 12 = 24$ on serving store 2 (its only cheaper assignment), so the cheaper center is not worth opening. Matches the book's Selected Solution; verified with PuLP.

Exercise 15.9 (Coloring a Five-Node Graph)

★★

(a) With at most $n = 5$ colors, $x_{ij} = 1$ if vertex i gets color j and $w_j = 1$ if color j is used:

$$\begin{aligned} \min \quad & \sum_{j=1}^5 w_j \\ \text{s.t.} \quad & \sum_{j=1}^5 x_{ij} = 1 \quad i \in \{A, \dots, E\} \\ & x_{ij} + x_{kj} \leq w_j \quad \{i, k\} \in E, j = 1, \dots, 5 \\ & x_{ij}, w_j \in \{0, 1\}. \end{aligned}$$

(b) The solver returns $\sum_j w_j = 3$, so the chromatic number is 3. One optimal coloring (relabelling the solver's colors): $A \rightarrow 1$, $B \rightarrow 2$, $C \rightarrow 3$, $D \rightarrow 2$, $E \rightarrow 3$; each of the six edges AB, BC, CD, DE, EA, AC joins different colors.

(c) The chord $\{A, C\}$ makes A, B, C pairwise adjacent, a triangle, so any proper coloring needs at least 3 colors. The coloring in (b) attains 3, so the bound is tight. (A 5-cycle alone already needs 3 colors, being an odd cycle; the triangle is simply the quickest certificate.) Matches the book's Selected Solution; verified with PuLP.

Exercise 15.10 (Either-Or Constraints)

★★

Introduce $\delta \in \{0, 1\}$ with $\delta = 0$ selecting mode A and $\delta = 1$ selecting mode B:

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 10 + M_1\delta \\ & x_1 + 3x_2 \leq 12 + M_2(1 - \delta) \\ & x_1, x_2 \geq 0, \quad \delta \in \{0, 1\}. \end{aligned}$$

Any sufficiently large constants work (the book's Selected Solution uses $M_1 = M_2 = 20$); Exercise 15.13 shows the tightest choices are $M_1 = 14$ and $M_2 = 18$. As a check, the optimum is $\delta = 1$ (mode B) with $x = (12, 0)$ and objective 60; note $2x_1 + x_2 = 24 = 10 + 14$, so the relaxed mode A constraint is exactly what $M_1 = 14$ allows. Verified with PuLP for both $M = 20$ and $M = 10^6$.

QA flag

The statement says “Exactly one of these two constraints must hold,” but the standard big- M construction (and the section it references) enforces *at least* one; solutions satisfying both constraints, such as the origin, remain feasible, and excluding them is neither possible in a useful sense nor intended. Suggest rewording to “at least one of these two constraints must hold (the machine operates in one mode)” to match the inclusive-or semantics of the Either Or Constraints subsection.

Exercise 15.11 (Fixed-Charge Production)

★★

(a) Let $x \geq 0$ be units produced and $y \in \{0, 1\}$ indicate startup:

$$\max 15x - 8x - 500y = 7x - 500y \quad \text{s.t.} \quad x \leq 200y, \quad x \geq 0, \quad y \in \{0, 1\}.$$

Optimum: $y = 1$, $x = 200$, profit $7(200) - 500 = \$900$.

(b) The linking constraint $x \leq 200y$ does double duty. If $y = 0$ it forces $x = 0$: no production without paying the fixed charge, which is exactly the fixed-cost logic (without it, the model would produce while claiming $y = 0$ and skip the \$500). If $y = 1$ it becomes the capacity constraint $x \leq 200$. The constant 200 is the tightest valid big- M here because it is the true capacity. Matches the book's Selected Solution; verified with PuLP.

Exercise 15.12 (Why Rounding the Relaxation Fails)

★★

(a) The knapsack example has weights $(12, 2, 1, 1, 4)$ and values $(4, 2, 2, 1, 10)$ with capacity 15. Value per kg: $1/3, 1, 2, 1, 5/2$. Greedy fills items 5, 3, 2, 4 completely (total weight 8, value 15), then puts the remaining 7 kg into item 1: $x_1 = 7/12$, all other $x_i = 1$, LP value

$$15 + \frac{7}{12} \cdot 4 = \frac{52}{3} \approx 17.33,$$

as the exercise states. Verified with PuLP ($x_1 = 0.5833$, value 17.333).

(b) Rounding x_1 up gives total weight $12 + 8 = 20 > 15$: infeasible. Rounding down gives $x = (0, 1, 1, 1, 1)$, weight 8, value 15, which happens to equal the true integer optimum (PuLP confirms the IP optimum is 15).

(c) The luck in (b) is not generic. Three failure modes: (i) rounding can be *infeasible*, as rounding up just showed, and with equality constraints it almost always is: in the min-coins model $p + 5n + 10d + 25q = 83$, rounding a fractional LP solution changes the left side and the equation no longer holds, so the rounded point is simply not a way to make 83 cents; (ii) rounding can be feasible but far from optimal, since the nearest integer point to the LP optimum can be a poor solution, and in fact the integer optimum can sit far from the LP optimum in both position and value; (iii) for binary decision variables like the terminal choices y_i of Exercise 15.5, a relaxation value $y_i = 0.5$ carries no meaning at all: one cannot half-build a terminal in Halifax, and rounding logical variables can violate linked constraints such as $y_1 + y_2 \geq 1$ or $x_3 \leq x_1$. These are exactly the reasons branch and bound, rather than rounding, is the workhorse for integer programs.

Exercise 15.13 (When is Big-M Too Big?)

★★

(a) With $M = 10^6$, a fractional $\delta = 10^{-5}$ adds $M\delta = 10$ units of slack to the mode A constraint: the relaxation can behave as if the right-hand side were 20 instead of 10 while paying an integrality violation of only 10^{-5} . The LP relaxation therefore sails far outside the true feasible region and its bound on the optimum is very weak, so branch and bound must do more work to close the gap. (Tolerance interaction is worse still: solvers accept δ within about 10^{-6} of integer, and $10^{-6} \times 10^6 = 1$, a full unit of phantom slack in an “integer feasible” solution.)

(b) The constant M_1 must bound $2x_1 + x_2 - 10$ whenever the mode B constraint holds. Maximizing $2x_1 + x_2$ subject to $x_1 + 3x_2 \leq 12$, $x \geq 0$ puts everything on x_1 : the maximum is 24 at $(12, 0)$, so $M_1 = 24 - 10 = 14$ suffices. Symmetrically, maximizing $x_1 + 3x_2$ subject to $2x_1 + x_2 \leq 10$, $x \geq 0$ gives 30 at $(0, 10)$, so $M_2 = 30 - 12 = 18$. Both maxima were confirmed by LP, and both are attained, so no smaller constants are valid.

(c) The text's Choosing Big-M Values paragraph names numerical instability: very large coefficients mixed with small ones produce ill-conditioned bases and floating-point roundoff, and (as computed in (a)) they let integrality tolerances translate into materially violated logic.

Exercise 15.14 (Reading the Kidney Exchange Model) ★★

(a) $\sum_{c \ni i} x_c \leq 1$ says each patient-donor pair joins at most one selected cycle. Medically, a donor has one kidney to give: if the constraint were dropped, the optimizer could schedule the same donor in two different swap cycles, and one of the two recipients would be prepped for a transplant whose kidney does not exist.

(b) A selected cycle c performs exactly $|c|$ transplants, one per pair around the cycle (each patient in c receives a kidney from the previous pair's donor). Because part (a)'s constraint prevents any pair from being counted in two cycles, $\sum_c |c|x_c$ adds up transplants without double counting.

(c) All surgeries in a cycle must run simultaneously so no donor can back out after their partner has received a kidney; a length- k cycle ties up $2k$ operating rooms and surgical teams at once, which hospitals can staff only for $k \leq 3$. Raising the cap makes $|C|$ explode combinatorially: the number of candidate cycles of length k grows roughly like the number of ordered k -subsets of pairs (order $|P|^k$), which is why long-cycle and long-chain variants need column generation rather than full enumeration.

(d) Add a set A of altruistic donors and a set H of feasible *chains*: sequences that start at some altruist $a \in A$ and pass through distinct pairs, each donor compatible with the next patient, up to a length cap L . Introduce $x_h \in \{0, 1\}$ for each chain, extend the packing constraints so that each pair is in at most one selected cycle *or* chain, add $\sum_{h \ni a} x_h \leq 1$ for each altruist, and credit each chain with its number of patient transplants (the number of pairs it contains) in the objective:

$$\max \sum_{c \in C} |c|x_c + \sum_{h \in H} n_h x_h \quad \text{s.t.} \quad \sum_{c \ni i} x_c + \sum_{h \ni i} x_h \leq 1 \quad \forall i \in P, \quad \sum_{h \ni a} x_h \leq 1 \quad \forall a \in A.$$

Exercise 15.15 (Piecewise-Linear Production Cost) ★★★

(a) Accumulating the segment costs: $C(0) = 0$, $C(10) = 50$, $C(20) = 50 + 30 = 80$, $C(30) = 80 + 80 = 160$. The marginal costs run 5, 3, 8: they decrease and then increase, so C is not convex (concretely, $C(10) = 50 > 40 = \frac{1}{2}(C(0) + C(20))$). The standard LP epigraph trick $y \geq$ each segment line only works when the cost is convex, so this problem genuinely needs the SOS2 or binary machinery.

(b) SOS2 formulation with weights on the four breakpoints:

$$\begin{aligned} \max \quad & 6x - y \\ \text{s.t.} \quad & x = 0\lambda_1 + 10\lambda_2 + 20\lambda_3 + 30\lambda_4 \\ & y = 0\lambda_1 + 50\lambda_2 + 80\lambda_3 + 160\lambda_4 \\ & \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1, \quad \lambda \geq 0 \\ & \{\lambda_1, \lambda_2, \lambda_3, \lambda_4\} \text{ SOS2.} \end{aligned}$$

Equivalent binary formulation from the text, with $z_i = 1$ selecting segment i :

$$\sum_{i=1}^3 z_i = 1, \quad \lambda_1 \leq z_1, \quad \lambda_2 \leq z_1 + z_2, \quad \lambda_3 \leq z_2 + z_3, \quad \lambda_4 \leq z_3, \quad z_i \in \{0, 1\}.$$

(c) Solving (PuLP with the binary formulation) gives $x = 20$, $C = 80$, profit $6(20) - 80 = \$40$, with $\lambda_3 = 1$, $z_2 = 1$. The marginal picture explains the stopping point: on the discounted middle segment each extra ton earns $6 - 3 = 3$, so the plant produces through it, but on the overtime segment each ton loses $6 - 8 = 2$, so production stops exactly at the segment boundary $x = 20$. (Endpoints check: profit is 0, 10, 40, 20 at $x = 0, 10, 20, 30$.)

Exercise 15.16 (Two Jobs, Three Machines)

★★★

(a) Let $s_{jk} \geq 0$ be the start of job j on machine k , with processing times p_{jk} from the table. Precedence within each job (both jobs route $M_1 \rightarrow M_2 \rightarrow M_3$):

$$s_{j,k+1} \geq s_{j,k} + p_{j,k}, \quad j = 1, 2, \quad k = 1, 2.$$

(b) For each machine k let $\delta_k = 1$ if job 1 precedes job 2 on k :

$$s_{2k} \geq s_{1k} + p_{1k} - M(1 - \delta_k), \quad s_{1k} \geq s_{2k} + p_{2k} - M\delta_k.$$

A valid M from the data is the sum of all processing times, $M = 3 + 2 + 4 + 2 + 4 + 3 = 18$: no start time exceeds 18 in any schedule without idle padding, so the inactive constraint is always released.

(c) Minimizing $C_{\max}$ with $C_{\max} \geq s_{j3} + p_{j3}$ for both jobs, CBC returns an optimal makespan of **12**, with job 1 first on all three machines:

	M_1	M_2	M_3
Job 1	0–3	3–5	5–9
Job 2	3–5	5–9	9–12

The schedule is tight everywhere: job 2 follows job 1 immediately on each machine and finishes at 12. Putting job 2 first instead yields makespan 13 (job 1 would finish M_3 at 13), and mixed orders are worse, so 12 is optimal, as the exercise promises. A quick lower bound for discussion: each job alone needs 9 hours, and M_2 cannot start before time 2, so 12 leaves little slack. Verified with PuLP.